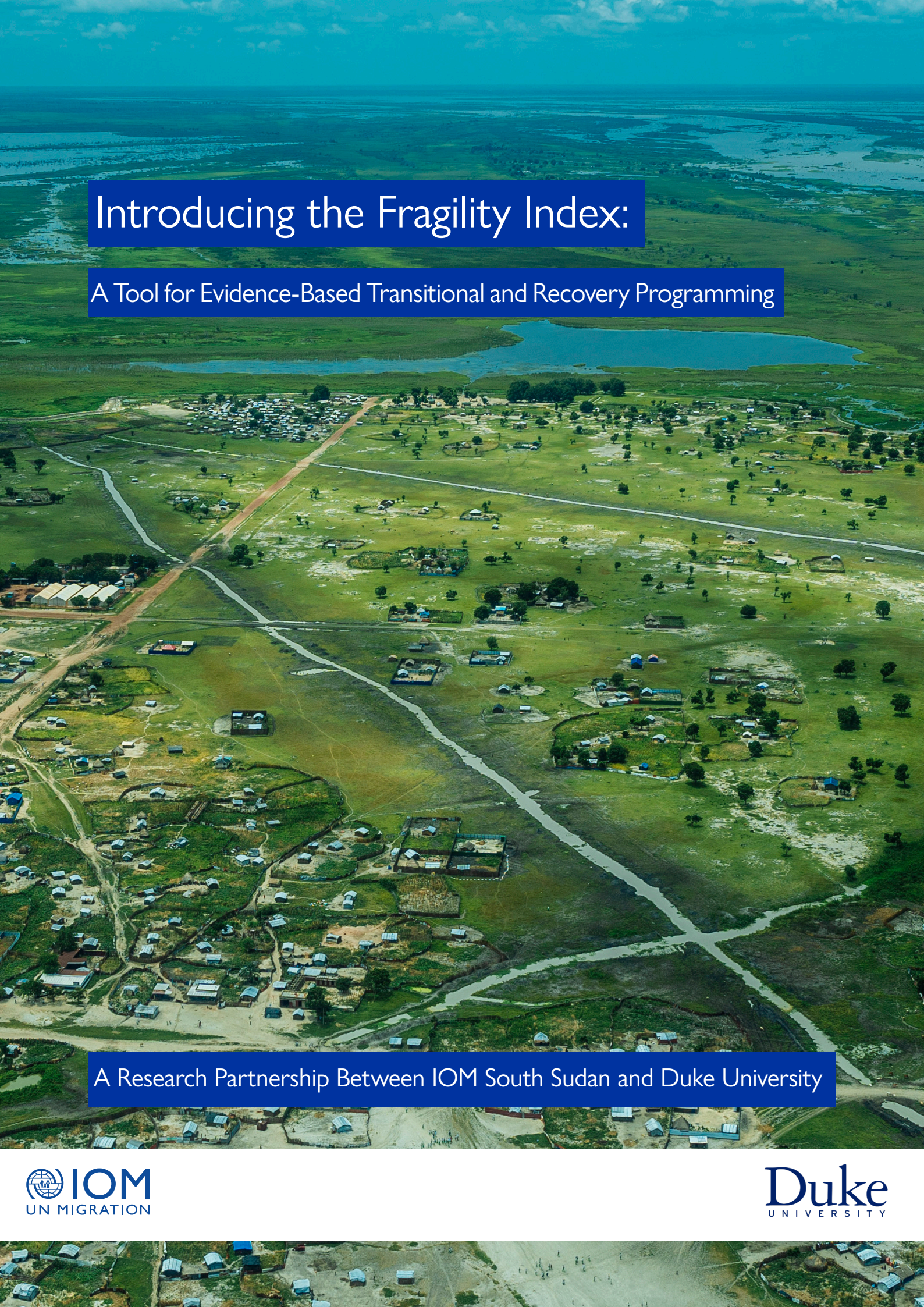


An aerial photograph of a rural settlement in South Sudan. The landscape is a mix of green fields and brown, possibly flooded or eroded, areas. A large body of water is visible in the background. A dirt road runs through the settlement, and several small clusters of buildings are scattered across the landscape.

Introducing the Fragility Index:

A Tool for Evidence-Based Transitional and Recovery Programming

A Research Partnership Between IOM South Sudan and Duke University

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A Research Partnership Between IOM South Sudan and Duke University

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⁴ Olga Aymerich, Mara Revkin and Samuel Akau, "[Understanding Multidimensional Fragility in South Sudan](#)," (Juba, IOM DTM, 2023).

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LIST OF ACRONYMS

CRA	Compensation and Reparation Authority
CTRH	Commission for Truth, Reconciliation and Healing
D4S	Data for Solutions
DSI	Durable Solutions Index
DTM	Displacement Tracking Matrix
EU DGINTPA	European Union Directorate General for International Partnerships
GBV	Gender-Based Violence
GDP	Gross Domestic Product
GPAA	Greater Pibor Administrative Area
HDP	Humanitarian-Development-Peace
HDPN	Humanitarian-Development-Peace Nexus
IDPs	Internally Displaced Persons
IMF	International Monetary Fund
IOM	International Organization for Migration
IPC	Integrated Food Security Phase Classification
MIT	Migration Integration Toolkit
NAS	National Salvation Front
NFI	Non-Food Items
OECD	Organisation for Economic Co-operation and Development
POC	Protection of Civilians
R-ARCSS	Revitalised Agreement on the Resolution of the Conflict in the Republic of South Sudan
RSF	Rapid Support Forces
RSRTF	South Sudan Reconciliation, Stabilization, and Resilience Trust Fund
SAF	Sudan Armed Forces
SI	Stability Index
SMI	Solutions Mobility Index
SPLM	Sudan People's Liberation Movement
SPLM-IO	Sudan People's Liberation Movement-in-Opposition
SSDM/A-Cobra Faction	South Sudan Defence Movement/Army – Cobra Faction
SSP	South Sudanese Pound
SSPDF	South Sudan People's Defence Forces
TGoNU	Transitional Government of National Unity
TRU	Transition and Recovery Unit
UN	United Nations
UNHCR	United Nations High Commissioner for Refugees
WASH	Water, Sanitation and Hygiene
WFP	World Food Programme

Note from Chief of Mission



South Sudan stands at a crossroads. In 2025, the country faces overlapping crises: the devastating spillover from Sudan's conflict, historic flooding that has displaced hundreds of thousands, and an economic collapse that has eroded livelihoods and access to basic services. Yet, amid these challenges, communities across South Sudan continue to demonstrate extraordinary resilience and determination to recover.

This report – a collaboration between IOM South Sudan and Duke University – comes at a pivotal moment. It introduces the Fragility Index, an innovative tool that transforms how we understand and respond to crisis and recovery dynamics in South Sudan. By measuring fragility across six interlinked dimensions – economic,

human, environmental, political-legal, security, and societal – the Index provides detailed perception data that can inform where, how, and with whom we should and can engage to maximize impact.

For IOM, this represents a practical step toward making evidence-based programming a cornerstone of transition and recovery efforts. The Index allows us to identify geographic and thematic priorities for investment; to understand where market-oriented and other resilience-based interventions are feasible; and to adapt programming over time as local contexts evolve. It strengthens our area-based approach, enabling comparative analysis across counties, payams, and bomas to better align needs, capacities, and opportunities for durable solutions.

For the wider UN system, government and partners, this research provides an important opportunity to ground collective planning in data. It highlights not only where fragility is most acute, but also where there is potential for progress. It reinforces that displacement cannot be addressed by humanitarian assistance alone. Durable solutions require investments in resilience, livelihoods, and governance, and they must be informed by the realities that households themselves face every day.

The findings in this report also challenge us to think differently about aid targeting. They show that vulnerability does not always map neatly onto displacement status; many host communities and returnees experience equal or greater fragility. This evidence reinforces the need for area-based, inclusive, and government-led approaches – approaches that reduce dependency, strengthen local systems, and rebuild trust between communities and institutions.

The Fragility Index is a shared resource for all actors working across the Humanitarian-Development-Peace Nexus. Its continued use and updating will enable us to monitor progress, adjust interventions as contexts shift, and contribute to accountability for results.

As South Sudan continues its path toward recovery and resilience, IOM remains committed to working alongside government, communities and partners to translate this data into actionable strategies that help bridge emergency response and long-term development contributing to real, lasting change for the people of South Sudan.

Vijaya Souri
IOM South Sudan Chief of Mission

FRONT MATTER

Executive Summary

The International Organization for Migration (IOM) in South Sudan has partnered with an interdisciplinary research team at Duke University to develop a novel Fragility Index which is designed to support humanitarian and development actors in moving beyond emergency response programming towards transitional and recovery programming. The Fragility Index, a new addition to IOM's global Data for Solutions (D4S) initiative,⁵ is a valuable tool for evidence-based programming that can be used at design, targeting, and impact evaluation stages. It measures six different dimensions of fragility (economic, environmental, human, political-legal, security, and societal) with household-level survey data. The Index allows for geographic disaggregation of the six kinds of fragility as well as analysis of sub-groups within the general population based on rural or urban residence, displacement status, and more. This report presents the findings of a proof-of-concept survey, which gathered data from a random sample of 3,685 respondents across nine counties in South Sudan between April and May 2024.⁶ The survey built on a smaller pilot study conducted in June-July 2023.⁷ This report expands on a summary of preliminary findings published in October 2024.⁸ The results suggest that, although South Sudan ranks as one of the most fragile countries in the world, there is important sub-national variation between counties with higher and lower levels of fragility. Further, the six dimensions of fragility do not all operate similarly; areas that are high in one kind of fragility may be low in other kinds. The report demonstrates how the Fragility Index can be used to inform programmatic and policy recommendations for recovery and development efforts in South Sudan. It also provides guidance for adapting and replicating the methodology for use by humanitarian and development actors working in other fragile contexts.



⁵ International Organization for Migration Displacement Tracking Matrix (IOM DTM), "[Data for Solutions](#)," (Geneva, IOM).

⁶ See Appendix D for a full list of sampled bomas.

⁷ Olga Aymerich, Mara Revkin and Samuel Akau, "[Understanding Multidimensional Fragility in South Sudan](#)," (Juba, IOM DTM, 2023).

⁸ Gabriella Levy and Mara Revkin, "[The Fragility Index: A Tool for Evidence-Based Transitional and Recovery Programming](#)," (Juba, IOM DTM, 2024).

1. INTRODUCTION

IOM in South Sudan has partnered with an interdisciplinary research team at Duke University to develop a novel Fragility Index which is designed to support humanitarian and development actors in moving beyond emergency response programming towards transitional and recovery programming. The Fragility Index, a new addition to IOM's global D4S initiative, is a valuable tool for evidence-based programming that can be used at design, targeting, and impact evaluation stages. This report details the results of a proof-of-concept pilot survey, which gathered data from a random sample of 3,685 respondents across nine counties in South Sudan between April and May 2024, building on a smaller pilot study conducted in June-July 2023.⁹

Objectives

The concept of “fragility” is defined by the Organisation for Economic Co-operation and Development (OECD) as “the combination of exposure to risk and insufficient coping capacities ... to manage, absorb and mitigate those risks.”¹⁰ Fragility has received growing attention in the context of the United Nations (UN) Secretary-General's 2017 calls for prioritizing conflict prevention and connecting the pillars of development, peace and security, and human rights¹¹ as well as amidst the World Bank's revised 2020-2025 strategy on fragility, conflict, and violence. This strategy emphasizes a “pivot” to prevention.¹² However, multidimensional fragility – as defined in this study – has previously been measured only at the country level, obscuring important subnational variation.

The Fragility Index measures six dimensions of fragility — societal, economic, political/legal, security, environmental, and human — with household-level survey data, providing nuanced snapshots of local vulnerabilities and capacities at different geographical scales, such as the household, the village, and the county levels. In doing so, it generates high-quality data for solutions and informs strategic, evidence-based decisions that support equitable resource allocation and foster resilience in South Sudanese communities. Each of the six

dimensions is measured using five simple questions about the respondent's household, each of which is designed to capture a distinct concept, for a total of 30 questions. This module of 30 Index questions was embedded in a longer questionnaire that included descriptive questions about respondents' perceptions and lived experiences. For example, the human fragility index includes a question about access to clean water. Additional questions on the survey inquire about potential barriers to accessing clean water, including distance from the nearest water source, insecurity, poor water quality, insufficient water, and lack of infrastructure. The survey also included several open-ended questions that enabled respondents to describe in their own words what they perceive to be the most important unmet needs in their communities. These open-ended questions provide valuable qualitative data that we use to interpret the quantitative survey data and to inform our recommendations for policy and programming.

By providing fine-grained data on variation in different dimensions of fragility (economic, environmental, political/legal, security, and societal), the Index can support evidence-based programming in line with the Humanitarian-Development-Peace Nexus (HDPN) in the following ways:

⁹ Olga Aymerich, Mara Revkin and Samuel Akau, “[Understanding Multidimensional Fragility in South Sudan](#),” (Juba, IOM DTM, 2023).

¹⁰ The Organisation for Economic Co-operation and Development (OECD), [States of Fragility 2022](#), (Paris, OECD, 2022).

¹¹ United Nations Security Council, “[Secretary-General, in First Address to Security Council Since Taking Office, Sets Restoring Trust, Preventing Crises as United Nations Priorities](#),” (New York, Security Council Meetings Coverage, 2017). See also United Nations News, “[Security Council has critical role in addressing fragility and conflict: UN chief](#),” (New York, United Nations News, 2021).

¹² World Bank, [World Bank Group Strategy for Fragility, Conflict, and Violence 2020-2025](#), (Washington, D.C., World Bank, 2020).

- It provides a comprehensive **assessment of vulnerabilities and capacities** among the general population that **can be disaggregated** by household displacement status, rurality, and more;
- Informs the **design** of context-specific, coordinated, multi-sectoral programmes covering the pillars of recovery, stabilization, and reintegration;
- Informs decisions about how to **target** programming based on objective, evidence-based criteria in support of all three major targeting approaches (area-based, status-based, or household-based);
- Measures changes in **perceptions of different dimensions of fragility over time**;
- Can be **adapted for data collection in contexts beyond South Sudan**,¹³
- Can be linked to a range of different outcomes and objectives across the pillars of peace and security, human rights, and development, potentially providing the foundations for **baseline and endline impact evaluations** of specific programmatic interventions that seek to reduce one or more dimensions of fragility.



¹³ While each question corresponds to a concept which is universal, not all question wording is applicable to all contexts. See notes regarding where question wording may need to be adjusted in Appendix A.

Defining and Measuring Multidimensional Fragility

This study adapts the OECD multidimensional fragility framework for use at the community level and tailors it for the context of South Sudan.¹⁴ These different dimensions are defined as follows:

1. **Economic Fragility** relates to the vulnerabilities and challenges within the economic systems of a country or region. In the Fragility Index, this component includes measures of education, financial satisfaction, the ability to pay for expected and unexpected large costs, and dependency on aid.
2. **Environmental Fragility** concerns the challenges and risks associated with the environment and natural resources. In the Fragility Index, this component includes climate change's impact on income, climate change's impact on food consumption, climate change's impact on mobility, climate change's impact on health, and the ability to act on early warning information to ensure household safety.
3. **Human Fragility** refers to people's "well-being and...ability to live healthy, long and prosperous lives."¹⁵ The Fragility Index focuses on factors relevant for health: food access, water access, medical treatment access, shelter, and exposure to sexual violence.¹⁶
4. **Political and Legal Fragility**¹⁷ captures the instability, weaknesses, and challenges in political systems and governance structures. In the Fragility Index, this component encompasses trust in state government, trust in national government, trust in informal authorities, participation in democracy without fear, and freedom of speech.
5. **Security Fragility** refers to threats, conflicts, and violence within a society. In the Fragility Index, this component includes the degree to which individuals are affected by crime, war, land conflict, resource conflict, and local armed militias.
6. **Societal Fragility** focuses on societal divisions, inequalities, and social cohesion. In the Fragility Index, it includes experiences of acceptance, experiences of discrimination, social cohesion, network embeddedness, and migrant-local cohesion.

¹⁴ While each question corresponds to a concept which is universal, not all question wording is applicable to all contexts. See notes regarding where question wording may need to be adjusted in Appendix A.

¹⁵ The Organisation for Economic Co-operation and Development (OECD), [States of Fragility 2022](#), (Paris, OECD, 2022).

¹⁶ The Organisation for Economic Co-operation and Development (OECD), ["Adding the Human Dimension to the OECD Fragility Framework"](#) (Paris, OECD, 2022). OECD includes a range of health indicators as well as human capital indicators, such as school enrollment.

¹⁷ OECD's fragility framework does not include a standalone dimension for "legal" fragility. This project has modified the category of "political fragility" to "political and legal fragility" to capture important aspects of the legal context including rule of law and preferences for dispute resolution among alternative providers of justice including state courts and customary tribal or religious authorities.

Complementarities with Existing Assessment Tools and Added Value

The Fragility Index is a unique tool that utilizes household-level survey data to provide a representative mapping of vulnerabilities and needs at the population level within a particular area. It complements but is distinct from several existing assessment tools within IOM's D4S initiative. The Migration Integration Toolkit (MIT) facilitates status-based targeting of aid for migrants.¹⁸ Although MIT provides valuable individual-level data on several dimensions of integration (social, political, navigational, economic, psychological and linguistic), it is limited to a particular sub-group: migrants. In contrast, the Fragility Index uses random sampling of the population as a whole, providing insight into migrants as well as the larger communities in which they live. Two other tools, the Stability Index (SI) and the Solutions Mobility Index (SMI),¹⁹ use perception data to operationalize varying components of stability, depending on the country — these components are (1) safety and security, (2) livelihoods and basic services, (3) social cohesion, and (4) disaster adaptation. Unlike the Fragility Index, both tools are based on interviews with non-random samples of “key informants”²⁰ who represent important social groups or other constituencies, including “community leaders, aid workers, and other community representatives.”²¹ Although key informant interviews provide valuable qualitative insights into local dynamics, they can be susceptible to biases that reflect the demographics and positionality of the key informants, who are often disproportionately older male elites.²² For example, the Solutions and Mobility Index Round 1 in Ghana conducted 395 key informant interviews, of which 91 per cent were with men.²³ The Fragility Index relies on representative random samples in which all eligible adults living in a particular area have an equal probability of being selected for participation. These two sources of information — qualitative data from key informant interviews and quantitative data from randomized representative surveys — provide different but complementary insights into communities.

Using the Index to Improve the Targeting of Assistance

The Fragility Index was developed in the context of calls for more evidence on the effectiveness of different targeting approaches and how they are perceived by beneficiaries. In recent years, several UN authorities and agencies have raised concerns about potential unintended harmful consequences of status-based targeting approaches, in which aid or programming is provided to internally displaced persons (IDPs), women, children, or other sub-groups based on their membership in that category rather than their individual needs. One concern is that status-based aid can exacerbate intergroup tensions and competition. For example, the 2019 report of the UN Secretary General on South Sudan indicated that providing assistance to people in Protection of Civilian (PoC) sites based solely on their residential status “raises questions of equity.”²⁴

Previous research in other contexts including Zimbabwe, Ethiopia, and Somalia has similarly found that status-based assistance “can exacerbate tensions through perceived favorable treatment of certain groups.”²⁵ A second concern is that status-based targeting may not lead to the most efficient or fair use of resources in contexts of extreme need and limited funding. In light of these concerns, World Food Programme (WFP) and United Nations High Commissioner for Refugees (UNHCR) in South Sudan recently announced plans to shift from status-based food and non-food items (NFI) assistance to targeted and prioritized assistance based on refugee households' vulnerability levels to ensure an efficient use of the limited resources available.²⁶

¹⁸ International Organization for Migration (IOM) and Stanford Immigration Policy Lab, “[Migrant Integration Measurement Toolkit: Offering a Multi-Dimensional Approach to Measure Migrant Integration](#),” (Geneva, IOM).

¹⁹ International Organization for Migration Displacement Tracking Matrix (IOM DTM), “[Solution and Mobility Index - Ghana Round 1](#),” (Accra, DTM, 2023). International Organization for Migration Displacement Tracking Matrix (IOM DTM), “[Solutions and Mobility Index in Libya](#),” (Libya, DTM, 2024).

²⁰ International Organization for Migration Displacement Tracking Matrix (IOM DTM), “[Stability Index - DTM Report](#),” (Dakar, IOM, 2020).

²¹ International Organization for Migration Displacement Tracking Matrix (IOM DTM), “[Stability Index - Nigeria December 2022-January 2023](#),” (Asokoro, DTM, 2023).

²² A previous IOM assessment in Iraq comparing the responses of key informants with those of a random sample found significant discrepancies in the opinions of these two populations, likely driven by the fact that the vast majority of key informants were men between the ages of 35-59. See Trigwell, R., J. Phillippo-Holmes, E. Zambrano, J. Bahn, P. Hirani and E. Griesmer, “[Validating Humanitarian Data Through Collective Intelligence: A Pilot Study](#),” (Geneva, IOM, 2022).

²³ International Organization for Migration Displacement Tracking Matrix (IOM DTM), “[Solution and Mobility Index - Ghana Round 1](#),” (Accra, DTM, 2023).

²⁴ United Nations Secretary General, “[S/2019/741 Future planning for the protection of civilians sites in South Sudan](#),” (New York, United Nations Security Council, 2019).

²⁵ Sam Grundy and Sarah Zingg, “MRS No. 66 - Community Stabilization—An approach for facilitating progress towards durable solutions and operationalizing the Humanitarian—Development—Peace Nexus: Lessons from Zimbabwe, Ethiopia and Somalia,” (Geneva, IOM, 2020).

²⁶ United Nations High Commissioner for Refugees (UNHCR) and World Food Programme (WFP), “[Prioritisation of assistance in South Sudan: Community consultation report](#),” (Geneva, UNHCR, 2023).

In this context of ongoing debates about how to improve aid targeting amidst limited resources, the Fragility Index can directly inform all three leading approaches to assessing needs and allocating aid:

- 1. Geographic/Area-Based:** By allowing for comparisons of relative levels of a particular dimension of fragility across different geographic areas, the Fragility Index can strengthen area-based targeting. For example, areas where human fragility scores are highest should have the greatest need for life-saving humanitarian aid including Water, Sanitation, and Hygiene (WASH), food assistance, and shelter. Conversely, areas with the lowest scores on the economic fragility dimension may be most prepared to benefit from development programmes such as loans for small and medium-sized enterprises.
- 2. Status-Based:** The Fragility Index provides detailed data on variation in the vulnerabilities of individuals within a particular status-based group that can be used to improve the precision of targeting within that group. For example, it can indicate which IDPs are most in need of shelter. It can also provide concrete evidence in support of or against the need for status-based targeting for certain groups of potential beneficiaries.
- 3. Household-Based:** The Fragility Index provides a comprehensive but simple measure of household needs and capacities that can be used to allocate individualized support such as cash assistance.

Key Findings

- Although the levels of fragility experienced by South Sudanese are likely higher than those experienced by residents of many other countries, there is important sub-national variation between the levels of economic, social, human, environmental, political/legal, and security fragility experienced by residents of different counties within South Sudan. These differences speak to the importance of targeting aid subnationally.
- The six dimensions of fragility do not all operate similarly; areas that are high in one kind of fragility may be low in another kind. For example, Bor South has the highest level of security fragility out of the surveyed counties, but the lowest level of societal fragility. This finding suggests that thinking of a county as having a given level of “fragility” and thus requiring a given level of support is a missed opportunity to tailor combinations of different forms of aid.
- There are not many statistically significant differences in the levels of fragility experienced by 1) IDPs outside of the camp context and host communities members or by 2) returnees and host community members. As such, the provision of aid based on displacement status does not necessarily provide support to those who need it most.

2. METHODOLOGY AND OVERVIEW OF RESULTS

The Fragility Index uses survey questions administered at the household level to measure perceptions of the six dimensions of fragility. This section describes the Index methodology in more detail.

Survey Locations and Sampling Procedure

The survey of 3,685 respondents was conducted between April and May 2024 in nine counties in five of South Sudan's ten states: Wau in Western Bahr el Ghazal State, Rubkona in Unity State, Malakal in Upper Nile State, Bor South and Pibor in Jonglei State, and Yei, Jonglei, Lainya, and Kajo-keji in Central Equatoria State.²⁷ The nine counties were chosen because these are locations where IOM is currently implementing programming and where DTM has access and the ability to recruit and train local enumerators.²⁸ Because this is a convenience rather than a random sample of counties, the results cannot generalize to all counties in South Sudan.

The survey did reach some populations which were displaced from unsampled counties but are currently residing in counties which were a part of the survey. By design, however, the survey excludes populations from unsampled locations that had no need to move or were too vulnerable to relocate. Thus, the findings from the fragility index do not necessarily generalize to populations from unsampled counties which faced structural barriers to displacement, such as the disabled, elderly, or killed-in-conflict.²⁹

IOM's DTM team surveyed approximately 400 respondents per county, totalling 3,685 respondents. Each of the nine counties was stratified into urban and rural bomas.³⁰ Each of the nine counties was stratified into urban and rural bomas. In each county, the research team randomly selected two urban bomas (within the county capital) and two rural bomas (outside the county capital), with a target of 100 respondents per boma.³¹ Bomas were selected from the list included in the IOM DTM Round 14 Baseline. Prior to data collection, backup urban and rural bomas were also randomly selected. Where a selected boma was inaccessible or had fewer than 100 households, enumerators completed as many interviews as possible and conducted the remaining surveys in a pre-selected backup boma.³² Where bomas did not have 100 households, enumerators completed as many surveys as feasible in the selected bomas and then completed the rest of the surveys in a backup boma.

Within each boma, enumerators followed a random-walk procedure. They first confirmed with a community leader or local authority how many dwellings were in the area and then divided that by 100 in order to calculate the appropriate sampling interval. Beginning at a location which enumerators themselves choose, they would then select dwellings separated by the sampling interval to survey. In general, the first identified adult at each selected dwelling was asked to participate. However, enumerators were instructed to oversample men or women as needed within dwellings to achieve a sample of approximately 50 per cent each per boma. Participation was anonymous and voluntary, and respondents were able to skip any question.³³ Enumerators were provided with the questionnaire

²⁷ See Appendix D for a full list of sampled bomas.

²⁸ South Sudan Central Bureau of Statistics, "5th Sudan Population and Housing Census – 2008: Priority Results," (Juba, South Sudan Central Bureau of Statistics, 2009).

²⁹ P. Adhikari, "Conflict-Induced Displacement, Understanding the Causes of Flight", *American Journal of Political Science*, 57(1), 2013, 82–89; C. Horst and K. Grabska, "Flight and Exile: Uncertainty in the Context of Conflict-Induced Displacement," *Social Analysis*, 59(1), 2015, 1–18.

³⁰ Bomas are the lowest-level administrative unit in South Sudan.

³¹ See Appendix B for a full list of sampled bomas.

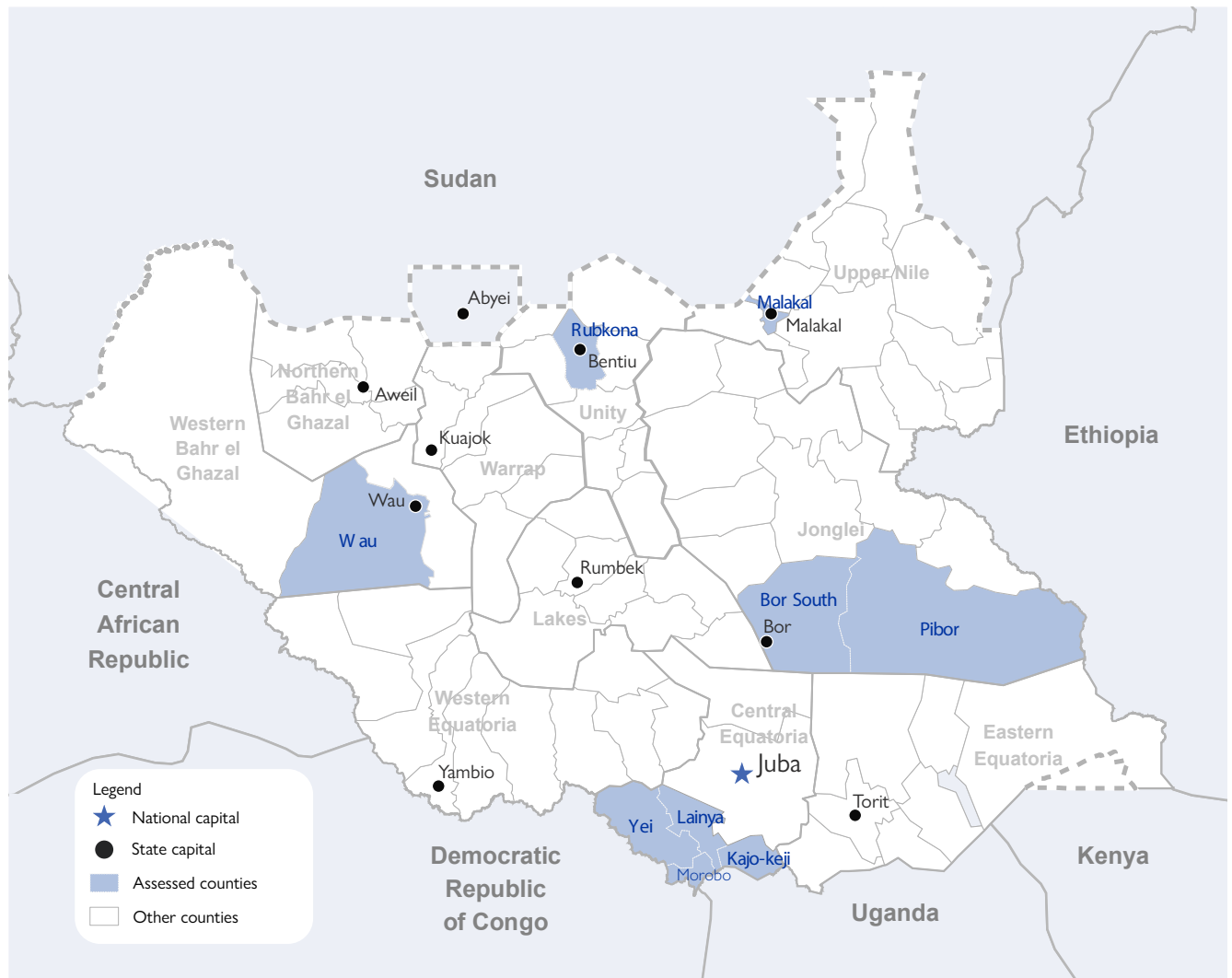
³² There were deviations from this approach, in which replacement bomas were not randomly selected, in two out of nine counties. In Pibor, all of the selected bomas were flooded; residents had moved to other, local bomas. The two replacement urban bomas were geographically proximate to each other, but the two replacement rural bomas were geographically proximate to the originally.

³³ The survey began with a detailed informed consent process in which enumerators explained that declining to participate in the survey would have no impact on their likelihood of receiving assistance from IOM. selected bomas. In Rubkona, the two selected rural bomas were not accessible. The replacement rural bomas were geographically proximate to each other.

³⁴ This approach introduced heterogeneity in the survey instrument presented to respondents. However, translating the survey into every local language before enumeration would be prohibitively expensive

Enumerator training with county focal points was conducted remotely by IOM's DTM team; the focal points then conducted in-person training with their teams. Each county had 10 enumerators, all of whom DTM had previously engaged with. DTM selected a gender-balanced team of enumerators, and all of them were proficient in both English and the predominant local languages. Prior to enumeration, in each county, DTM staff consulted with government authorities to obtain their permission for access to the survey locations.

Figure 1. Surveyed Counties and Payams



Constructing the Fragility Index

While “fragility” is an abstract and complex concept, each of the six dimensions is assessed with five concrete survey questions. Each captures lived experiences relevant to the underlying concept. For example, “Human Fragility” is measured with five survey questions about the respondent household’s: (1) access to food, (2) access to water, (3) access to medical treatment, (4) type of shelter, and (5) exposure to sexual violence.

The core of the questionnaire consists of 30 questions forming the six fragility indices. The full text of these questions can be found in Appendix A. To construct each of the six indices, each indicator value was normalized to range from 0-1; 0 indicates the lowest possible level of fragility and 1 indicates the highest possible level of fragility.³⁵ Then, six fragility scores were calculated for each subject; each fragility score is a simple additive index of the five questions which make up that score.³⁶ In other words, each question is weighted equally in the indices.



As a final step, the indices were rescaled (normalized) to range from 0-100; 0 indicates the lowest observed fragility score and 100 indicates the highest observed fragility score.³⁷ Although normalization facilitates visualization of the results on a common scale, it is important to recognize that different types of fragility capture fundamentally different concepts that are not easily comparable. As such, the relative numerical scores across dimensions do not imply substantive comparisons between the levels of different kinds of fragility. For example, Yei County’s security and economic fragility scores are both 53. Despite the similarity in scores, the underlying challenges associated with each type of fragility differ significantly, preventing any conclusion that economic and security fragility levels are equivalent in this context. Similarly, in Lainya county, the data should not be interpreted as indicating that the level of environmental fragility (environmental fragility score: 51) is substantively greater than the level of human fragility (human fragility score: 40).

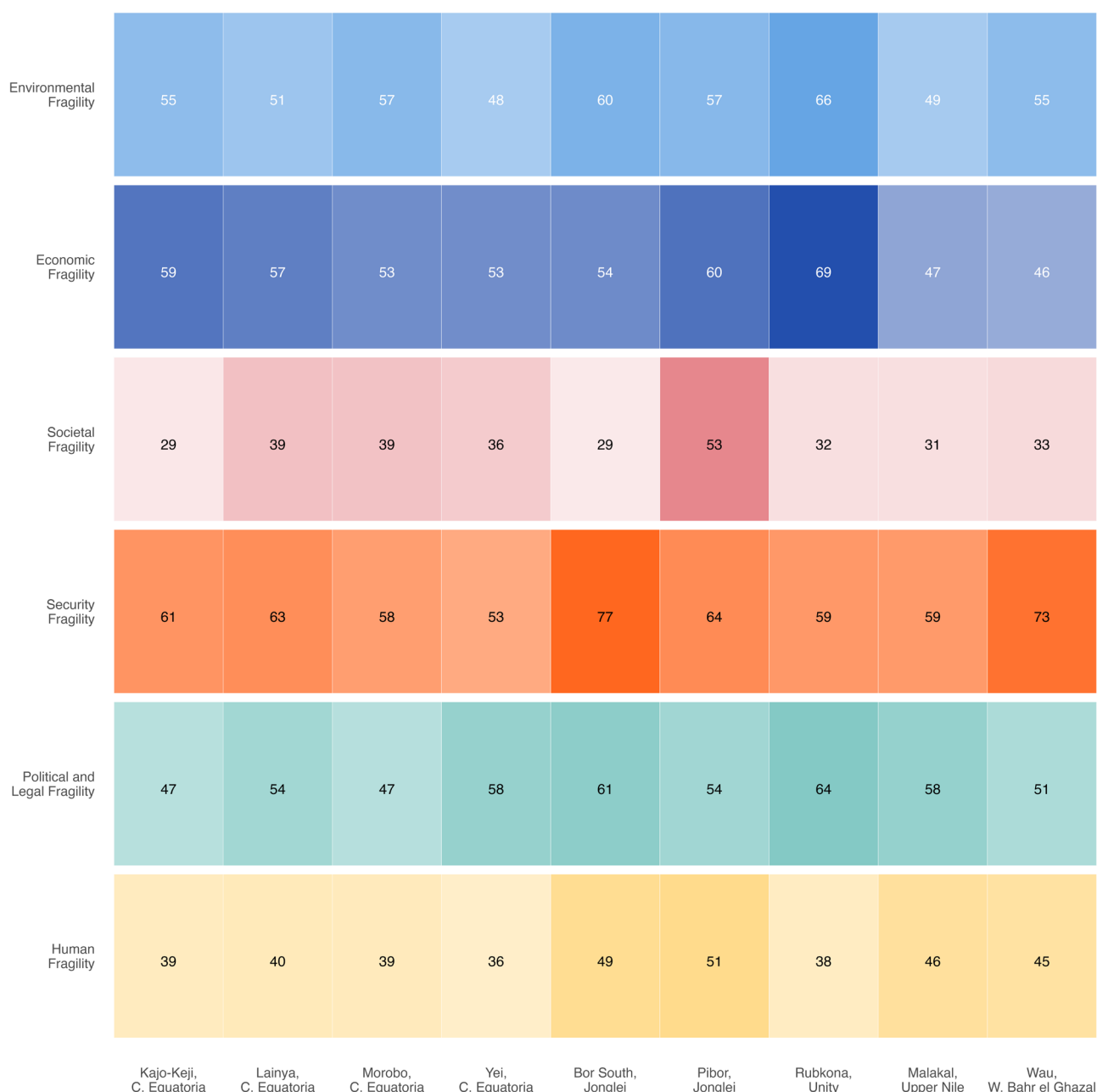
³⁵ More formally, the normalization formula is as follows: $(\text{value} - \text{minimum possible value}) / (\text{maximum value} - \text{minimum possible value})$. Note that all responses with negative values (e.g. “-2. Prefer not to answer”) were changed to NA before this normalization. To ensure that higher values indicate higher levels of fragility, many variables were reverse coded; for details, see Appendix A.

³⁶ To avoid scores that were artificially lowered due to missing values, for all questions which contributed to indices, the research team imputed missing values with the median response values to those questions before building the additive indices. As a result, fragility scores contain imputed data. However, analysis of all individual survey questions uses unimputed versions of the variables.

³⁷ More formally, this normalization and rescaling formula is as follows: $[(\text{score} - \text{minimum actual score}) / (\text{maximum actual score} - \text{minimum actual score})] \times 100$. One way to evaluate the indices is to consider whether the variables provide similar information. Appendix C takes two approaches to doing so. Cronbach’s Alpha is a measure of the reliability, or internal consistency, of a set of variables. It measures the amount of shared variance between the variables and ranges from 0 to 1, where 0 means that the variables are not correlated. Principal Component Analysis (PCA) is a machine learning method which reduces the dimensions in a dataset by transforming the variables into a smaller set of essential features (i.e. “principal components”). PCA provides information about what percentage of overall variance is explained by each component; the more similar the variables are, the more (ctnd p.17)

The Fragility Index effectively compares variations within a single dimension of fragility across different counties. For example, Lainya's level of environmental fragility (51) is lower than Rubkona's (66). The Fragility Index can be used to make other comparisons that can inform all three of the primary approaches to aid targeting, discussed below. The heat map below presents the main results of the Fragility Index; it contains fragility scores for each dimension along the vertical Y-axis and the nine counties along the horizontal X-axis. Darker shades indicate higher fragility scores on a normalized scale from 0 to 100.

Figure 2. Mean Fragility by County



(ctnd from p.16) variance the first component will explain. These analyses suggest that the Security and Environmental Fragility indices are reliable, but the rest require further development. For more information on these approaches, see for example: Labrín, Caterina and Urdinez, Francisco. "Principal Component Analysis." In R for Political Data Science. 2020. Taylor & Francis. Flake, Jessica K, Jolynn Pek, & Eric Hehman. "Construct Validation in Social and Personality Research: Current Practice and Recommendations." *Social Psychology and Personality Science* 8(4): 370-378.

Respondent Demographics

Table 1 summarizes the demographics of the sample. Respondents are 61 per cent female and 39 per cent male. As a point of comparison, estimates suggest that the population of South Sudan consists of 51 per cent women and 49 per cent men.³⁸ Some possible explanations for this marked gender imbalance in respondents include: women may be more likely to be at home during the day when enumerators arrive; men were killed at higher rates than women during the war;³⁹ and previous surveys indicate that women outnumber men among South Sudanese IDPs.⁴⁰ The largest ethnic groups in our sample are Bari (19%), Dinka (18%), Kakwa (13%), Nuer (11%), and Murle (10%). However, there is great ethnic heterogeneity across counties but relative ethnic homogeneity in the sample within many of them. For example, all except two respondents in Bor South were Dinka, 97 per cent of respondents in Rubkona were Nuer, and 98 per cent of respondents in Lainya were Bari. Kajo-Keji and Malakal were the only two counties in which the second largest ethnic group consisted of more than one third of the sample.

29 per cent of our sample has, at some point in their lives, been forcibly displaced i.e. moved for the purposes of “security or survival.” 24 per cent of respondents are returnees and 4 per cent are IDPs. The sample includes a small population of refugees who indicate that they were displaced from locations outside of South Sudan. 16 are from Sudan, 13 are from Uganda, 4 are from the Democratic Republic of the Congo, 1 is from the Central African Republic, 1 is from Ethiopia, and 1 is from Kenya.

The relatively low percentage of IDPs, in comparison with other recent surveys in some of the same locations,

reflects a methodological decision to exclude camps and other informal residential sites for IDPs from this survey. The decision to limit the survey to non-camp locations was driven by several related considerations. First, IOM and other NGOs routinely conduct needs assessments in camps but have much less information about conditions in non-camp settings.⁴¹ Yet, less than 500,000 out of the over 1.8 million refugees and IDPs in South Sudan



are living in camps.⁴² Second, populations living outside camps tend to receive less attention and funding from international donors than those living in camps. Third, understanding conditions in non-camp settings is essential for developing evidence-based peacebuilding and development programmes. In contrast, humanitarian programmes are often designed for beneficiaries who are living in or transiting between camps.

³⁸ World Bank Group Data. “Population, female (% of total population) - South Sudan.” (Washington, DC, World Bank, 2024).

³⁹ Francesco Checchi, Adrienne Testa, Amy Gimma, Emilie Koum-Besson, and Abdihamid Warsame. “A method for small-area estimation of population mortality in settings affected by crises,” *Population Health Metrics* 20, no. 1 (2022): 4.

⁴⁰ Oxfam International, “No Simple Solutions: Women, Displacement, and Durable Solutions in South Sudan,” p. 7 (noting that up to 80% of displaced households in South Sudan are female-headed).

⁴¹ See, for example, IOM DTM South Sudan, “IDP Site Multi-Sector Needs and Vulnerabilities Survey (FSNMS+): Malakal PoC Site,” International Organization for Migration (July, 2022), p. 5 (an IOM survey of 404 households living in the Malakal Protection of Civilians site).

⁴² UNHCR, “IGAD Region Population of concern to UNHCR as of 31 Mar 2025,” (Nairobi, Kenya, UNHCR, 2025). United Nations Peacekeeping, “In South Sudan, Protection of Civilians sites are transitioning into Internally Displaced Persons Camps,” (Juba, South Sudan, UN Peacekeeping, 2023).

Table 1. Demographics

State	County	Number of Respondents	Gender Distribution (observed by enumerator)	% of IDPs of Refugees ⁴³	% Returnees	3 Largest Ethnicities (% of respondents) ⁴⁴
Central Equatoria	Kajo-Keji	398	53% Female 47% Male	1%	44%	Kuku (60%), Bari (38%), Bviri (1%)
Central Equatoria	Yei	410	50% Female 50% Male	3%	16%	Kakwa (57%), Baka (11%), Other (5%)
Central Equatoria	Lainya	414	41% Female 59% Male	8%	27%	Bari (98%), Kakwa (<1%), Bviri (<1%)
Central Equatoria	Morobo	404	51% Female 49% Male	4%	54%	Kakwa (55%), Bari (23%), Other (18%)
Jonglei	Bor South	429	71% Female 29% Male	5%	29%	Dinka (100%), Baka (<1%), Mundari (0%) ⁴⁵
Jonglei	Pibor	407	66% Female 34% Male	4%	12%	Murle (93%), Anyuak (3%), Dinka (1%)
Western Bahr el Ghazal	Wau	402	64% Female 36% Male	2%	11%	Fertit (52%), Other (30%), Bviri (5%)
Upper Nile	Malakal	421	73% Female 27% Male	11%	22%	Shilluk (52%), Dinka (42%), Nuer (5%)
Unity	Rubkona	400	84% Female 16% Male	1%	4%	Nuer (97%), Other (2%) ⁴⁶
Total (all 9 counties)		Total: 3,685⁴⁷	61% Female 39% Male	4%	24%	Bari (19%), Dinka (18%), Kakwa (13%)

⁴³ IDPs are defined as having moved from one county to another "for security or survival" and not currently living "in the county in which [they] were born." Returnees moved for the same reason but are currently living in the county in which they were born. Defining IDPs and returnees in this fashion means that those who were displaced from their homes but moved to another location (such as a POC site) within the same county are not included. In addition to not sampling in POC sites, this aspect of the definition could be a reason why the percentage of IDP respondents in the survey is low. Members of host communities are defined as anyone who does not identify as either an IDP or returnee.

⁴⁴ Respondents were provided with the following options: Acholi, Anyuak, Azande, Baka, Bari, Bongo, Bviri, Didinga, Dinka, Dungotona, Fertit, Kakwa, Kuku, Lango, Lndi, Mundari, Murle, Ndogo, Nuer, Shilluk, and Other. Less than 1 per cent of respondents declined to provide their ethnicity.

⁴⁵ There was one Baka respondent and one Mundari respondent in Bor South.

⁴⁶ The Murle, Shilluk, and Lndi groups each represent just 0.25 per cent of respondents in the county.

⁴⁷ There are 13 respondents for whom the location was not recorded. These are not included in the total number of respondents and have been removed from the data and analysis.

Using the Fragility Index to Improve Evidence-Based Targeting of Assistance

Comparing relative levels of specific dimensions of fragility across different counties provides valuable insights for programming and targeting. For example, as Figure 2 indicates, Pibor county has the highest level of societal fragility, and Rubkona county has the highest level of political and legal fragility. Wau has the lowest level of economic fragility, and Yei has the lowest level of security fragility. This data would be helpful in the context of deciding which areas to allocate funds designated for a certain purpose.⁴⁸

Similarly, the results can help to identify areas of lower vulnerability where certain types of programming are more likely to succeed. For example, financing programmes for small businesses are more likely to succeed in areas with lower economic fragility, where beneficiaries are more likely to have the necessary skills, resources, and infrastructure to take advantage of the resources. Malakal, for example, has a relatively low level of economic fragility (its economic fragility score is 47 when the range for the nine counties is 46-69);

this makes sense given that it was once South Sudan's second largest city. However, transition and recovery programming should also consider other dimensions of fragility. For example, although the county has faced periodic conflicts between different ethnic groups driven by competition over land or other resources, which have disrupted local markets, Malakal's current societal fragility score is relatively low (its societal fragility score is 31 when the range for the nine counties is 29-53).⁴⁹ Malakal does have a moderate level of security fragility (its security fragility score is 59 when the range for the nine counties is 53-77). This is likely driven in part by concerns about organized crime and gang violence, both of which present risks for business owners.⁵⁰ In other words, the design and targeting of programming should take into account not only the most directly relevant dimension of fragility (such as economic fragility in the context of programming to support small businesses) but also other dimensions of fragility (e.g., security, societal, and environmental) that may interact with economic fragility.

Subgroup Differences

The Fragility Index results can be analysed for variation in the same dimension of fragility between different subgroups (e.g., women, IDPs, refugees, persons with disabilities) to inform status-based criteria for allocating aid and programming.

Table 2 presents the results of t-tests, a simple test of statistically significant differences between the means of two groups. IDPs and returnees report higher levels of environmental fragility than respondents who have never been displaced ("host communities").⁵¹ However, IDPs are not significantly more fragile than host communities in any other dimension. IDPs are also not more fragile than returnees on any dimension of fragility. These tests provide concrete evidence suggesting that ongoing targeting based on IDP status may not be providing support to the most vulnerable residents of South Sudan. Table 2 also suggests that people living outside of county capitals are more environmentally and economically fragile; this finding points to the urgent need to ensure that programming reaches people living in more rural areas.

⁴⁸ Given the lack of reliable population data in South Sudan, the research team refrains from providing suggested sample sizes at the county level in order to achieve a given confidence interval. The Peace and Conflict Resolution Evidence Platform, in a survey enumerated by Detcro, has done the same. See, for example, Jan Pospisil, Melissa Johnston, Adul Alaak Garang, Nyachangkuoth Rambang Tai, "[Bring Enough Cows to Marry: Brideprice, Conflict, and Gender Relations in South Sudan](#)," (Coventry and Edinburgh, The University of Edinburgh and PeaceRep and Coventry University, 2024). However, it should be noted that it is not necessary to sample every boma to establish the level of fragility in a given payam, county, or state. What is necessary to achieve reasonably representative samples is to rely on clustered, stratified random sampling. For primers on such methodologies, see, for example, Groves et al, *Survey Methodology*, (Hoboken, Wiley, 2009). Anna Wilke, "[10 Things to Know About Sampling](#)," (Berkeley, Evidence in Governance and Politics). Abdul Latif Jameel Poverty Action Lab (J-Pal), "[How to Randomize](#)," (Cambridge MA, J-PAL).

⁴⁹ Radio Tamazuj, "[Malakal traders confirm resumption of business, market stability](#)," (Juba, Radio Tamazuj, 2024).

⁵⁰ Michael Daniel, "[100 Youth Arrested on Gang Suspicion in Malakal](#)," (Juba, Eye Radio, 2024).

⁵¹ These analyses can also be replicated at more disaggregated geographic levels.

The Fragility Index could also provide data on variation in the vulnerabilities of individuals *within* a particular status-based group, helping improve the precision of targeting within that group. For example, it could help identify *which* groups of women in a given area are most vulnerable to risks such as gender-based violence and would benefit from additional protection assistance.

Table 2. Differences in Means Across Subgroups⁵²

Dimension	IDPs/Host	Returnees/ Host	IDPs/Returnees	County Capital/ Not
Human Fragility	43.74 / 42.66	42.38 / 42.66	43.74 / 42.38	43.09 / 42.17 °
Political /Legal Fragility	57.61 / 54.94 °	53.88 / 54.94	57.61 / 53.88	55.32 / 54.24 °
Security Fragility	61.35 / 63.22	62.48 / 63.22	61.35 / 62.48	63.44 / 62.43
Societal Fragility	35.42 / 35.74	35.30 / 35.74	35.42 / 35.30	35.56 / 35.69
Economic Fragility	54.90 / 55.31	55.29 / 55.31	54.90 / 55.29	54.56 / 56.03
Environmental Fragility	52.67 / 56.20 **	53.30 / 56.20 ***	52.67 / 53.30	54.81 / 55.90 *

⁵² The following indicate that the differences between the means of the two groups are statistically significant to the following levels: ***, p<0.001; **, p<0.01; *, p<0.05; °, p<0.1.

3. ECONOMIC FRAGILITY

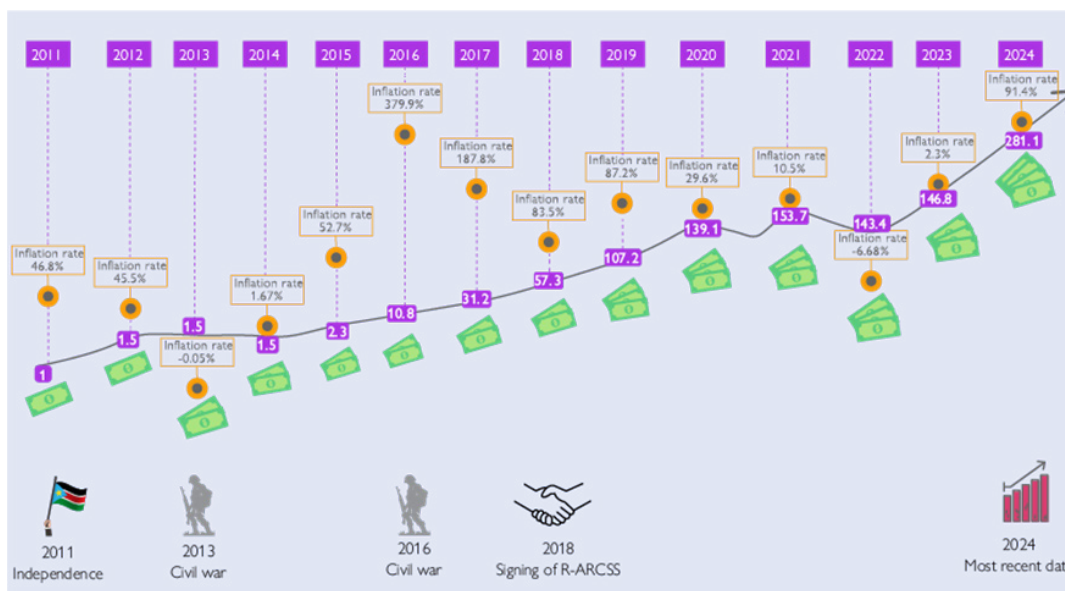
Key Findings

- **Aid gap:** Results indicate that current levels of humanitarian assistance and aid may be insufficient to cover basic needs for households. Only 8 per cent of households in the sample are either highly or fully dependent on aid. Conversely, 82 per cent indicate that the impact of an unexpected death on their household's financial situation would be not severe or not severe at all. Further, 72 per cent of respondents are dissatisfied or very dissatisfied with their household's financial situation.
- **Continued reliance on agriculture:** A majority of South Sudanese (51%) continue to rely on agriculture as their household's primary source of income.
- **Mixed livelihood strategies:** Agricultural activity is largely subsistence-based: in most farming households (79%), crops are grown primarily for household consumption. Even so, around half (47%) also sell some of their produce for profit, suggesting a hybrid strategy in which many households are trying to move beyond mere subsistence to engage in market activity.

Overview of Economic Fragility in South Sudan

Economic fragility and conflict are mutually reinforcing in South Sudan. Economic fragility is a significant driver of both inter-communal conflicts and grievances against the government, hindering peacebuilding efforts. At the same time, persistent cycles of conflict are a barrier to economic development. In 2025, South Sudan is experiencing its worst economic crisis since Independence. Currently, South Sudan faces a multifaceted economic crisis that includes cuts to international aid, declining oil prices alongside damages to the pipeline that brings South Sudanese oil to market, the rapidly depreciating value of the South Sudanese Pound (SSP), shortages of hard currency, and dependence on imports.⁵³

Figure 3. Inflation rates since 2011



Inflation rates since 2011. Purple boxes show how many dollars were required in a given year to purchase what \$1 could buy in 2011.

Source: Inflation, consumer prices (annual %), South Sudan, International Financial Statistics database, International Monetary Fund (IMF), Accessed 1 July 2025, <https://api.worldbank.org/v2/en/indicator/FP.CPI.TOTL.ZG?downloadformat=csv>.

⁵³ International Crisis Group, "South Sudan on Edge as Its Neighbor's War Disrupts Oil Exports," International Crisis Group (May 15, 2024), <https://www.crisisgroup.org/africa/horn-africa/south-sudan/south-sudan-edge-its-neighbours-war-disrupts-oil-exports>. International Monetary Fund African Department, "Republic of South Sudan: 2023 Article IV Consultation, and First and Second Reviews under the Staff-Monitored Program with Board Involvement-Press Release; Staff Report; and Statement by the Executive Director for Republic of South Sudan," (June 10, 2024), <https://www.elibrary.imf.org/view/journals/002/2024/160/article-A001-en.xml>.

In December 2024, a basket of consumer goods increased in price by 14.9 per cent compared to the previous month.⁵⁴ More than 80 per cent of the population lives below the poverty line.⁵⁵ The unemployment rate is around 11 per cent, with youth unemployment as high as 50 per cent.⁵⁶

Oil and agriculture remain South Sudan's two most critical economic sectors. Oil accounts for approximately 95 per cent of exports and 90 per cent of fiscal budget revenue,⁵⁷ making the economy highly susceptible to fluctuations in global oil prices and security threats to oil infrastructure. South Sudanese oil reaches the international market through a pipeline to Port Sudan. In 2011, South Sudan produced approximately 350,000 barrels of oil per day. In February 2024, damage to the pipeline had cut production to 45,000 barrels per day.⁵⁸ The damage occurred when the Rapid Support Forces (RSF) in Sudan took control of a pumping station on the pipeline that delivers about 60 per cent of South Sudan's oil to market from the Upper Nile oil fields.⁵⁹ In April 2025, South Sudan's Ministry of Petroleum declared that the pipeline was functional again. However, the Sudan Armed Forces is trying to renegotiate oil transit fees with the South Sudanese government and the RSF is also demanding a fee of its own – potentially causing further disruptions to revenue.⁶⁰



Agriculture, although secondary to oil in terms of Gross Domestic Product (GDP) contribution, is the primary livelihood source for the majority of the South Sudanese population. According to some estimates, 85 per cent of people in South Sudan rely on non-wage work that involves rainfed subsistence agriculture.⁶¹ In recent years, conflict and climate-related disasters have severely impacted agricultural production, contributing to economic vulnerability and food insecurity, particularly in rural areas.

South Sudan is the least electrified country in the world; as of 2020, only 8 per cent of the population has access to electricity.⁶² Thermal power plants provide limited electricity for Juba, but South Sudan lacks a national electric grid that would be necessary to transmit electricity generated in Juba to other parts of the country. Renk County in northern South Sudan relies on imported electricity from Sudan.⁶³

⁵⁴ Joshua Craze, 'On the Brink: The Politics of Violence in South Sudan', HSBA/Small Arms Survey Briefing Paper, March 2025; International Crisis Group, 'South Sudan on the Precipice of Renewed Full-blown War', 7 March 2025, <https://www.crisisgroup.org/africa/horn-africa/south-sudan/south-sudan-precipice-renewed-full-blown-war>.

⁵⁵ Amina Lahreche and Niko Alfred Hobdari, 'Four Things to Know About How Fragile States Like South Sudan Are Coping With COVID-19', International Monetary Fund African Department (Nov. 20, 2020), <https://www.imf.org/en/News/Articles/2020/11/19/na112020-four-things-to-know-about-how-fragile-states-like-south-sudan-are-coping-with-covid19>.

⁵⁶ 'Vocational Training Increases Employability Opportunities For Young South Sudanese | United Nations Development Programme,' U.N. Development Programme (Jul. 14, 2022), <https://www.undp.org/south-sudan/stories/vocational-training-increases-employability-opportunities-young-south-sudanese>.

⁵⁷ International Monetary Fund African Department, 'Republic of South Sudan: 2023 Article IV Consultation, and First and Second Reviews under the Staff-Monitored Program with Board Involvement-Press Release; Staff Report; and Statement by the Executive Director for Republic of South Sudan,' (June 10, 2024), <https://www.elibrary.imf.org/view/journals/002/2024/160/article-A001-en.xml>.

⁵⁸ International Monetary Fund African Department, 'Republic of South Sudan: 2023 Article IV Consultation, and First and Second Reviews under the Staff-Monitored Program with Board Involvement-Press Release; Staff Report; and Statement by the Executive Director for Republic of South Sudan,' (June 10, 2024), <https://www.elibrary.imf.org/view/journals/002/2024/160/article-A001-en.xml>.

⁵⁹ United Nations Security Council, 'S/2025/442 Report of the Panel of Experts on South Sudan,' para 42.

⁶⁰ 'South Sudan on Edge as Its Neighbour's War Disrupts Oil Exports,' International Crisis Group, May 2024, <https://www.crisisgroup.org/africa/horn-africa/south-sudan/south-sudan-edge-its-neighbours-war-disrupts-oil-exports>.

⁶¹ 'S/2024/855 Interim Report of the Panel of Experts on South Sudan.'

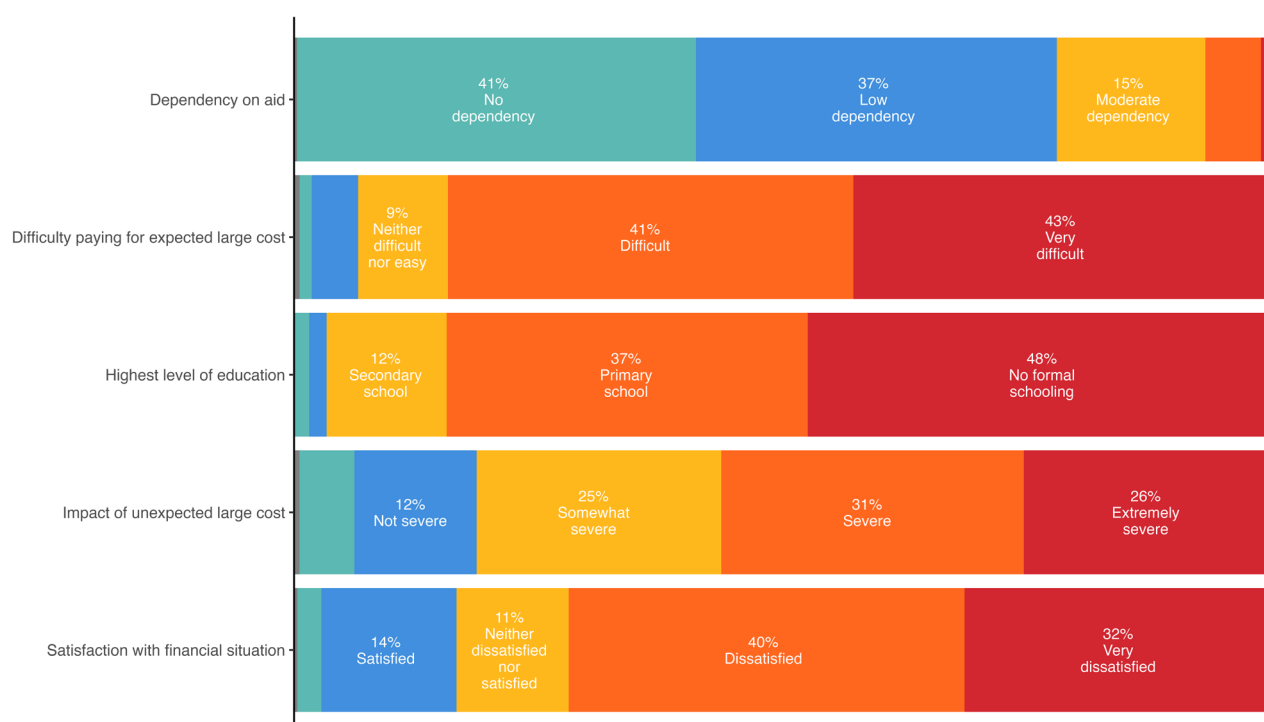
⁶² NUPI, 'Climate, Peace and Security Fact Sheet: South Sudan,' March 2025, <https://www.nupi.no/news/climate-peace-and-security-fact-sheet-south-sudan3>; Mayen, J. V., Wood, E. and Frazier, T. G., 'Practical flood risk reduction strategies in South Sudan', Journal of Emergency Management, vol. 20, no. 8 (July 2022).

⁶³ World Bank, 'Accelerating Sustainable & Clean Energy Access Transformation in AFE Region Multi-Phase Programmatic Approach' (May 30, 2023): 6, <https://documents1.worldbank.org/curated/en/099070723133032794/pdf/P18054700d8958090a1020bb11eeca12a7.pdf>.

⁶⁴ Eugene Chen, Flora McCrone, and David Mozersky, 'Renewable Energy and the United Nations: A Green Spark for Peace in South Sudan,' 2023. The Stimson Center.

Index Component Results: Economic Fragility

Figure 4. Responses to Index Questions: Economic Fragility



The economic fragility index is built from five questions: dependency on aid, difficulty paying for an expected large cost (in this case, a bride price), highest level of education, the impact of an unexpected large cost (in this case, an unexpected death) on the household's financial situation, and satisfaction with the household's financial situation. The figure above detailing responses to the questions which make up the economic fragility index makes clear that most households are not dependent on "aid from NGOs, the UN, or the government." As such, their income primarily comes from other sources; in fact, only 3 per cent of respondents indicate that their primary source of income is UN/NGO support. Yet, most respondents (72%) are either dissatisfied or very dissatisfied with their household's financial situation. Indeed, it would be difficult or very difficult for 84 per cent to pay a bride price,⁶⁵ and an unexpected death would have a severe or extremely severe impact on the economic situation of 57 per cent of households. This disparity between economic precarity and dependence on aid may reflect the insufficiency of current assistance levels in comparison with needs. South Sudan has one of the lowest rates of adult literacy in the world: Only around a third of adults (34.5%) are literate according to the World Bank, with a significant gender disparity. Women are much less likely to be literate (28.9%) than men (40.3%).⁶⁶ Low levels of education, particularly among women and girls, youth, and displaced persons, are closely linked with poverty and unemployment in South Sudan.⁶⁷

Key Issue: Sources of Income

Agriculture plays a central role in livelihoods, with more than half of respondents (51%) identifying it as their household's primary source of income. An even larger majority (76%) report living in households that engage in at least some farming activity. Agricultural activity is largely subsistence-based: in most farming households (79%), crops are grown primarily for household consumption. Even so, around half (47%) also sell some of their produce for profit, suggesting a mixed livelihood strategy in which many households are trying to move beyond mere subsistence to engage in market activity. Notably, only 31 per cent of farming households report that members

⁶⁵ Bride prices in South Sudan legally finalize a marriage under customary law. They are typically paid by the groom's family to the male members of the bride's family. Bride prices are often paid in cattle, cash, or other livestock. Paying bride prices can impose financial strains on South Sudanese families; in Greater Bahr el-Ghazal, Greater Upper Nile, and Greater Equatoria, the average price is between 30 and 50 heads of cattle i.e. between \$17,000 and \$22,000 USD. For more information, see Pospisil, Jan, Melissa Johnston, Adut Alaak Garang, and Nyachangkuoth Rambang Tai. "Bring Enough Cows to Marry: Brideprice, Conflict, and Gender Relations in South Sudan." (Juba, PeaceRep, 2024). <https://peacerep.org/wp-content/uploads/2024/02/Bring-Enough-Cows-to-Marry-BridePrice-Conflict-and-Gender-Relations-in-South-Sudan-DIGITAL.pdf>.

⁶⁶ UN Women, *Gender, Education, and Literacy in South Sudan* (Jan. 2025).

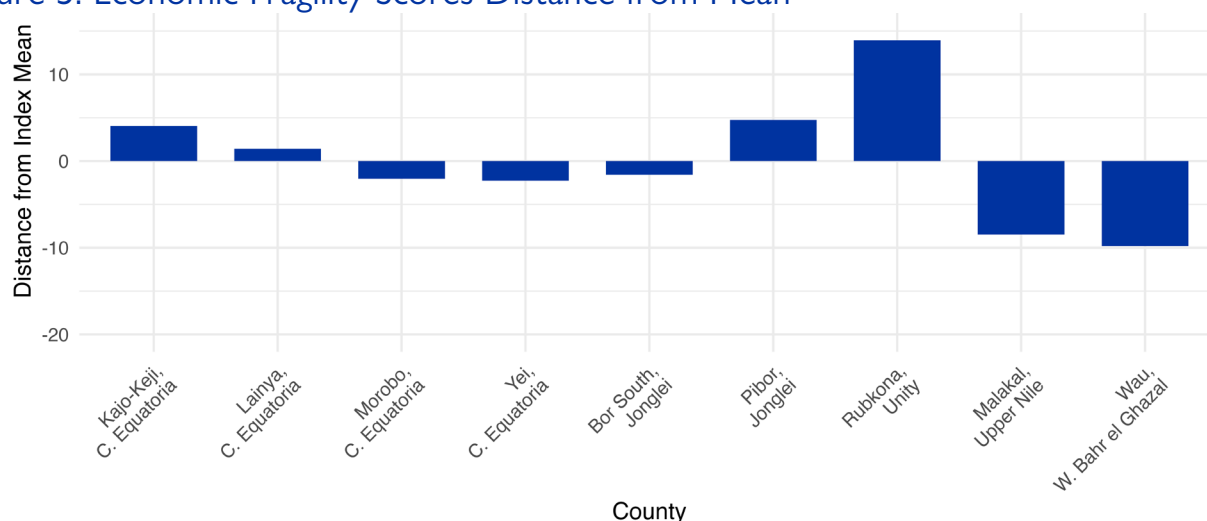
⁶⁷ Sida, "Multidimensional Poverty Analysis: South Sudan," (May 11, 2024), p. 21.

are paid for their role in crop production, underscoring the informal and unpaid nature of much of this labor. The African Development Bank has identified several key challenges for the agricultural sector in South Sudan that hinder farmer's abilities to produce enough crops to sell at a large scale, including the use of low-yield seeds, a lack of roads, limited microfinancing, and unclear land tenure.⁶⁸

While livestock rearing is rarely the primary source of household income (reported by just 4% of respondents), it is much more common as a supplemental livelihood activity: 42 per cent of respondents live in households that own livestock or farm animals. Fishing similarly plays a supporting role in household economies, with 29 per cent of respondents reporting that their households engage in it. Outside of agriculture and natural resource-based livelihoods, smaller portions of the population rely on non-agricultural work as their primary source of money: 12 per cent of respondents say their household's main income is from owning a business, 11 per cent from casual labor, and 6 per cent from employment in someone else's business. These patterns suggest that, while the economy remains overwhelmingly agrarian, many households combine subsistence production with partial market participation and diversify their income sources across farming, livestock, fishing, and informal sector work. It is worth noting, however, that (as determined by a chi-squared test on the frequency of reports about income source) such participation does differ between urban and rural areas. In particular, for example, less than 1 per cent of rural respondents report working for a business, while in urban areas it is almost 5 per cent.

Index Results By County

Figure 5. Economic Fragility Scores Distance from Mean



Rubkona: Highest Economic Fragility

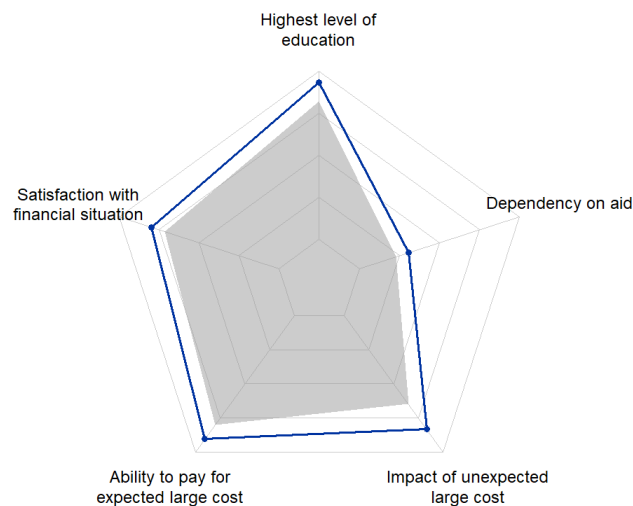
Rubkona has the highest level of economic fragility of the nine surveyed counties. On every question that comprises the economic fragility index, respondents in Rubkona have more difficulty than the average respondent. These challenges reflect both current and historical dynamics. Historically, Bentiu was completely destroyed during the first phase of civil war with limited reconstruction since then. The area has received significant returns from Sudan since April 2023, many of whom have struggled to find livelihood opportunities. Severe flooding in Rubkona in recent years has harmed property, agriculture, infrastructure, and roads, isolating the county from the national economy and impeding the delivery of humanitarian aid. Further, Bentiu town houses a significant displaced population originally from rural Rubkona, who have lost property, livestock and other forms of income over four years of historic flooding.⁷⁹ This qualitative evidence of the adverse effects of flooding on livelihoods is consistent with a key finding of the Environmental Fragility analysis below: 28 per cent of households identified flooding as a significant challenge for their household's ability to produce crops.⁷⁰

⁶⁸ African Development Bank Group, "South Sudan: An Infrastructure Action Plan." (Abidjan, African Development Bank Group, 2013). <https://www.afdb.org/en/countries/east-africa/south-sudan/infrastructure-action-plan-in-south-sudan-a-program-for-sustained-strong-economic-growth>.

⁶⁹ REACH, 'Integrated Public Health Rapid Assessment of Internally Displaced Persons in Rubkona County, Unity State', July 2024, https://repository.impact-initiatives.org/document/repository/e67d9595/REACH-SSD_Integrated-Public-Health-Rapid-Assessment_Rubkona_July2024.pdf.

⁷⁰ See "Environmental Fragility," p. 27.

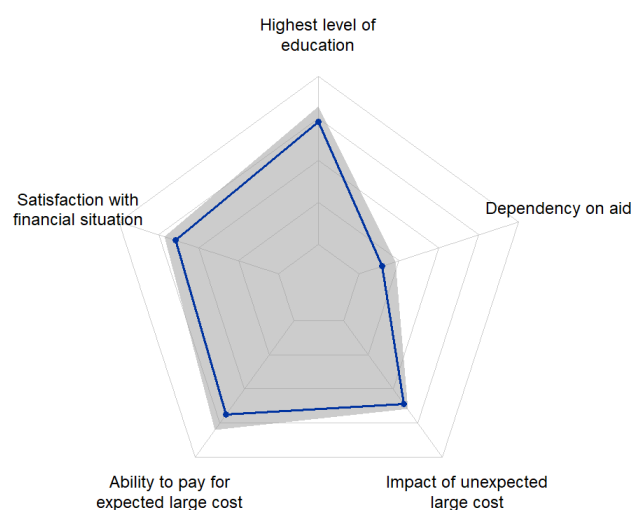
Figure 6. Rubkona Economic Fragility



Wau: Lowest Economic Fragility

Wau stands out as having the lowest level of economic fragility. Respondents from Wau show a higher satisfaction with their household's financial situation than in any other county. Further, across the other four questions that make up the index, respondents from Wau are less fragile than the sample average. These descriptive statistics may reflect the relatively high level of urbanization in Wau county; Wau city is one of the largest towns in the country. This higher level of infrastructure development is also reflected in the lower dependence respondents from Wau report on aid from NGOs, the UN, or the government. Wau town lies at the intersection of significant transportation routes and is the location of the principal market that feeds goods to communities in Northern Bahr-el-Ghazal and Warrap States. This trade in goods supports a lower reliance on subsistence agriculture compared to other areas in South Sudan. Furthermore, levels of education are in Wau are higher than the South Sudanese average due to the prevalence of secondary schools and the University of Western Bahr-el-Ghazal, one of South Sudan's few public universities.⁷¹

Figure 7. Wau Economic Fragility



⁷¹ Wau County Profile, Conflict Sensitivity Resource Facility, South Sudan, accessed 12 August 2025, https://csrf-southsudan.org/county_profile/wau/.

4. ENVIRONMENTAL FRAGILITY

Key Findings

- **Awareness of climate change:** A substantial majority of respondents have observed a change in the timing of seasonal rains over the past 5 years (78%), and a slightly smaller majority (65%) report that water shortages during the dry season have become more frequent.
- **Extreme weather is a significant threat to agriculture:** On average, around a quarter of all respondents reported that droughts (23%), changes in temperature (26%), and flooding (28%) are a “large challenge” for their household’s ability to produce crops.
- **Urban and rural differences:** Respondents in rural bomas report slightly higher levels of environmental fragility as compared to respondents in urban bomas; this difference is statistically significant (Table 2).
- **County differences:** Of the 9 surveyed counties, residents of Rubkona and Bor South experience the highest levels of environmental fragility, whereas inhabitants of Malakal and Yei have the lowest environmental fragility (Figure 2).
- **Climate Change’s Impact on Maternal and Infant Health:** A majority of respondents indicate that climate change has had a severe or extremely severe impact on the health of pregnant women and infants in their community.

Overview of Environmental Fragility in South Sudan

South Sudan is ranked globally as one of the most vulnerable countries to natural hazards including droughts and flooding.⁷² Climate change is most clearly evidenced by more frequent and intense flooding and droughts since the 1960s.⁷³ Further, for the last decade, the temperature in South Sudan has been increasing at twice the global average, while the country has also experienced record-breaking floods.⁷⁴ South Sudan contains the Sudd, one of the world’s largest wetlands, which is fed by Lake Victoria and the Ethiopian Highlands. Rising precipitation and water levels in these areas have significantly increased flood risks in central areas of South Sudan, particularly in Unity State.⁷⁵ Climate change has disproportionately affected rural communities that rely heavily on subsistence farming and pastoralism.⁷⁶ According to UNOCHA, in 2024 alone, 379,000 people across 22 counties in South Sudan fled their homes due to flooding.⁷⁷ The floods and resulting displacement have

reduced access to food, production of food, increased prevalence of agricultural pests, and driven outbreaks of diseases including cholera and malaria.⁷⁸

Droughts and flooding exacerbate tensions between communities in at least three ways. First, limited pastures for grazing and water for cattle have led cattle herders to change their routes, moving beyond agreed cattle migration corridors and impinging on land used for farming. These changes drive violent confrontations between sedentary farmers and nomadic herders.⁷⁹ Second, a lack of water amidst drought can lead to livestock death or force pastoralists to sell cattle that they cannot support at very low prices to merchants and elites who accumulate power by amassing large, consolidated cattle herds.⁸⁰ Forced displacement from floods can also lead families to lose their cattle through drowning or abandonment. Such climate stresses have led some families which have lost their cattle and income to

⁷² International Rescue Committee, “[Emergency Watchlist 2023: South Sudan](#)” (Geneva, International Rescue Committee, 2023).

⁷³ Borgomeo, Edoardo, Claire Chase, Nicolas Salazar Godoy, and Victor Osei Kwadwo, *Rising from the Depths: Water Security and Fragility in South Sudan* (Washington, D.C., World Bank Publications, 2023).

⁷⁴ Famine Early Warning Systems Network (FEWS NET), “[South Sudan: Food security outlook, June 2024–January 2025](#)” (Washington, D.C. USAID Famine Early Warning Systems Network, 2024); Shelley Knowles, “[Entire villages are quickly becoming islands: Dara Johnston from UNICEF explains why climate change is a threat we can't ignore](#)”, UNICEF (24 November 2022).

⁷⁵ Anyier Chaat Paul, “[Facing the Floods](#)”, UN International Children’s Fund (UNICEF) (8 August 2024).

⁷⁶ Z. Ndebele Z. and M. Zenda, “Climate Change Impact and Adaptation of Agro-Pastoral Farmers in Awerial County, Lakes State of South Sudan”, *South African Journal of Agricultural Extension*, 49(4):135–160 (2023).

⁷⁷ UN Office for the Coordination of Humanitarian Affairs (OCHA), “[South Sudan: Floods Snapshot \(As of 8 November 2024\)](#)” (8 November 2024).

⁷⁸ World Food Programme, “[WFP at Hand in South Sudan as Relentless Floods Threaten Lives, Land and Livestock](#)” (18 April 2022).

⁷⁹ Borgomeo, Edoardo, Claire Chase, Nicolas Salazar Godoy, and Victor Osei Kwadwo, “[Rising from the Depths: Water Security and Fragility in South Sudan](#)”, (Washington, D.C., 2023).

⁸⁰ Alex Humphrey et al., “Discussion Paper: Impacts of Disruptions to Livestock Marketing in South Sudan”, SPARC - Supporting Pastoralism and Agriculture in Recurrent and Protracted Crises (July 2021); Ifat Idriss, “Livestock and Conflict in South Sudan”, Helpdesk Report: Knowledge, Evidence & Learning for Development, U.K. Department for International Development (29 October 2018).

engage in cattle raiding.⁸¹ Third, water scarcity exacerbates other stresses, producing grievances ripe for political elites to mobilize. Indeed, only 10 per cent of South Sudanese households have access to sanitation, and 15 per cent of households travel over two hours round-trip for water access during the dry season.⁸²

In addition to floods and drought, South Sudan has experienced environmental degradation from illegal logging and artisanal gold mining. Illegal logging activities increased in intensity after the intensification of conflict in 2016. As refugees fled to northern Uganda, the illegal logging industry moved into the vacuum caused by displacement, resulting in rapid deforestation. When these refugees have returned home, there have been tensions between returnees and those who stayed and oversaw the illegal logging activities.⁸³ Gold mining in South Sudan has also contributed to Sudan's ongoing civil conflict and helped fund South Sudan armed groups.⁸⁴

Populations attempt to cope with climate change using tactics such as the adoption of flood resistant farming methods, livelihood diversification, and weather forecasting and preparedness passed down through indigenous knowledge.⁸⁵ However, some strategies for livelihood diversification, such as the collection of

firewood and charcoal production for fuel, may have adverse environmental impacts such as deforestation. These changes also have gendered implications. For example, women and girls disproportionately bear the burden of collecting firewood (as well as water) for their households, limiting their opportunities for education and employment and increasing their risk of exposure to gender-based violence as discussed in more detail under Human Fragility.⁸⁶



Index Component Results: Environmental Fragility

The first four questions that comprise the environmental fragility index asked respondents to rate the impact of any of the following climate-related changes on household income, food consumption, mobility, and health: “too much rain, too little rain, temperatures that are too high, temperatures that are too low, pests and crop diseases, and deforestation and erosion.” The fifth question in the index assesses the household’s ability to act on early warnings of impending environmental threats such as flooding. See *Figure 8. Responses to Index Questions: Environmental Fragility* below for descriptive statistics regarding the index questions.

⁸¹ Tonny Muwangala, “UNMISS Begins Campaign to Preempt Deadly Clashes Between Cattle Camps in Lakes Region”, U.N. Mission in South Sudan (20 January 2020). It is important to note that it is not only environmental stressors which contribute to cattle raiding. In fact, the practice has a long history in South Sudan. For example, see Wild, Hannah, Jok Maduk Joj, and Ronak Patel. 2018. “The militarization of cattle raiding in South Sudan: how a traditional practice became a tool for political violence.” 3(2): 1-11. .

⁸² Borgomeo, Edoardo, Claire Chase, Nicolas Salazar Godoy, and Victor Osei Kwadwo, *Rising from the Depths: Water Security and Fragility in South Sudan* (World Bank Publications, 2023).

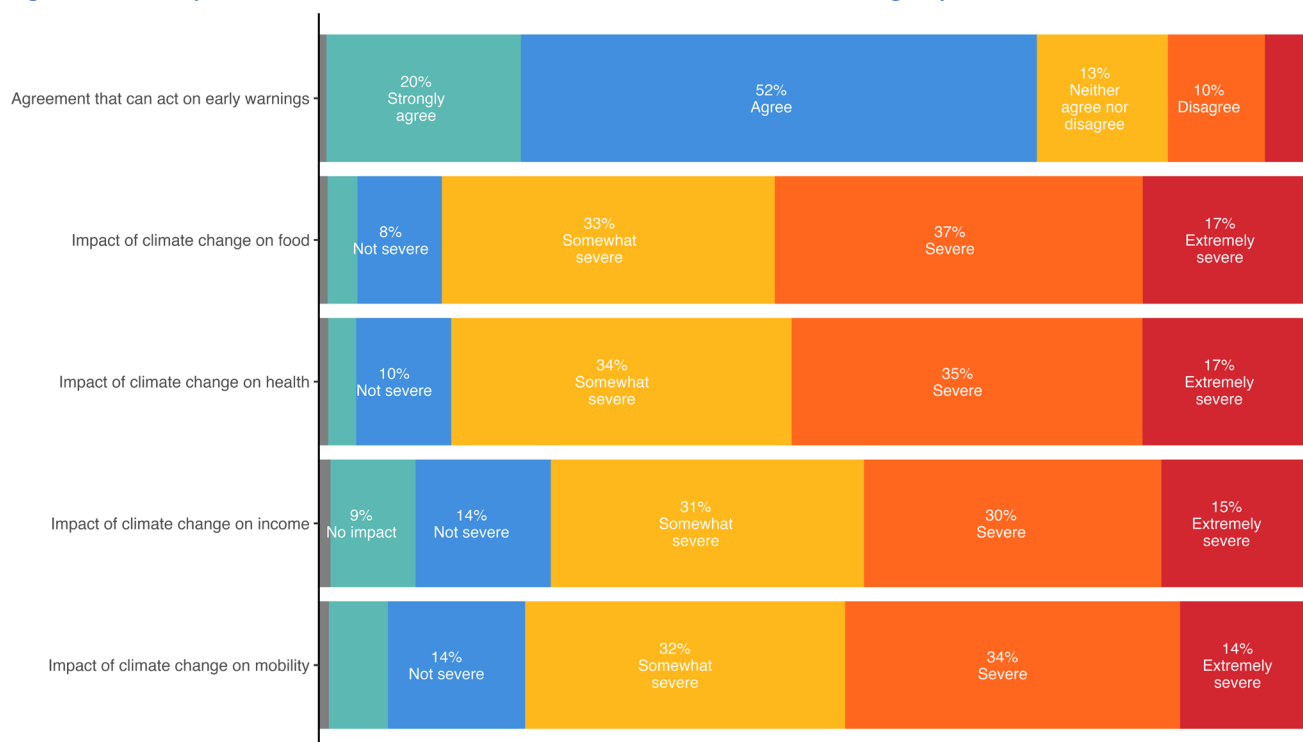
⁸³ South Sudan Council of Churches & UK Aid, “The Impact of Logging Activities on Local Communities” (2020): 8.

⁸⁴ Dylan Walsh, “The Gold Rush at the Heart of a Civil War,” New York Times (11 December 2024) ; Joshua Craze, “On the Brink: The Politics of Violence in South Sudan,” (Norway, Norwegian Ministry of Foreign Affairs, 2025); Marcena Hunter and Ken Opala, “Tarnished Hope: Crime and Corruption in South Sudan’s Gold Sector,” (Switzerland, Global Initiative Against Transnational Organized Crime, 2023)

⁸⁵ Shisanya, Chris A., Joy A. Obando (2024). Stories from the Front Line: Coping Strategies for Flood Disasters among Dinka Community of Bor County, South Sudan. Open Access Library Journal 11(9): 1-21.

⁸⁶ UN Humanitarian, “South Sudan: ‘Living as a woman is a fight we go through daily,’” (Juba, UN Humanitarian, 2023).

Figure 8. Responses to Index Questions: Environmental Fragility



Respondents perceive household income as being less affected by climate change than food consumption and health. This suggests that the economic consequences of climate change may be viewed as secondary to more urgent day-to-day needs. Respondents may not be aware of how climate change's impact on income may in turn shape their longer-term access to food, access to healthcare, and overall health. The severity of climate change's impact on food and health highlights the vulnerability of subsistence-based livelihoods, where even small disruptions in weather patterns can rapidly escalate into food shortages or health crises. Ultimately, these findings suggest that climate change is not only an economic issue but also a direct threat to survival.

The results regarding mobility underscore a structural dimension of environmental fragility; climate-related pressures are degrading people's autonomy over where they live, contributing to displacement. In other words, environmental degradation is reshaping settlement patterns in South Sudan, likely forcing people to abandon their traditional environmental coping strategies.

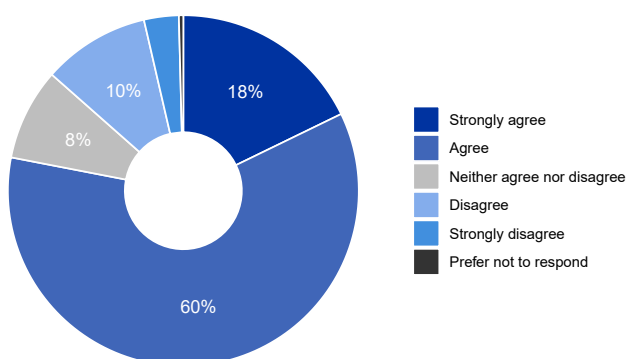
However, the strong agreement that respondents have in their ability to protect their household "if [they] knew about [an] incoming flood before the water arrived," an indicator of capacity to respond to early warnings, points to a critical entry point for building resilience. Indeed, communities are aware, willing, and able to take protective action when equipped with timely information.

Key Issue: Perceived Impact of Climate Change

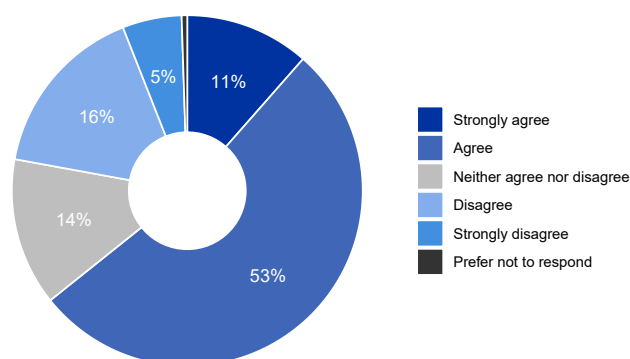
Awareness of climate change in South Sudan is high. A substantial majority of respondents expressed a clear awareness of shifts in local weather patterns, particularly noting changes in the timing of seasonal rains and increased frequency of dry seasons.⁸⁷ Other sources of data provide objective evidence of the changes that are subjectively perceived by survey respondents. Satellite imagery confirms a decrease in the amount and frequency of rainfall over the last 50 years.⁸⁸

Figure 9. Perceptions of Shifts in Local Weather Patterns

Agreement: “In the last 5 years, the timing of the seasonal rains has changed”



Agreement: “In the last 5 years, water shortages in the dry season have become more frequent”



These perceived shifts in climate and weather patterns are not seen in isolation; they are linked to broader concerns about environmental degradation including declining soil fertility and diminishing wildlife diversity.⁸⁹ Looking ahead, there is a strong sense of apprehension about what continued climatic changes could bring. People expect more frequent droughts and floods, but what stands out most is the fear of worsening food insecurity. A majority believe that these changes could lead to more food shortages and potentially famines.⁹⁰ Other studies provide evidence to corroborate this perceived relationship between changes in rainfall and food insecurity.⁹¹

Many also anticipate increased displacement as a result of climate change. There is a growing belief that, if current trends continue, migration may become a necessity rather than a choice for a significant portion of the population.⁹² The perceived link between migration and climate change is discussed further in the next section.

⁸⁷ 78 per cent of respondents agreed or strongly agreed that the timing of seasonal rainfall has changed over the past five years, and 65 per cent agreed or strongly agreed that the water shortages in the dry season have become more frequent.

⁸⁸ Andu Zakaria Lukwasa et al., “Spatio-temporal Rainfall Variability and Its linkage with Large Scale Climate Oscillations Over the Republic of South Sudan,” *Climate Services* 28 (2022).

⁸⁹ 52 per cent agree or strongly agree that wildlife diversity has diminished over the past five years, and 53 per cent of respondents agree or strongly agree that land fertility has declined.

⁹⁰ 41 per cent of respondents think that environmental changes will cause more droughts and water shortages over the next five years, and 42 per cent think that the changes will cause more floods. 58 per cent of respondents think that these changes will cause more famines and food shortages.

⁹¹ Hilda Muteshi & Laban Ashioya Etindi, “Too much and too little rain: food insecurity among displaced and host communities in South Sudan,” *Humanitarian Practice Network* 84 (2024).

⁹² 52 per cent expect that their household’s need to migrate would moderately or significantly increase “if changes in the climate or weather patterns continue to worsen at the same rate that they are changing now.” 36 per cent believe that these climatic changes will increase the number of displaced persons over the next five years.

Key Issue: Climate-Related Movements

Climate displacement is not only a threat in the future, but also a reality in the present. The data highlight how climate-related factors are emerging as significant drivers of displacement and decision-making among both IDPs and returnees. According to DTM, natural disaster is the largest trigger for displacement in South Sudan, with 31.6 per cent of IDPs listing disaster as the cause of their displacement.⁹³ While floods stand out as the most commonly cited reason for displacement among IDPs in the sample, other environmental stressors including droughts, wildfires, and temperature shifts are also perceived as drivers.⁹⁴

Climate conditions are not just triggering displacement; they are also shaping whether and where people feel comfortable returning to their place of origin. Over a third of IDPs (36%) reported that weather patterns

play a major role “when deciding whether to return to your area of habitual residence or relocate to a different location.” For 49 per cent of returnees in the sample, similarly, “weather conditions” were very important considerations when their household decided to return to their county of origin.

These findings suggest that climate is a central consideration in how displacement patterns are unfolding as well as an important factor shaping how people recover from displacement. For solutions to displacement to be effective, they must factor in how environmental risk perceptions and evolving mobility preferences are linked to climate vulnerability. displacement to be effective, they must factor in how environmental risk perceptions and evolving mobility preferences are linked to climate vulnerability.

Key Issue: Climate Change and Maternal/Infant Health

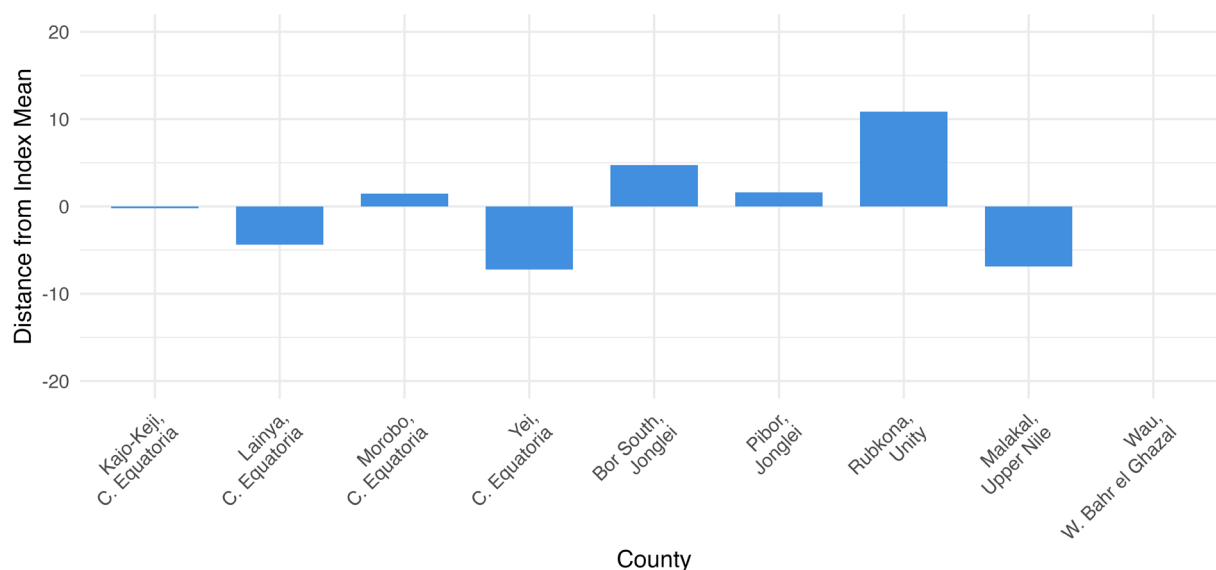
Climate change has had an impact on the health of pregnant women; 51 per cent of respondents indicate that the impact of climate change on “the health of pregnant women in your community writ large” has been either severe or extremely severe. Similarly, a majority of respondents (54%) indicate that the impact of climate change “the health of infants in your community writ large” has been severe or extremely severe. Given that climate change has affected access to health care facilities, and flooding can exacerbate the spread of diseases such as cholera, it is unsurprising that respondents indicate that climate change has a negative impact on the health of some of the most vulnerable members of their communities.

⁹³ IOM, Displacement Tracking Matrix, Round 16, April 2025.

⁹⁴ Among the IDPs in our sample, 35 per cent say that floods played “a large role” in their displacement, 22 per cent say that droughts did, and 14 per cent indicated that wildfires did. 20 per cent noted that “changes in temperature” played such a role.

Index Results By County⁹⁵

Figure 10. Environmental Fragility Scores Distance from Mean



Rubkona: Highest Environmental Fragility

Rubkona stands out as the county with the highest level of environmental fragility, which is unsurprising given the extreme flooding that has affected the area over the past four years. According to an IOM conflict analysis in 2023, floods in the preceding two years had affected more than 80 per cent of the land in the county.⁹⁶ During the wet season, dry land remains extremely limited, and many flee to higher-ground areas such as the IDP sites in Rubkona town and the IDP camp.⁹⁷ Further, the dikes surrounding Bentiu town, which were built in order to protect populations from flood water, remain vulnerable.⁹⁸

Many of the impacts of environmental degradation manifest as economic and societal fragility. In Rubkona, years of floods have weakened households' social bonds. Cattle and livestock in rural areas have died in large numbers due to flooding, disease, and a lack of pastures, devastating livelihoods.⁹⁹ As a result, formerly-rural populations have crowded into the IDP sites in the town but have few livelihood options because their skills are in traditional farming, there is limited land available for farming in more urban areas, and there are limited non-farming livelihood opportunities within camps.¹⁰⁰ The consequences are far-reaching. For example, according to an IOM Inter-Sectoral Needs Assessment, over half of the urban population of Rubkona is concerned about early marriage for women; early marriage is often used as a negative livelihood coping strategy. Further, about a third of people in the urban areas of Rubkona are concerned about sexual violence against children.¹⁰¹

⁹⁵ Note that the average environmental fragility score for Wau is within .1 of the sample mean.

⁹⁶ IOM South Sudan, "Rubkona Village Assessment Survey," (Juba, IOM, 2023). Kwajok Tongun, "Roriak Livestock and Flood Assessment," (Juba, Veterans Without Borders, 2022).

⁹⁷ REACH, "Integrated Public Health Rapid Assessment of Internally Displaced Persons in Rubkona County, Unity State," (Geneva, REACH, 2024).

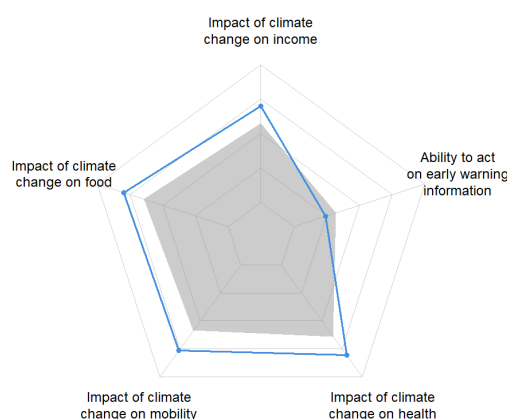
⁹⁸ Mark Townsend, "A drowning town: are Bentiu's dykes high enough to save it from disaster?" (London, The Guardian, 2024).

⁹⁹ Lieth Nyak, "Floods Escalating Livestock Casualties in South Sudan," (Juba, Juba Echo, 2021). Kwajok Tongun, "Roriak Livestock and Flood Assessment," (Canada, Veterinarians Without Borders, 2022).

¹⁰⁰ REACH, "Integrated Public Health Rapid Assessment of Internally Displaced Persons in Rubkona County, Unity State," (Geneva, REACH, 2024).

¹⁰¹ IOM DTM, "Inter Sectoral Needs Assessment 2022 Rubkona County," (Juba, IOM, 2022).

Figure 11. Environmental Fragility in Rubkona



Malakal and Yei: Lowest Environmental Fragility

Malakal and Yei stand out as the counties with the lowest levels of environmental fragility. Malakal is less likely to flood than other counties and lies at a fertile point on the Nile. In fact, people from the Nuer ethnic group from Northern Jonglei who were displaced by floods have travelled hundreds of miles and settled in Malakal without plans of return to their areas of origin.¹⁰² Similarly, Yei is a trading hub at the center of one of the most fertile agricultural areas in South Sudan. Its geography makes it less susceptible to climate stresses. Indeed, according to the World Bank, drought risks are low in the part of the Upper Nile in which Malakal is located as well as in Central Equatoria State, where Yei County is located.¹⁰³ Central Equatoria is also one of only a few southernmost states in South Sudan which the World Bank indicates contains at least some area outside the highest risk factor for hot day heat risk.¹⁰⁴

Despite the overall low levels of environmental fragility in these two counties, in both Malakal and Yei there is a statistically significant difference in the level of environmental fragility between rural and urban areas. These differences are also substantively significant. While rural residents of Malakal have an average environmental fragility score of 52, urban residents have a mean score of only 46. In Yei, the difference is between 47 and 49.

Figure 12. Environmental Fragility in Malakal

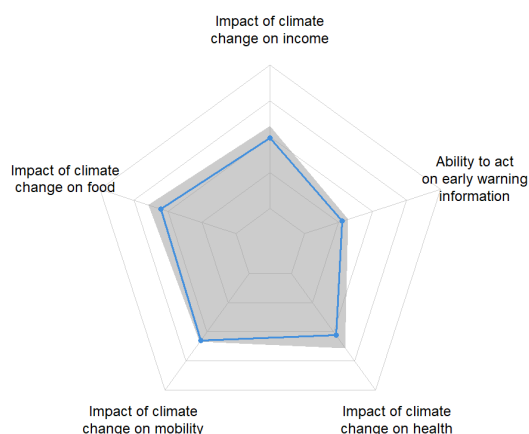
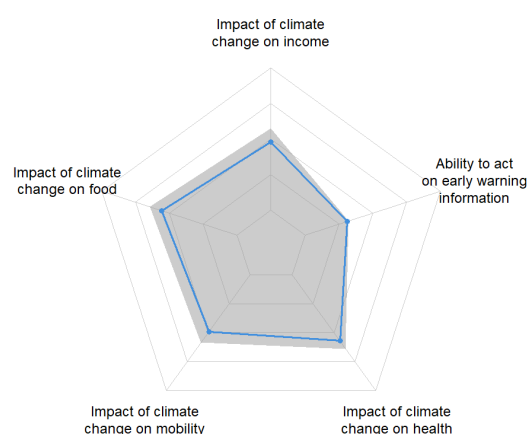


Figure 13. Environmental Fragility in Yei



¹⁰² Focus groups interviews with authors and IOM in July 2024 with Nuer flood displaced at the Old Fire Brigade in Malakal Town.

¹⁰³ World Bank Group, Water: *Rising from the Depths: Water Security and Fragility in South Sudan*, (Washington, D.C., World Bank, 2003).

¹⁰⁴ World Bank Climate Change Water Portal. *Heat Risk*, (Washington, D.C., World Bank, 2021).

5. HUMAN FRAGILITY

Key Findings

- **The burden of water collection falls heavily on women and girls:** Women and girls are most likely to be responsible for water collection, increasing their exposure to risks including gender-based violence.
- **A large proportion of households experience challenges with food security:** A majority of households (60%) report “never” or “just once or twice” having gone without food in the past year. However, 40 per cent report they have gone without food at least “several times.”
- **Consumption tradeoffs leading to health exposure:** Many households report having to forgo health treatment due to an inability to pay for such healthcare; less than a quarter of respondents have “never” gone without necessary healthcare in the past year. This finding implies that households are more likely to forgo healthcare than other essentials such as food.
- **Fear and risk of gender-based violence:** Most people in the sample (61%) report being frightened or very frightened of gender-based violence, and women are more fearful than men.

Overview of Human Fragility in South Sudan

Human fragility refers to risk factors that threaten the health, well-being, and livelihoods of individuals. Human fragility encapsulates access to food, clean water, healthcare, and shelter as well as exposure to the risk of sexual violence. The study follows the OECD in including gender-based violence as a component of human fragility.¹⁰⁵

The Integrated Food Security Phase Classification (IPC) predicted that 57 per cent of South Sudan’s population will face crisis-level or worse acute insecurity during the 2025 lean season, which lasts from April through July.¹⁰⁶ Food insecurity is due to a range of factors, including conflict and unpredictable climatic events, limited use of modernized farming techniques or technology, disruptions in trade caused by the war in Sudan, and a limited ability to move food around the country.¹⁰⁷ The population experiencing food insecurity has increased after the influx of populations fleeing the conflict in Sudan. As of April 2025, returnees fleeing Sudan now account for half of those suffering from catastrophic hunger levels in South Sudan.¹⁰⁸ Further, as of December 2024, 5 million people were in need of WASH support.¹⁰⁹ 37 per cent of the population relies on unsafe sources of water, and three quarters of households lack access to handwashing facilities. Sanitation challenges are closely related to health challenges.¹¹⁰ Indeed, in July 2024, the World Health Organization identified malaria, maternal and child health risks, non-communicable diseases, and water-borne illnesses as all posing very high risks for South Sudan. For example, malaria is the leading cause of death and illness in the country, accounting for about 30 per cent of deaths. Although non-communicable deaths (NCD) cause 28 per cent of all deaths in South Sudan, only 10 per cent of primary health care centers provide NCD services.

¹⁰⁵ OECD, “How fragile contexts affect the well-being and potential of women and girls,” *OECD* (March 9 2023), [https://one.oecd.org/document/DCD\(2022\)12/en/pdf](https://one.oecd.org/document/DCD(2022)12/en/pdf). OECD, “Adding the human dimension to the OECD fragility framework,” *OECD Development Cooperation Directorate* (March 9 2022), [https://one.oecd.org/document/DCD\(2022\)11/en/pdf](https://one.oecd.org/document/DCD(2022)11/en/pdf).

¹⁰⁶ IOM, “*South Sudan Humanitarian Needs and Response Plan*,” IOM (Juba, IOM, 2024).

¹⁰⁷ UNICEF, “Hunger and Malnutrition Being Driven by Climate Crisis and Conflict in South Sudan,” October 16, 2024, <https://www.unicef.org/press-releases/hunger-and-malnutrition-being-driven-climate-crisis-and-conflict-south-sudan>.

¹⁰⁸ United Nations, “*South Sudan: Conflict and hunger push millions to the brink*,” (New York, UN News, 2025).

¹⁰⁹ Humanitarian Country Team, “*Humanitarian Needs and Response Plan South Sudan 2025*,” (Juba, OCHA, 2024).

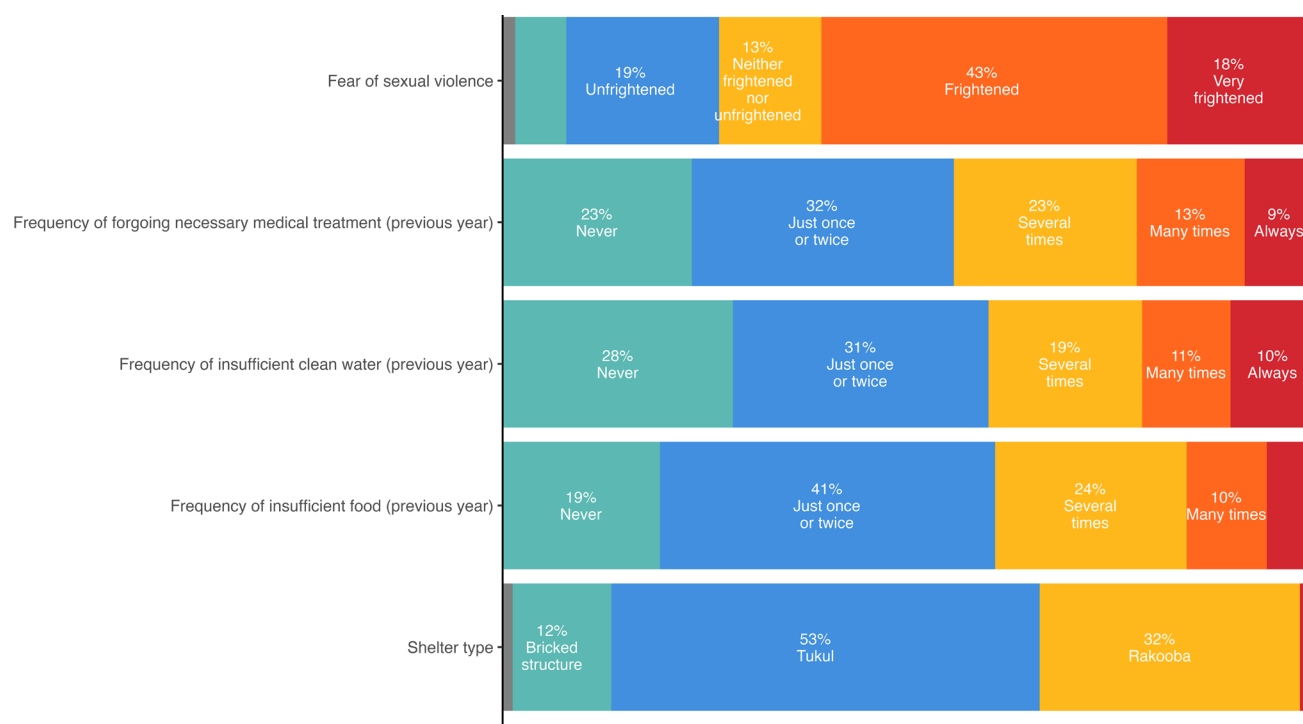
¹¹⁰ World Food Programme, UNICEF, and Food and Agriculture Organization of the United Nations, “*Food Security and Nutrition Monitoring System Round 30*,” (Juba, South Sudan, WFP, FAO, UNICEF, 2025).

South Sudan also has an extremely high maternal mortality rate, with an estimated 789 maternal deaths per 100,000 live births. 64 infants out of every 1,000 births die before they turn 1 year old.¹¹¹ Water-borne illnesses are linked to inadequate shelter. Indeed, IOM has noted that over 4 million people in South Sudan will require emergency shelter and non-food item assistance (SNFI) in 2025.¹¹²

In South Sudan, women and girls experience widespread gender-based violence (GBV).¹¹³ The main causes include women's underrepresentation in positions of authority, their economic dependence on men, and strong gender norms. Constitutionally, South Sudan has 35 per cent gender quotas for positions in government – but these have been poorly implemented since the Revitalized Agreement on the Resolution of the Conflict in South Sudan (R-ARCSS).¹¹⁴ Although GBV is stigmatized and therefore under-reported, a 2022 UN Human Rights report concluded that it is “widespread.”¹¹⁵ A 2023 study indicated that 34 per cent of women in South Sudan between the ages of 15-64 have experienced physical violence, and 13.5 per cent have suffered from sexual violence.¹¹⁶ The vast majority of victims of GBV in South Sudan are women and girls, but boys and men have also been targeted.¹¹⁷

Index Component Results: Human Fragility

Figure 14. Responses to Index Questions: Human Fragility



¹¹¹ Health Cluster, World Health Organization, “South Sudan: Public Health Situation Analysis,” *World Health Organization* (July 15 2024), <https://reliefweb.int/report/south-sudan/south-sudan-public-health-situation-analysis-phsa-15-july-2024>.

¹¹² OCHA, “South Sudan Humanitarian Needs and Response Plan,” (Juba, OCHA, 2024).

¹¹³ “Steven Chimwemwe Iphani, “A boy should be a fighter: Addressing harmful masculinities driving cattle-related violence,” IOM South Sudan (2021), <https://publications.iom.int/system/files/pdf/A-Boy-Should-be-a-Fighter.pdf>.

¹¹⁴ South Sudan Women's Coalition, Norwegian Embassy, Sweden, and UNDP, “Implementation of the 35 per cent provision R-ARCSS,” (South Sudan Women's Coalition, Juba, 2020), <https://www.undp.org/sites/g/files/zskgke326/files/migration/ss/Gender-Matrix-RARCSS-South-Sudan-April-2020.pdf>.

¹¹⁵ “UN Human Rights Council, “Conflict-Related Sexual Violence Against Women and Girls in South Sudan,” A/HRC/49/CRP.4 (Mar. 21, 2022): 1.

¹¹⁶ United Nations Population Fund (UNFPA), “South Sudan Ranks Second in GBV Prevalence Rate in East Africa – New Study Indicates,” November 27, 2024, <https://southsudan.unfpa.org/en/news/south-sudan-ranks-second-gbv-prevalence-rate-east-africa-%E2%80%93-new-study-indicates>.

¹¹⁷ IOM South Sudan, “Gender-Based Violence Knowledge, Attitudes and Practices Survey in South Sudan,” (2019): 5, <https://publications.iom.int/system/files/pdf/south-sudan-gender-based-kap.pdf>.

Responses to the index questions on human fragility demonstrates that access to food, clean water, healthcare, and shelter varies greatly across households. A majority of respondents have uninterrupted or almost continuous access to food, although a significant minority (39%) have gone without food at least “several times” in the past year. Similarly, a majority of respondents have uninterrupted or almost continuous access to clean water. Among those respondents who indicate that clean water is not available in their community, the most common reasons cited were distance from the nearest water source (41%) and not enough water (24%).¹¹⁸

Access to healthcare and adequate shelter is more limited than access to food and clean water. About one quarter have sacrificed necessary health care “always” or “many times” over the past year, and only about a quarter have “never” gone without necessary health services. Given that healthcare is required with less frequency than food or water, these responses suggest that a significant majority of households are not receiving critical treatment. Further, while very few respondents do not have any shelter to live in, about a third are living in rabookas i.e. temporary shelters built with available local materials. Inadequate shelter likely contributes to healthcare needs; individuals living in informal housing and crowded conditions face health risks from poor sanitation, toxic building materials, unvented cooking facilities, and the spread of communicable disease.¹¹⁹ Most concerning, a clear majority of respondents fear sexual violence in their community. Men and women respond to this question differently; t-tests indicate that women are more afraid of such violence than men. While the question does not inquire about various forms of sexual violence, South Sudan struggles with widespread conflict-related sexual violence¹²⁰ as well as abuses perpetrated by civilians.¹²¹



Key Issue: Water Collection

When asked which member(s) of the household are primarily responsible for collecting water, a majority of respondents indicate that the task exclusively falls on women and young girls.¹²² Heavy reliance on women and girls for water collection increases their exposure to the risk of gender-based violence. The UN's Food Security and Nutrition Monitoring System found in 2025 that 25

per cent of women and girls in South Sudan felt unsafe when accessing WASH facilities.¹²³ According to UNICEF, less than half of households in South Sudan have access to improved water sources within 30 minutes.¹²⁴ As such, water collection is a time consuming task which limits girls' and women's educational and employment opportunities.

¹¹⁸ Question 11.3: “Reasons why water is not available to everyone?” was asked only to the 1,091 respondents who previously reported that water is not available to everyone in their county.

¹¹⁹ World Health Organization, “[WHO Housing and Health Guidelines](#),” (Geneva, World Health Organization, 2018), . 4.

¹²⁰ UN Human Rights Council, “[Conflict-related sexual violence against women and girls in South Sudan](#),” (New York, UN Human Rights Council, 2022).

¹²¹ ICRC, “[Addressing Sexual Violence in South Sudan: Highlights from 2025](#),” (Geneva, ICRC, 2024)

¹²² For example, 22 per cent rely exclusively on an “adult woman” and 36 per cent rely exclusively on an “adult woman” and a “female child.”

¹²³ Humanitarian Country Team, “[Humanitarian Needs and Response Plan South Sudan 2025](#),” (Juba, OCHA, 2024).

¹²⁴ UNICEF, “[Progress on Household Drinking Water, Sanitation and Hygiene 2000-2022](#),” (New York, UNICEF, 2023).

Key Issue: Gender-Based Violence and Child Marriage

In addition to the index question on fear of GBV, the survey included a question on the social acceptability of two common forms of GBV: intimate partner violence and child marriage.¹²⁵ Overall, 10 per cent of respondents believe that it is always acceptable for a man to hit a woman and another 23 per cent believe it is sometimes acceptable under certain conditions – for example, within a relationship, during war, or to protect the family's honor. Surprisingly, men and women do not differ in a statistically significant fashion in their responses to this question despite the fact that women are disproportionately harmed by GBV. There is also not a statistically significant difference in responses between residents of county capitals and rural areas.

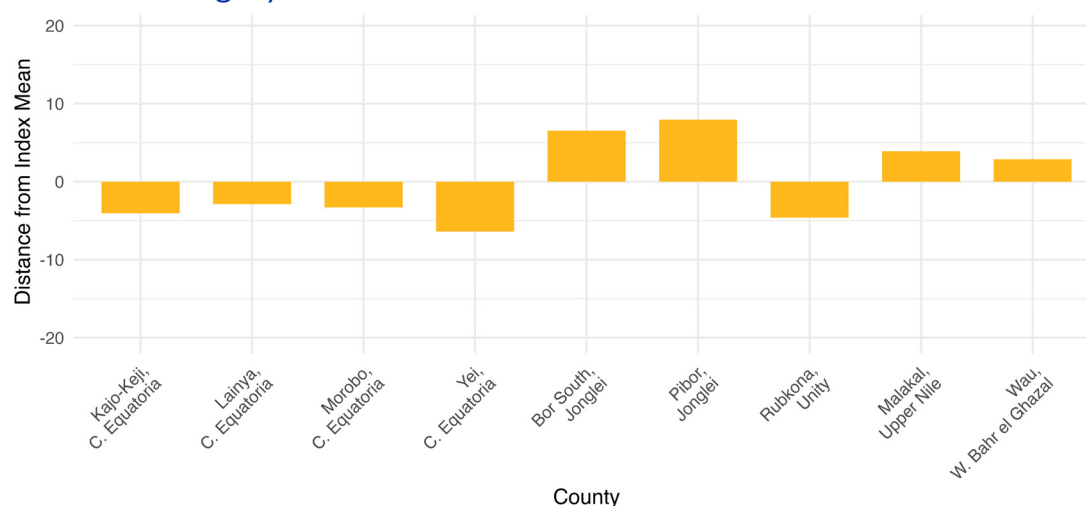
50 per cent of respondents indicate that it is “never acceptable” for children to marry before the minimum legal age of 18. However, 34 per cent of respondents believe that it is “sometimes acceptable, such as to resolve a dispute between families, gain money/cows for bride price, or feed one's household.” 14 per cent of respondents believe that child marriage is “always acceptable.” T-tests indicate that men and women responded to this question differently; women are more supportive of child marriage than men. There are also statistically significant differences between how rural and urban respondents answer; rural South Sudanese in the sample are more supportive of child marriage than those living in county capitals.

Key Issue: Human Fragility & Dependence on Aid

None of the measures of human fragility used in the index are correlated with a question about household dependence on aid.¹²⁶ One interpretation of this finding is that humanitarian organizations are accurately targeting those with the most need of lifesaving aid, successfully elevating their access to food, water, shelter, and healthcare to the average level of South Sudanese. An alternative – more likely – interpretation is that provision of aid makes little difference to those who receive it, meaning that it is insufficient to address their full range of critical needs. In July 2025, the World Food Programme could only reach 30 per cent of South Sudanese facing severe hunger.¹²⁷ In 2025, UN agencies faced a funding gap of 46 per cent in South Sudan.¹²⁸ These gaps between need and assistance have widened progressively since the redirection of international priorities with the war in Ukraine, and are not purely the consequence of cuts to US foreign aid.

Index Results by Country

Figure 15. Human Fragility Scores Distance from Mean



¹²⁵ Question 6.13: “In your opinion, is it ever acceptable for a man to hit a woman?” and Question 6.15: “The minimum age of marriage according to South Sudan's laws is 18 years. In your opinion, is it ever acceptable for children to marry before the minimum legal age?” UN Human Rights Council, “Conflict-related sexual violence against women and girls in South Sudan,” (New York, UN Human Rights Council, 2022).

¹²⁶ None of the correlations have an absolute value greater than 0.1.

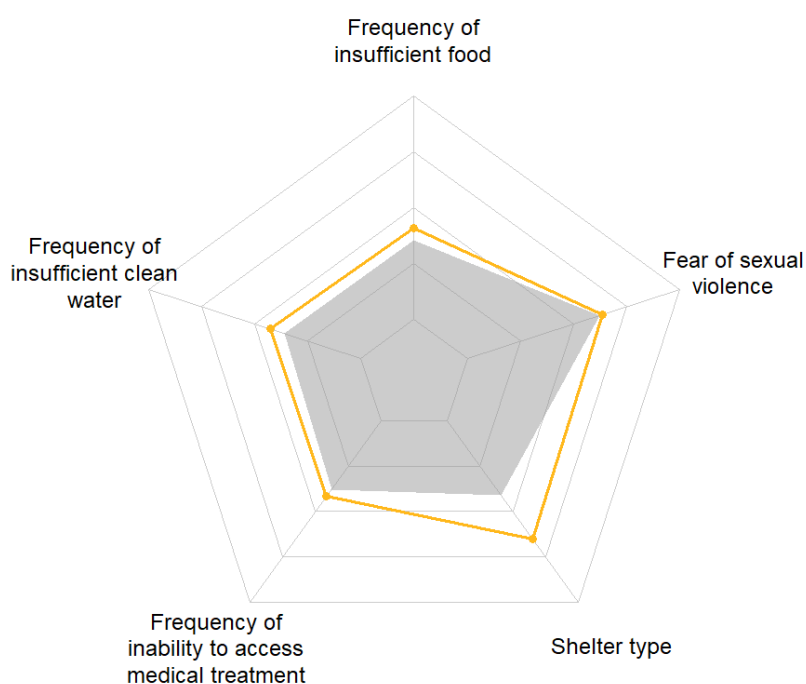
¹²⁷ World Food Programme, “Millions risk losing access to humanitarian food assistance amid funding slowdown in South Sudan”, July 2025, <https://www.wfp.org/news/millions-risk-losing-access-humanitarian-food-assistance-amid-funding-slowdown-south-sudan>

¹²⁸ “2025 STRATEGY ON PRIORITY ENGAGEMENTS OF OFFICE OF THE DSRSG/RC/HC, SOUTH SUDAN,” May 2025: p2.

Pibor: Highest Human Fragility

Pibor stands out as the county with the highest level of human fragility. At the end of 2024, the IPC assessed that 80 per cent of the residents of the Greater Pibor Administrative Area (GPAA) were facing severe acute food insecurity – the highest level of any administrative unit in South Sudan.¹²⁹ This human fragility has both political and climate causes. The largest ethnolinguistic group in GPAA, the Murle, fought against the Sudan People's Liberation Movement (SPLM) and experienced marginalization at the hands of other groups and the government in Juba – until the Murle insurgent movement, the South Sudan Defence Movement/Army – Cobra Faction (SSDM/A-Cobra Faction) won special autonomy in 2014 through the creation of GPAA from a territory that used to be part of Jonglei State.¹³⁰ However, since 2014, the GPAA has received little assistance from the government, and, as the UN Panel of Experts on South Sudan has documented, been subject to “violent asset-stripping”.¹³¹ This has occurred through the medium of apparent ‘inter-communal’ disputes between the Murle and Lou Nuer and is exacerbated by messages from national politicians, causing violence that has both disrupted agriculture and aid distributions.¹³² Overlaying these political drivers of human fragility, floods have damaged clean water sources, contributing to disease outbreaks.¹³³ Severe flooding in 2021 led to diarrhea, malaria, and acute malnutrition in children.¹³⁴ Sanitation is clearly an issue in the county, as reflected in the lack of clean water (55% at least going without clean water “several times” in the past year) or medicines (56% at least “several times” in the last year) respondents report.

Figure 16. Human Fragility in Pibor



¹²⁹ United Nations Security Council, “S/2025/442 Report of the Panel of Experts on South Sudan,” para 121; IPC, “[South Sudan: Acute Food Insecurity Situation for September - November 2024 and Projections for December 2024 - March 2025 and for April - July 2025](#),” (IPC, Geneva, 2024).

¹³⁰ For a history of Murle insurgency and marginalization, see: Edward Thomas, *South Sudan: A Slow Liberation* (London: Zed Books, 2015).

¹³¹ United Nations Security Council, “S/2025/442 Report of the Panel of Experts on South Sudan,” para 121.

¹³² PC, “[South Sudan: Acute Food Insecurity Situation for September - November 2024 and Projections for December 2024 - March 2025 and for April - July 2025](#),” (IPC, Geneva, 2024).

¹³³ Sudan: Cholera Outbreak Situation Report,” *World Health Organization* (2025).

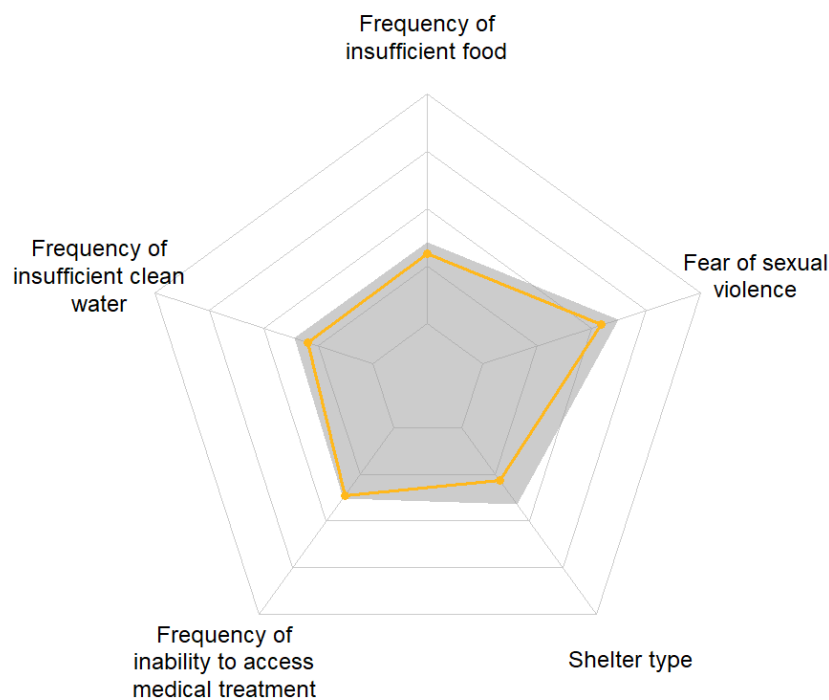
¹³⁴ Willemot, Yves. “[Will we be able to care for the growing number of children in need?](#)” In Pibor, the floods and intercommunal violence have led to an explosion of malnutrition in children.” (UNICEF, South Sudan, 2021); Oxfam International, “[South Sudan: People in Pibor County Resort to Eating Wild Vegetables as Hunger Crisis Deepens](#),” (July 29, 2024).

Yei: Lowest Human Fragility

Large-scale infrastructure projects to improve sanitation may have had an impact in Yei, as indicated by its low level of human fragility.¹³⁵ Several hospitals have also been constructed in the county.¹³⁶ Indeed, respondents in Yei are amongst the least likely to have to go extended periods of time without medicine or medical treatment, according to our results. 18 per cent of respondents in Yei also live in a brick shelter as opposed to a more rugged structure.

The low level of human fragility in Yei may also be explained by its favorable geography and climate; the county has long been considered the agricultural “breadbasket” of South Sudan, benefiting from fertile soils and higher-than-average rainfall conducive to farming.¹³⁷ Yei has experienced acute periods of insecurity that have periodically disrupted agriculture – it was the location of attacks by the Lord’s Resistance Army in the early 2000s and became the epicenter of the civil war after 2015, when the National Salvation Front (NAS) and other rebel groups conducted military offensives in the surrounding areas.¹³⁸ However, Yei has a favorable position on the tripartite border between South Sudan, Uganda and the DRC. This means that populations have been able to move back and forth fluidly during periods of conflict, and that there has been a steady flow of trade through the Yei town.¹³⁹

Figure 17. Human Fragility in Yei



¹³⁵ ReliefWeb, “Groundbreaking for Yei’s drinking water supply system” (2014)

¹³⁶ Namely, “His House of Hope and Faith Hospital” in 2012 (according to the Harvesters Reaching the Nations website); a second health institute in the county opened in 2024 (Radio Tamazuj, “New health institute launched in Yei River County” (Yei, South Sudan, 2024)

¹³⁷ UN Peacekeeping, “A tale of an empty bread basket in Yei, the once land of abundance” (Jan. 25, 2019), <https://peacekeeping.un.org/en/tale-of-empty-bread-basket-yei-once-land-of-abundance>.

¹³⁸ International Crisis Group, “South Sudan’s Other War: Resolving the Insurgency in Equatoria” (2021); Paul Aufiero and Nyagoah Tut Pur, “South Sudan at a Crossroads: Challenges and Hopes after 10 Years of Independence,” Human Rights Watch (2021).

¹³⁹ Yei County Profile, Conflict Sensitivity and Resource Facility, South Sudan, https://csrf-southsudan.org/county_profile/yei/

6. POLITICAL-LEGAL FRAGILITY

Key Findings

- **Low confidence/trust in formal authorities:** Results imply that respondents have more confidence in “informal authorities, such as councils of elders” than they have in their state governments, which administer South Sudan’s 10 different states.¹⁴⁰ Further, a majority of respondents (54%) believe that “politicians in Juba make [their] community” either less safe or much less safe.
- **Limited faith in respect for political rights:** A majority of respondents (53%) feel unsafe or very unsafe speaking out about politically sensitive issues, thereby exercising their freedom of speech.
- **Desire for elections:** While respondents are committed to engaging in the democratic process and voting in elections, they are less optimistic about the nature of the election that would be held by the South Sudanese government.

Overview of Political and Legal Fragility in South Sudan

In 2018, to the SPLM-in-Government, the Sudan People’s Liberation Movement-in-Opposition (SPLM-IO) and several other political parties signed the R-ARCSS. This resulted in the formation of the Transitional Government of National Unity (TGoNU), with non-signatories to R-ARCSS, such as the NAS, continuing military offensives and alternative negotiations. The TGoNU has since struggled to unify the various factions of the military, write a new constitution, and prepare for elections.¹⁴¹ The transitional government was extended in 2022 and again in 2024, meaning that elections are not currently scheduled to be held until 2026.¹⁴² Indeed, the country has yet to hold an election since it gained independence over a decade ago. The Tumaini Dialogue Initiative in 2024 – widely believed to have widened rifts between the government and opposition groups – and the March 2025 ceasefire violations followed by the house arrest of the First Vice President, Riek Machar, further undermined implementation of the peace agreement.¹⁴³

The R-ARCSS provided for the creation of several transitional justice mechanisms, including the Commission for Truth, Reconciliation, and Healing (CTRH), the Compensation and Reparation Authority (CRA), and the Hybrid Court for South Sudan.¹⁴⁴ While none were operational as of the beginning of 2025, the government passed two transitional justice laws in November 2024 establishing the CTRH and the CRA.¹⁴⁵ Despite this progress toward accountability for the past, as the United Nations Commission on Human Rights in South Sudan recently wrote, “armed conflict and gross human rights violations persist. Millions of people have been displaced, deprived and traumatized, and continue to live in fear” (p. 2).¹⁴⁶ Notably, a recent bill allows for warrantless arrests.¹⁴⁷

Land and property insecurity inhibits economic development and sustainable peace in South Sudan. Several factors have contributed to widespread uncertainty over property rights and frequent conflicts over land, housing, livestock, water, and other natural resources in South Sudan. These include delays in the adoption of necessary legal frameworks

¹⁴⁰ 48 per cent of respondents have confidence or a lot of confidence in state governments, whereas 57 per cent have confidence in informal authorities.

¹⁴¹ United Nations Security Council, “[Assessment of progress achieved on the key benchmarks established in paragraph 2 of resolution 2577 \(2021\)](#),” (New York, United Nations, 2024). United Nations Human Rights Council, “[Report of the Commission on Human Rights in South Sudan 26 February–5 April 2024](#),” (New York, United Nations, 2024).

¹⁴² George Obulutsa, “[South Sudan extends transitional government by two years](#),” Reuters (Aug. 4, 2022). Ammu Kannampilly, “[South Sudan postpones long-delayed election by two years, presidency says](#),” Reuters (Sep. 13, 2024).

¹⁴³ Joshua Craze, “Is This the End of the Peace Process?,” African Arguments, March 28, 2025, <https://africanarguments.org/2025/03/is-this-the-end-of-the-peace-process/>; Charles Onyango-Obbo, “South Sudan Faces a Bloody Collapse – Again,” The EastAfrican, July 14, 2025, <https://www.theeastafrican.co.ke/tea/news/east-africa/south-sudan-faces-a-bloody-collapse-again-5118074>.

¹⁴⁴ Revitalised Agreement on the Resolution of the Conflict in the Republic of South Sudan (“R-ARCSS”), Sept. 12, 2018.

¹⁴⁵ Radio Tamazuj, “[TNLA Passes Two Critical Bills](#),” Radio Tamazuj (3 September 2024). Radio Tamazuj, “[President Kiir Assents to Key Bills](#),” Radio Tamazuj (12 November 2024).

¹⁴⁶ United Nations Human Rights Council, “[Report of the Commission on Human Rights in South Sudan 26 February–5 April 2024](#),” (New York, United Nations, 2024). United States Department of State, “2023 Country Reports on Human Rights Practices: South Sudan,” (Washington, D.C., Department of State, 2024).

¹⁴⁷ Deng Machol, “[South Sudan’s Security Bill Allowing Warrantless Arrests Becomes Law](#),” Associated Press (15 August 2024).

and policies concerning housing, land, and property (HLP) ownership;¹⁴⁸ a lack of deeds of other legal documentation of property rights;¹⁴⁹ high levels of property destruction and displacement since independence; and the weakness of government institutions responsible for land management, registry, and dispute resolution.¹⁵⁰ After the 2018 peace agreement, many IDPs and refugees who fled to other countries returned home to find that their houses had been claimed by other occupants, or were severely destroyed or damaged.¹⁵¹ HLP disputes are a significant barrier to return and reintegration of displaced persons¹⁵² as well as a driver of inter-communal conflicts, particularly in areas where ethnic groups claim ancestral land rights.¹⁵³ Further, Land Matrix has documented almost 2 million hectares negotiated in large-scale land grabs.¹⁵⁴



Any discussion of political and legal fragility in South Sudan must take into account the country's long history of customary justice and dispute resolution by traditional authorities, including priests, tribal chiefs, and other community leaders.¹⁵⁵ South Sudan's 2011 Transitional Constitution recognizes the authority of customary law in several important areas, including marriage, divorce, and inheritance. Although customary courts do not formally have jurisdiction over criminal cases, they tend to adjudicate them nonetheless, in part because statutory courts are often inaccessible.¹⁵⁶ Estimates suggest that 80 to 90 per cent of disputes in South Sudan are resolved in customary courts.¹⁵⁷ Previous surveys indicate that many South Sudanese prefer customary courts and other traditional dispute resolution mechanisms including mediation by chiefs, elders, and religious authorities over the formal justice system not only because they may need to travel long distances to reach the nearest statutory court but also because the formal justice system is perceived by many as corrupt and inefficient.¹⁵⁸ However, some customary law practices are inconsistent with international law and/or South Sudanese law. For example, child marriage is illegal under the age of 18 in South Sudan. Nonetheless, UNICEF estimates that 52 per cent of girls in the country are married before the age of 18.¹⁵⁹ Similarly, although national law guarantees women the right to own land, customary law practices constrain women's land and property ownership throughout the country.¹⁶⁰ A related problem is that various ethnic and religious groups have different customary laws, resulting in different outcomes for similar cases; this runs afoul of the principle of equality that is enshrined in South Sudan's Constitution.¹⁶¹

¹⁴⁸ Housing, Land, and Property Technical Working Group, [HLP TWG Note: Housing, Land, and Property](#) (February 2021).

¹⁴⁹ IGAD, ["South Sudan Country Profiles: Assessment of Land Governance Framework, Training and Research Land Governance Institutions,"](#) (IGAD, Djibouti, 2018).

¹⁵⁰ Bernardo Ribeiro de Almeida and Janina M. Ubink, [Land and Justice Pathways in South Sudan](#), Van Vollenhoven Institute for Law, Governance and Society (Leiden, the Netherlands, 2024).

¹⁵¹ Joshua Craze, [Why the Return of Displaced People is Such a Thorny Issue in South Sudan](#), *The New Humanitarian* (Jan. 5, 2022).

¹⁵² Research and Evidence Facility and Samuel Hall, [South Sudan's Decades of Displacement: Understanding Return and Questioning Reintegration](#), REF (January 2023).

¹⁵³ Peter Hakim Justin and Han van Dijk, ["Land Reform and Conflict in South Sudan: Evidence from Yei River County,"](#) *Africa Spectrum* 52(2):3–28 (2017).

¹⁵⁴ Land Matrix, ["Map,"](#) (Rome, Italy, Land Matrix, 2025).

¹⁵⁵ Winnifred Bedigen, [Indigenous Peacebuilding in South Sudan: Delivering Sustainable Peace Through Traditional Institutions, Customs and Practices](#) (Routledge, 2023).

¹⁵⁶ United Nations Mission in South Sudan, ["Bridging the Gap in Post-Conflict South Sudan with Mobile Courts,"](#) (Juba, UNMISS, 2024).

¹⁵⁷ Agatha Ndonga, ["Some Level of Peace: Addressing Intercommunal Violence Through Customary Justice in South Sudan,"](#) (New York, International Center for Transitional Justice, 2024).

¹⁵⁸ David K. Deng, ["Challenges of Accountability: An Assessment of Dispute Resolution Processes in Rural South Sudan,"](#) (South Sudan Law Society, March 2013).

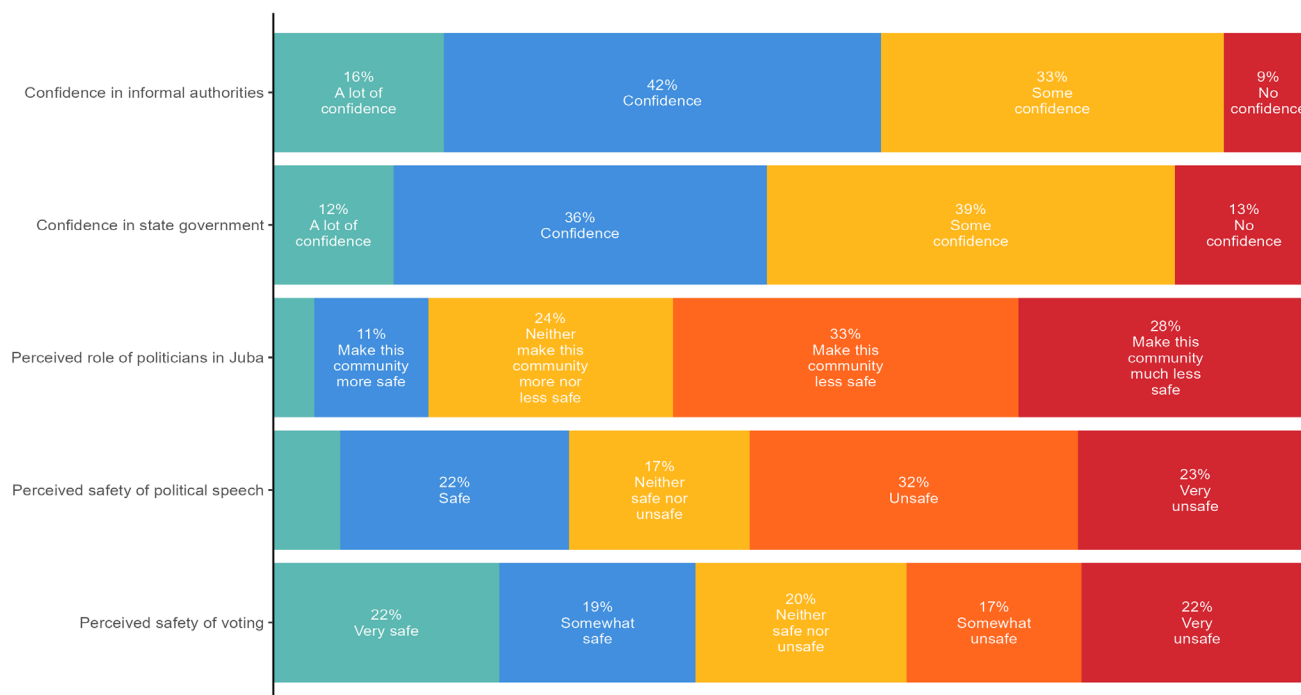
¹⁵⁹ UNICEF, ["Ending child marriage should not be a choice but a necessity: Young voices calling for an end to child marriage in South Sudan,"](#) (Juba, UNICEF, 2023).

¹⁶⁰ Cynthia M. Caron, ["Customary law, norms, practices and related factors that enable and constrain women's access to housing, land and property in South Sudan,"](#) (Juba, IOM, 2021).

¹⁶¹ Constitution of South Sudan (2011), Art. 14 reads "All persons are equal before the law and are entitled to the equal protection of the law."

Index Component Results: Political-Legal Fragility

Figure 18. Responses to Index Questions: Political-Legal Fragility



The chart above demonstrates that many South Sudanese have very limited confidence in formal governmental institutions. Over a quarter of respondents indicate that “politicians in Juba make [their] community less safe,” and another quarter note that such politicians make their communities “much less safe.” Importantly, a significant number of respondents (8%) preferred not to answer this question. This absence of trust may be attributed at least partly to a lack of respect for political rights on the part of the government.¹⁶² For example, only 28 per cent of respondents feel either safe or very safe when “speaking publicly on politically sensitive issues.” One respondent from Pibor said that it “causes death to talk [about] political issues especially when it’s [not] on the government side.”¹⁶³ More respondents feel safe when voting than when engaging in political speech; nonetheless, almost a quarter of respondents (22%) would feel very unsafe voting. These results are consistent with other recent surveys in South Sudan.¹⁶⁴ Importantly, respondents have a slightly more positive view of state governmental institutions than national ones; a slim majority (47%) have confidence or a lot of confidence in their state institutions.

Notably, there is a substantively small but statistically significant difference between male and female respondents to the questions about politicians in Juba, freedom of speech, and safe participation in elections. This gender disparity may reflect entrenched patriarchal norms in South Sudanese society and a prevalence of GBV.¹⁶⁵ There are also few women in government in South Sudan, despite a 35% quota established in the 2018 peace agreement.¹⁶⁶

¹⁶² Human Rights Watch. “World Report 2025: South Sudan Events of 2024.” (New York, Human Rights Watch, 2025). <https://www.hrw.org/world-report/2025/country-chapters/south-sudan>.

¹⁶³ Respondent #11247389.

¹⁶⁴ A June 2024 poll conducted by Friedrich-Ebert-Stiftung found that 44% of respondents felt unsafe or very unsafe speaking publicly about politically sensitive issues (68% of respondents in IDP settlements). David Deng, Jan Pospisil, Sophia Dawkins & Christopher Oringa, “Elections and Civic Space in South Sudan: Findings from the 2024 Public Perceptions of Peace Survey,” (Friedrich-Ebert-Stiftung South Sudan Office, Juba, 2024).

¹⁶⁵ IOM South Sudan, “Gender-Based Violence Knowledge, Attitudes and Practices Survey in South Sudan,” (IOM, Juba, 2019).

¹⁶⁶ South Sudan Women’s Coalition, Norwegian Embassy, Sweden, & UNDP, “Implementation of the 35% provision R-ARCSS,” (South Sudan Women’s Coalition, Juba, 2020).

Respondents have more confidence in informal authorities than they do in formal institutions; in fact, a majority (57%) have either confidence or a lot of confidence in informal authorities, “such as councils of elders.” There is no statistically significant difference between men and women’s levels of confidence in informal authorities, but T-tests indicate that residents of county capitals have less confidence in them compared to residents of rural areas.

Key Issue: Elections

Although the survey was enumerated prior to the postponement of the elections scheduled for the end of 2024,¹⁶⁸ it is nonetheless important to review respondents’ views on the elections. 64 per cent of respondents thought that the election should be held “on time” in December of 2024, and only 33 per cent believed it should either occur “after another delay” or “not at all.”¹⁶⁹ Indeed, 79 per cent of respondents intended to vote in the next national elections.¹⁷⁰ One respondent from Bor South, Jonglei explained the importance of elections as follows: “[A] referendum [on the constitution] should be done so that the unifying law is in the right place [and] so that citizens are known as important and powerful to vote [politicians] in and out.”¹⁷¹

While respondents were committed to engaging in the democratic process, they were less optimistic about the nature of the election that would be held. Only 57 per cent thought that the government was “capable [of holding] free and fair elections,” and only 51 per cent indicated that it would have a positive effect “on the stability of South Sudan.”¹⁷²

Key Issue: Transitional Justice

When asked whether the following peacebuilding and transitional justice mechanisms would make long-term peace in South Sudan less likely or more likely,¹⁷³ respondents had favorable views of all of the suggested mechanisms. Some of these mechanisms were delineated in the 2018 peace accord in South Sudan, whereas others are hypothetical. The mechanisms that respondents thought would have the most positive effect on peace were dialogues among civilian community members (84%) and increasing the participation of both youth (84%) and women (83%) in local peacebuilding and governance.¹⁷⁴ The high levels of support for dialogue among civilians (84%) as well as between civilians and representatives of the military and non-state armed forces (81%) is consistent with previous household surveys, including by IOM; this may reflect South Sudan’s long tradition of dispute resolution and mediation by local traditional authorities including elders, chiefs, and religious leaders.¹⁷⁵ The mechanism least expected to have a positive effect on peace was a hypothetical museum about the conflict (67%). Importantly, only small minorities of respondents expected any of the mechanisms to have a negative effect on the likelihood of peace.

¹⁶⁸ Ammu Kannampilly, “South Sudan postpones long-delayed election by two years, presidency says,” Reuters (Sep. 13, 2024).

¹⁶⁹ 3 per cent of respondents preferred not to answer.

¹⁷⁰ 20 per cent did not intend to vote, and 2 per cent preferred not to answer.

¹⁷¹ Respondent ID 10981224.

¹⁷² 36 per cent did not think that the government was capable of holding free and fair elections, and 7 per cent declined to answer. In terms of the effect of elections on stability, 26 per cent of respondents thought that they would have a “negative effect (less stability)”, 18 per cent indicated that elections would have “no effect either way,” and 5 per cent preferred not to answer.

¹⁷³ Q9.8: “How much do you think the following policies would contribute to long-term peace in South Sudan, on a scale from 1 through 5 where 1 means ‘makes peace much less likely’ and 5 means ‘makes peace much more likely.’”

¹⁷⁴ This was described in the survey as the “inclusion of more youth/women in dialogues, community leadership, or resource management committees.”

¹⁷⁵ Agatha Ndonga, “Some Level of Peace: Addressing Intercommunal Violence Through Customary Justice in South Sudan” (International Center for Transitional Justice, November 2024).

Table 3. Perceptions of Potential Mechanisms for Peace

Response	% Who Agree Mechanism Would Make Peace “More Likely” or “Much More Likely”
<i>Dialogues between community members</i>	84%
<i>Inclusion of more YOUTH in dialogues, community leadership, or resource management committees</i>	84%
<i>Inclusion of more WOMEN in dialogues, community leadership, or resource management committees</i>	83%
<i>Dialogues between representatives of the military and representatives of non-state armed groups</i>	81%
<i>A commission to discover the truth about the conflict</i>	80%
<i>Confessions by perpetrators of violence or crimes</i>	79%
<i>Apologies by perpetrators of violence or crimes</i>	79%
<i>Rehabilitation programmes for perpetrators of violence or crimes</i>	78%
<i>Criminal prosecutions for perpetrators of violence or crimes</i>	78%
<i>Compensation or other reparations for victims</i>	78%
<i>Draft a new permanent constitution</i>	77%
<i>A museum about the conflict</i>	67%

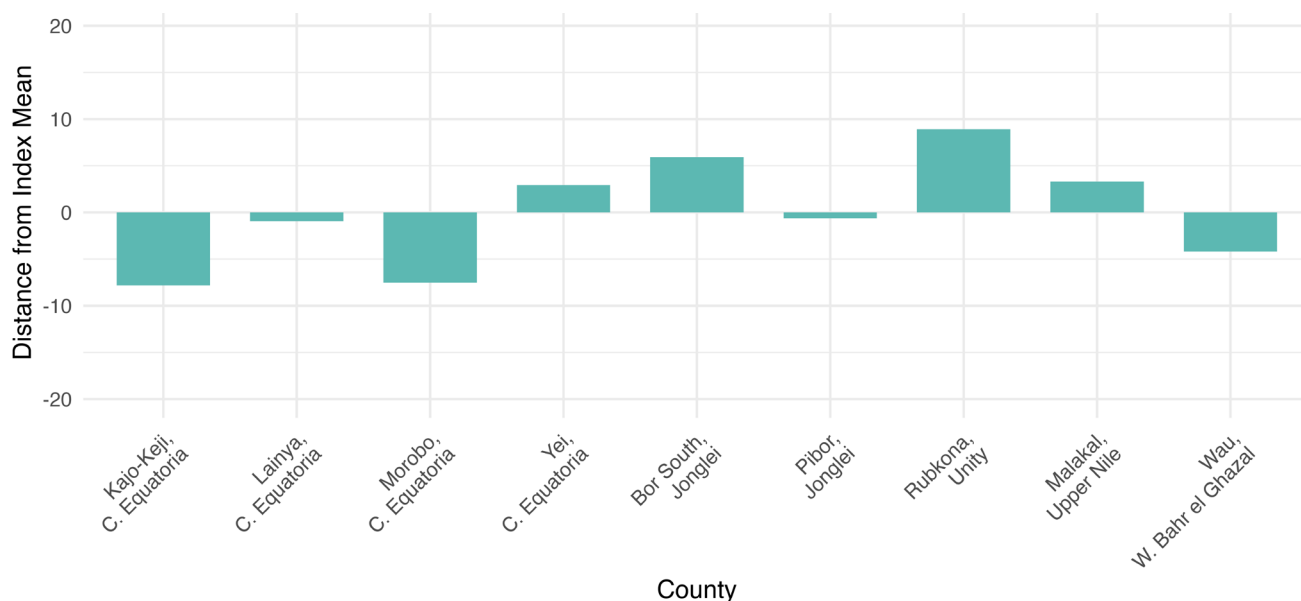
Pibor had the highest percentage of respondents who expected prosecutions of perpetrators to have “no effect” on peace, make peace “less likely,” or make peace “much less likely” (34%). Pibor is a majority Murle county; the Murle are an ethnic minority at the national level who experienced severe identity-based persecution and targeting by both SPLA forces and Lou Nuer tribal militias during South Sudan’s civil war, including massacres of civilians, looting, and forced displacement.¹⁷⁶ The Murle and Lou Nuer are longtime rivals who have clashed periodically over reciprocal cattle raids and child abductions in what has been described as a “cycle of revenge and counterattack.”¹⁷⁷ The lower level of support for prosecutions in Pibor in comparison with other counties could reflect lack of confidence in South Sudan’s formal justice system or concerns that Murle who have engaged in revenge killings and other crimes could be convicted. This history also helps us understand Pibor’s high level of societal fragility, as discussed below.

¹⁷⁶ See, e.g., Human Rights Watch, “[They Are Killing Us](#)”: Abuses Against Civilians in South Sudan’s Pibor County (HRW, September 2013); Nick Taylor, [South Sudan: 3,000 Dead in Pibor County Ethnic Massacre](#) (Global Post, May 13, 2017).

¹⁷⁷ Hereward Holland, [In South Sudan, Tribal Militias Exact Revenge](#) (Christian Science Monitor, Jan. 17, 2012).

Index Results by Country

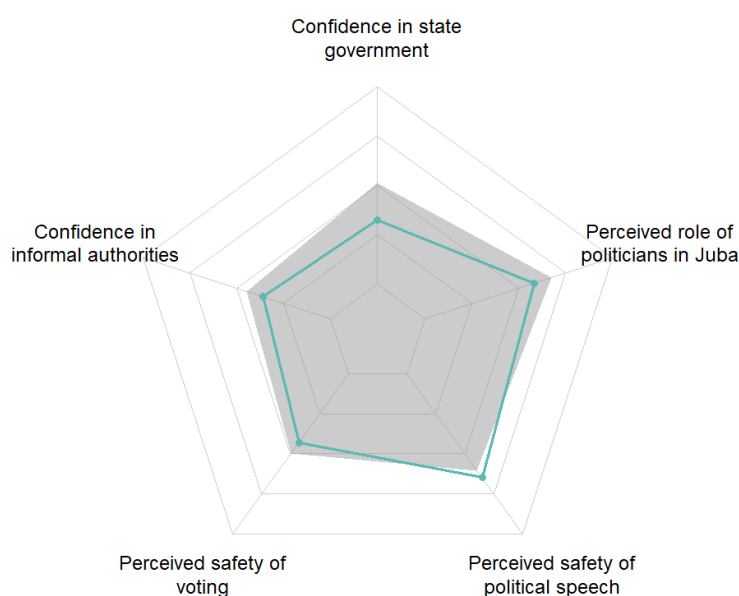
Figure 19. Political-Legal Fragility Distance from Index Mean



Kajo-Keji: Lowest Political/Legal Fragility

Respondents in Kajo-Keji have a markedly higher trust in their state government than respondents in other counties. In parallel, respondents also have higher trust in informal authorities. On the other hand, residents feel less safe talking about politically sensitive issues, perhaps reflecting fears due to recent political violence in the region. Interestingly, those surveyed also show higher trust in authorities in Juba, despite the Kuku, the main ethnic group in the area, having poor representation in national politics.¹⁷⁸

Figure 20. Political-Legal Fragility in Kajo-Keji



¹⁷⁸James Barnett, "War and Peace in South Sudan," *New Lines Magazine* (2024).

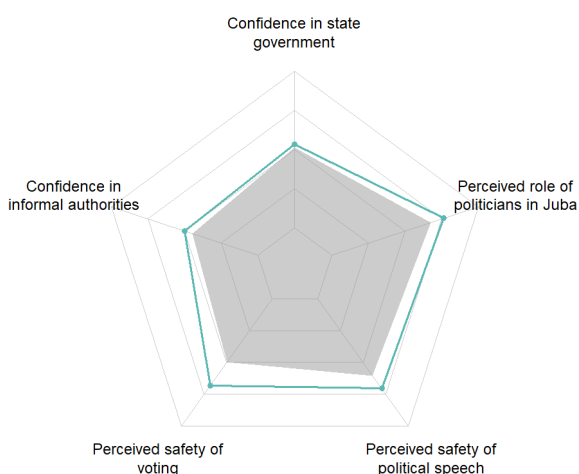
Rubkona and Bor South: Highest Political/Legal Fragility

Respondents in both Rubkona and Bor South show low trust in authorities in Juba. Both counties have been central to national-level political conflicts – changing hands between rebel and government control numerous times at the outset of the 2013 civil war. From 2013 to May 2024, when this survey data was gathered, Rubkona was under the governorship of the Bul Nuer leader, Nguen Monytil. President Kiir dismissed Monytil as governor in part due to his marginalization of and unpopularity among many other Nuer sub-groups in Unity State, whom Monytil is alleged to have denied safety and services.¹⁷⁹ Rubkona hosts a mixed Nuer population of groups that have been traditionally resident in the county, and groups that have come to the state capital due to their loss of land through floods and insecurity. The former governor's style of rule has correlated with limited infrastructure rebuilding in Bentiu following its extensive devastation during the initial civil war period of 2013 and 2014 (see section “Economic Fragility”).

Bor South also experienced substantial destruction at the start of the civil war that still scars the state capital.¹⁸⁰ Bor Town is the gateway of a significant transportation route that connects Juba to the Murle in the GPAA, to Lou Nuer communities in northeast Jonglei, and to Dinka and Nor in the east and west of the state. Bor Town also holds significance in Nuer-Dinka memory, since it was the location of a massacre in 1991 when Riek Machar mobilized Nuer supporters to kill 2,000 Dinka civilians.¹⁸¹ As such, it has been a center of political campaigning and violent disputes, in a context of limited rule of law. Distrust in the government may persist due to the use of Bor Town as a launching point for cattle raids between the Dinka and Murle.¹⁸² That has accompanied insecurity in Bor Town, which has included the murder of Murle traders and mob violence between different religious parishes.¹⁸³ While people may distrust the government due to these security issues, high trust in informal authorities in Bor South may be indicative of the high level of organization present amongst the Dinka there.¹⁸⁴

Figure 21. Political-Legal Fragility in Rubkona

Figure 22. Political-Legal Fragility in Bor South



¹⁷⁹ The Body Count: Controlling Populations in Unity State, Situation Update (Small Arms Survey, 2023), p3.

¹⁸⁰ Jacob Achiek, “Life Trickle Back in South Sudan’s Bor,” Voice of America English News (2014).

¹⁸¹ Bor County Profile, Conflict Sensitivity and Resource Facility, South Sudan, https://csrf-southsudan.org/county_profile/bor-south/

¹⁸² UNMISS/OHCHR. (2021). Armed Violence Involving Community-Based Militia in Greater Jonglei: January-August 2020.

¹⁸³ Radio Tamazuj, ‘Jonglei State investigates Bor revenge killings,’ 2021, <https://www.radiotamazuj.org/en/news/article/jonglei-state-investigates-bor-revenge-killings>; Radio Tamazuj, ‘ECSS leadership dispute leads to altercation: 27 individuals arrested,’ 2023, <https://www.radiotamazuj.org/en/news/article/ecss-leadership-dispute-leads-to-altercation-27-individuals-arrested>.

¹⁸⁴ Markus V. Hoehne, “Traditional Authorities and Local Government in South Sudan,” CSRF South Sudan (2008).

7. SECURITY FRAGILITY

Key Findings

- **Impact of cattle raiding:** Despite most respondents (65%) finding it an unacceptable practice, cattle raiding continues to be a major concern for many in South Sudan. The majority of respondents (62%) are frightened or very frightened about cattle raiding happening in their community.
- **Impact of conflict:** Respondents are most fearful about “armed conflict” and “local armed militias,” and a smaller majority of respondents are frightened or very frightened about land conflicts and resource conflicts (over cows).
- **Impact of crime:** Most respondents are afraid of crime to some degree in their community. This may be part of the reason why trust in formal authorities is so low, as described above; citizens see the government as unable to maintain the rule of law.

Overview of Security Fragility in South Sudan

In March 2025, the government-commanded South Sudan People's Defence Forces (SSPDF) and opposition SPLM-IO clashed around Nasir, prompting fears that South Sudan was descending into renewed civil war.¹⁸⁶ SSPDF and SPLM-IO had yet to integrate into a unified force – a central provision of R-ARCSS, pre-requisite for progress towards other milestones in the transition like elections. Armed groups, including non-signatories of R-ARCSS and SPLM-IO splinter groups, remain active in South Sudan, while so-called ‘intercommunal violence’ is both widespread and linked to national political conflicts.¹⁸⁷ State security forces, particularly the police, are widely viewed as ineffective in responding to the security needs of local populations.¹⁸⁸ In addition to a lack of capacity, ongoing mistrust between civilians and the police has undermined their effectiveness. Citizens who do not trust the police are less likely to share information and report problems, and they are more likely to try to solve problems on their own, including through organized vigilante groups.¹⁸⁹

In this context of limited peace agreement implementation, recent assessments of security in South Sudan have found declines in community perceptions of safety and increasing reliance on armed groups. For example, surveys by PAX in Jonglei State found that initial improvements in 2016–2018 were reversed in 2018–2020. While the overall number of reported incidents has decreased over the years, levels of communal violence and crime remain high.¹⁹⁰ Women and youth are particularly vulnerable to violence and insecurity. Although women are allowed to join the military or serve as police officers, they rarely do so because of strong cultural norms that discourage women from pursuing these and other traditionally male career paths.¹⁹¹

¹⁸⁶ United Nations Security Council, “S/2025/442 Report of the Panel of Experts on South Sudan,” Summary; Joshua Craze, “Is This the End of the Peace Process?,” African Arguments, March 28, 2025, <https://africanarguments.org/2025/03/is-this-the-end-of-the-peace-process/>; Charles Onyango-Obbo, “South Sudan Faces a Bloody Collapse – Again,” The EastAfrican, July 14, 2025, <https://www.theeastafrican.co.ke/tea/news/east-africa/south-sudan-faces-a-bloody-collapse-again-5118074>.

¹⁸⁷ United Nations Security Council, “S/2024/572 Situation in South Sudan Report of the Secretary-General,” United Nations Security Council (July 29 2024), <https://digitallibrary.un.org/record/4058910?ln=en&v=pdf>; Daniel Akech, “High Stakes for the South Sudan Talks in Kenya,” International Crisis Group (September 11 2024), <https://www.crisisgroup.org/africa/horn-africa/south-sudan-kenya/high-stakes-south-sudan-talks-kenya>.

¹⁸⁸ Small Arms Survey, “Policing in South Sudan: Transformation Challenges and Priorities,” HSBA for Sudan and South Sudan (Mar. 2017), <https://www.smallarmssurvey.org/sites/default/files/resources/HSBA-IB26-Policing-in-South-Sudan.pdf>.

¹⁸⁹ See International Crisis Group, “Double-edged Sword: Vigilantes in African Counter-insurgencies,” Africa Report No. 251 (Sep. 7, 2017), <https://icg-prod.s3.amazonaws.com/251-double-edged-sword.pdf>; Radio Tamazuj, “Bor youth announce vigilantism” (Feb. 2, 2022), <https://radiotamazuj.org/en/news/article/bor-youth-announce-vigilantism>.

¹⁹⁰ Residents of rural communities increasingly feel reliant on armed youth for their physical security as a result of insufficient numbers and quality of police and other government security forces. Anton Quist & Anita Hossain, “HSS 2016-2022: Trend Analysis of Jonglei, South Sudan,” PAX (Jun. 13, 2022), <https://protectionofcivilians.org/infographic/hss-2016-2022-trend-analysis-of-jonglei-south-sudan/>.

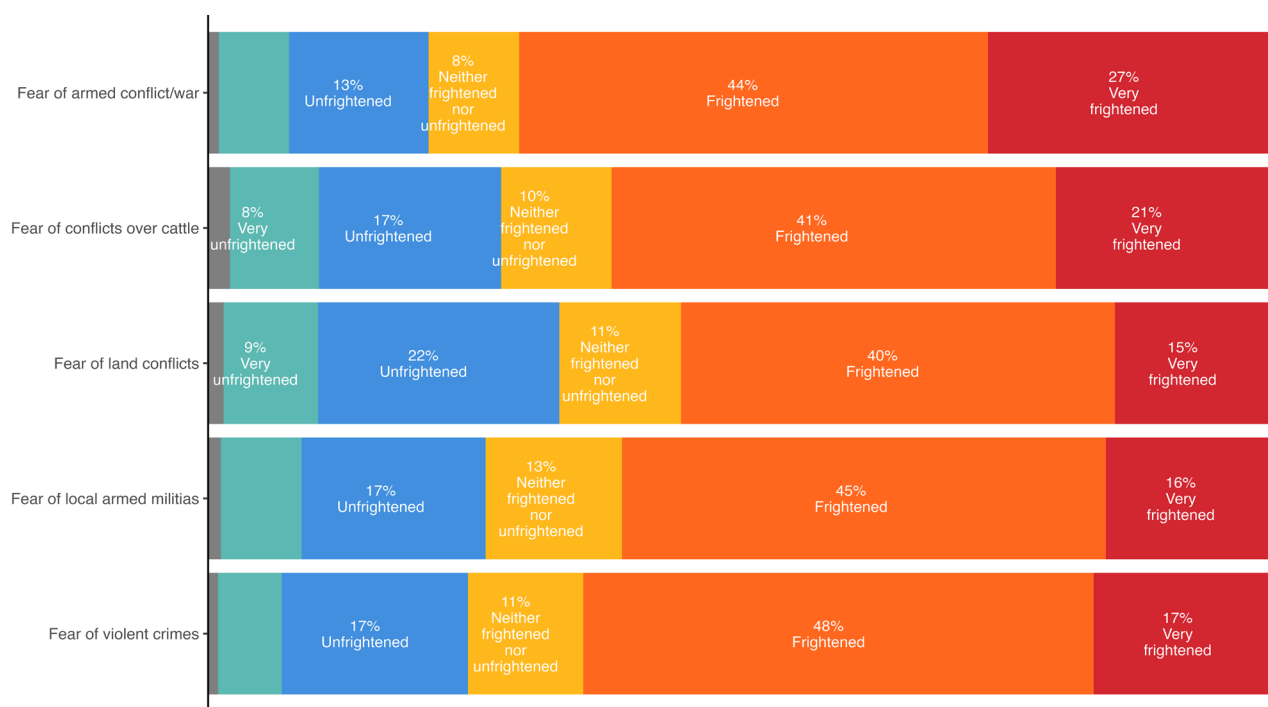
¹⁹¹ See, e.g., Eva Yayi Mawa, “Technology Not For Her?,” World Wide Web Foundation (Jul. 20, 2018), <https://webfoundation.org/2018/07/technology-not-for-her/>. Also International Civil Society Action Network, “‘We Rise for a Peaceful South Sudan’: The Role of Women in Shaping Post-War South Sudan,” (Jan. 30, 2024) and Nicki Kindersley and Alex Miskin, “Women in Authority in South Sudan: Connections, Challenges, and Strategies for Leadership,” Rift Valley Institute, (2024).

South Sudan's history of armed conflict has led to the proliferation of small arms and other light weapons, many of which remain in the hands of civilians due to lack of implementation of the peace agreement's provisions for disarmament.¹⁹² Furthermore, when the government does initiate disarmament campaigns, they tend to involve violence and the inequitable targeting of groups, dependent on perceived political affiliation.¹⁹³ Cross border trade in small arms and ammunition persists in places like the Abyei Administrative Area, sustaining widespread civilian armament in violation of the arms embargo.¹⁹⁴

The war in Sudan adds another security dimension. In January 2025, IOM recorded the millionth displaced person crossing the border into South Sudan from Sudan.¹⁹⁵ Overlaying this complicated flow of people, the Sudan Armed Forces (SAF) and RSF are recruiting South Sudanese men to fight as mercenaries in Sudan, and the SAF has provided material support to the SPLM-IO in South Sudan.¹⁹⁶ The South Sudanese government has sought to maintain friendly relations with both the Sudanese government and RSF to prevent either from turning against it.¹⁹⁷ However, this became more complicated in February 2025 when the RSF established an alliance with the Sudan People's Liberation Movement-North (SPLM-N) – to which President Kiir has remained close – prompting the SAF to retaliate by providing material support to SPLM-IO commanders within South Sudan, and to accuse Kiir of supporting the RSF.¹⁹⁸ In this context, gang activity in large displacement centers such as Bentiu and Malakal has become increasingly violent and undermined livelihood activities.¹⁹⁹

Index Component Results

Figure 23. Responses to Index Questions: Security Fragility



¹⁹² UNDP and Small Arms Survey, "A UNDP-supported baseline assessment of civilian firearm holdings, violence victimisation, and perceptions of security in South Sudan," (Juba, South Sudan, UNDP, 2017). Radio Tamazuj, "Allegations of targeted attacks in Malakal; police dispute claims," (Apr. 11, 2025)

¹⁹³ Radio Tamazuj, "Disarm or face force, army tells youth in Warrap, Mayom", June 2025, <https://www.radiotamazuj.org/en/news/article/disarm-or-face-force-army-tells-armed-youth-in-warrap-mayom>

¹⁹⁴ United Nations Security Council, "S/2025/442 Report of the Panel of Experts on South Sudan," Chapter IV.

¹⁹⁵ "South Sudan Hits Record One Million New Arrivals From Sudan Crisis," IOM, January 21, 2025, <https://eastandhornofafrica.iom.int/news/south-sudan-hits-record-one-million-new-arrivals-sudan-crisis>.

¹⁹⁶ Alan Boswell cited in Mat Nashed, "Is Sudan's War Merging with South Sudanese Conflicts?," Al Jazeera, March 29, 2025, <https://www.aljazeera.com/features/2025/3/29/sudans-war-risks-merging-with-conflict-in-south-sudan-say-analysts>.

¹⁹⁷ "South Sudan on Edge as Its Neighbour's War Disrupts Oil Exports."

¹⁹⁸ Nashed, "Is Sudan's War Merging with South Sudanese Conflicts?"

¹⁹⁹ International Crisis Group, "South Sudan on the Precipice of Renewed Full-Blown War," (Mar. 7, 2025).

The figure demonstrates the salience of a range of security threats. A majority of respondents are either frightened or very frightened of armed conflict, conflicts over cattle, land conflicts, and local armed militias happening in their communities. Of the four kinds of conflict, respondents are most afraid of armed conflict or war, and local armed militias. Importantly, a majority are also frightened or very frightened of violent crime.

Additional questions provide insight into perceived changes in security over time. Despite high levels of insecurity overall, a significant majority of respondents (69%) believed that safety in their community had improved in the past 12 months prior to the survey. Even in Bor South, the most security fragile county, 41 per cent of respondents indicate that their community was safer than a year before. Many respondents, when asked why they responded this way, mention dialogues between the military and civilians or reduced abuses by the military. As one individual from Lainya noted, the “rampant killing of civilian[s] by soldiers has slightly reduced.”²⁰⁰

It is important to note that Security Fragility may be more fluid than other fragility components. For example, studies that have measured improvements in perceptions of civilian safety saw reversals in 2025 to gains from the previous years.²⁰¹

Key Issue: Cattle Raiding

Cattle are a vital store of wealth across pastoral societies in South Sudan, defining power distributions and social statuses.²⁰² They also have spiritual significance to many communities.²⁰³ Cattle are central to dowry payments and herd size confers status and are central to perceptions of masculinity.²⁰⁴ Communities or individuals may raid cattle from others to pay bride price or as acts of revenge.²⁰⁵ These raids can escalate quickly, and the proliferation of small arms amongst cattle-camp youth can make cattle raids high-fatality events.²⁰⁶

Cattle raiding in South Sudan is intertwined with national political disputes.²⁰⁷ In the Second Sudanese Civil War between the early 1990s and 2005, warring factions started to use cattle raids as proxy conflicts for consolidating political power.²⁰⁸ Communities can perceive cattle raids as existential threats with a spiritual toll that goes deeper than a narrow financial loss. Therefore, indirect manipulation of cattle conflicts offers a mechanism to political elites to whip up civilian sentiments and mobilize people against rivals.²⁰⁹ For example, young Nuer men, similar to youth in other pastoral communities, live in cattle camps close to their villages, protecting the community's wealth and providing security. These camps have occasionally mobilized as the ad hoc ‘White Army’ which has perpetrated revenge killings for cattle raids, but also occasionally responded to the direct command of the SPLM-IO to support large military offensives. Some researchers have argued that militarized cattle keepers are central to the implementation of government-instigated campaigns of violence.²¹⁰

²⁰⁰ Survey Respondent 10168522, Bari, Male, Age 41-50, in Lainya County, in Central Equatoria State (April 19, 2024)

²⁰¹ See, for example, PeaceRep/Detcro, ‘Perceptions of Peace in South Sudan Dashboard’, <https://peacerep.org/perceptions-peace-south-sudan/>

²⁰² Jan Pospisil, Melissa Johnston, Adut Alaak Garang, Nyachangkuoth Rambang Tai, ‘Bring Enough Cows to Marry: Brideprice, Conflict, and Gender Relations in South Sudan,’ PeaceRep (2024).

²⁰³ Diana Felix da Costa, Naomi Pendle, Jérôme Tubiana, ‘What is happening now is not raiding, it’s war’, in Jean-Nicolas Bach (ed.), *The Routledge Handbook of the Horn of Africa* (2022).

²⁰⁴ Oxfam, Preventing Cattle Raiding Violence in South Sudan, Aug. 2016, oxfamlibrary.openrepository.com/bitstream/10546/617936/1/cs-cattle-raiding-south-sudan-040816-en.pdf.

²⁰⁵ Tom Richardson, “Pastoral Violence in Jonglei.” Cattle Raiding in Jonglei, South Sudan, Dec. 2011, mandalaprojects.com/ice/ice-cases/jonglei.htm.

²⁰⁶ Hannah Wild, Jok Madut Jok, and Ronak Patel. “The militarization of cattle raiding in South Sudan: how a traditional practice became a tool for political violence,” *Journal of International Humanitarian Action* 3, no. 1 (2018): 1-11.

²⁰⁷ Jana Krause, ‘Stabilization and Local Conflicts: Communal and Civil War in South Sudan,’ *Ethnopolitics* 18:5 2019.

²⁰⁸ Hannah Wild, Jok Madut Jok, and Ronak Patel. “The militarization of cattle raiding in South Sudan: how a traditional practice became a tool for political violence,” *Journal of International Humanitarian Action* 3, no. 1 (2018): 1-11.

²⁰⁹ Diana Felix da Costa, Naomi Pendle, Jérôme Tubiana, ‘What is happening now is not raiding, it’s war’, in Jean-Nicolas Bach (ed.), *The Routledge Handbook of the Horn of Africa* (2022).

²¹⁰ Naomi Pendle, “‘They Are Now Community Police’: Negotiating the Boundaries and Nature of the Government in South Sudan through the Identity of Militarised Cattle-keepers,” *International Journal on Minority and Group Rights* 22 (2015).

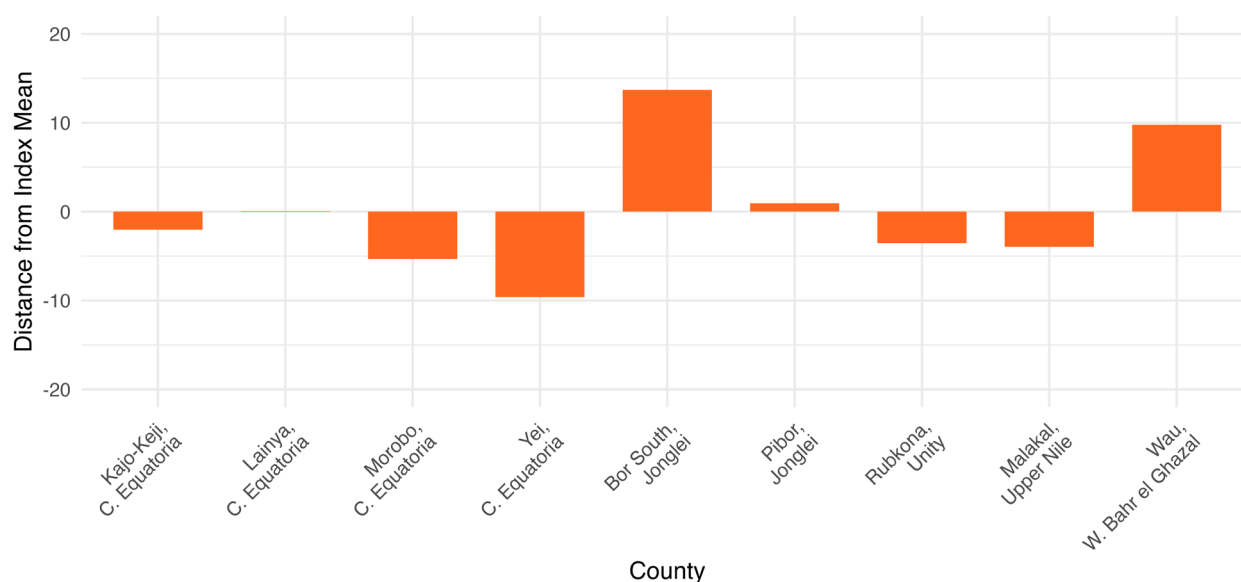
Policy Institute (2025, London), www.qmul.ac.uk/gpi/media/images/Policy-Report-Reviewing-the-Climate-Security-Nexus-FINAL.pdf.

Recent research suggests that climate change may influence patterns of cattle raiding through multiple distinct pathways. Climate change-induced fluctuations in weather patterns – such as frequent droughts and flooding – can harm livestock and agricultural production, exacerbating competition for scarce resources.²¹¹ Such competition then increases the profitability of cattle raiding. Additionally, climate-driven displacement and migration of pastoral communities into new areas can contribute to intercommunal tensions associated with cattle raiding.²¹² Finally, these environmental stressors interact with weak governance, youth unemployment, and easy access to small arms, driving the militarization of raiding groups, resulting in more frequent and violent raids.

Many respondents are frightened of cattle conflicts, and most respondents (65%) think the practice is not acceptable. Around one-sixth (16%) think it is acceptable “only in some circumstances.” The county where the highest percentage of people think cattle raiding is acceptable (39%) is Malakal. The most common reasons people provide for thinking cattle raiding is acceptable or sometimes acceptable are “for revenge” and “to gain money/cows for bride price.” Far fewer think that the need to “feed one’s household” justifies such violence.

Index Results by Country

Figure 24. Security Fragility Distance from Index Mean



Bor South: Highest Security Fragility

A plurality of respondents in Bor South (46%) perceived no change in community safety over the past year, 41 per cent perceived an improvement, and 10 per cent reported a decrease.²¹³ A majority of respondents (69%) in Bor South are concerned about the visibility of firearms in the community.²¹⁴ Despite fears about cattle raiding, slightly more than half of respondents in Bor (53%) believe that it is unacceptable. The remainder of respondents believe that it is always acceptable (16%), and another 28 per cent believe that it is acceptable but “only in some circumstances.” This high security fragility links to dynamics that also underly high political-legal fragility in Bor Town. The use of the town as a staging base for inter-communal attacks and periodic revenge killings may explain people’s ambivalence about their daily security.²¹⁶

²¹¹ Mobjörk, Krampe & Tarif, Climate, Peace and Security Fact Sheet: South Sudan (2025), <https://www.nupi.no/news/climate-peace-and-security-fact-sheet-south-sudan3>.

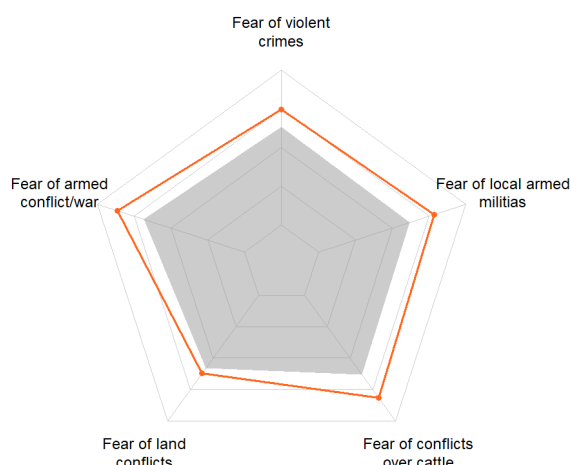
²¹² Peter Machieng Chan Gaduel, Reviewing the Climate-Security Nexus in Unity State, South Sudan: Insights for Improving Policy and Programming, Queen Mary University of London Global.

²¹³ Question 8.2, “Q8.2 Compared to 12 months ago, do you feel that the safety in the community has improved, there has been no change, or the safety in the community has worsened?”

²¹⁴ 69% agreed or strongly agreed with the statement, “The high visibility of firearms or weapons in the streets of my community is causing concerns for me” (Question 8.4.6).

²¹⁶ Radio Tamazuj, ‘Jonglei State investigates Bor revenge killings,’ 2021, <https://www.radiotamazuj.org/en/news/article/jonglei-state-investigates-bor-revenge-killings>; Radio Tamazuj, ‘ECSS leadership dispute leads to altercation: 27 individuals arrested’, 2023, <https://www.radiotamazuj.org/en/news/article/ecss-leadership-dispute-leads-to-altercation-27-individuals-arrested>

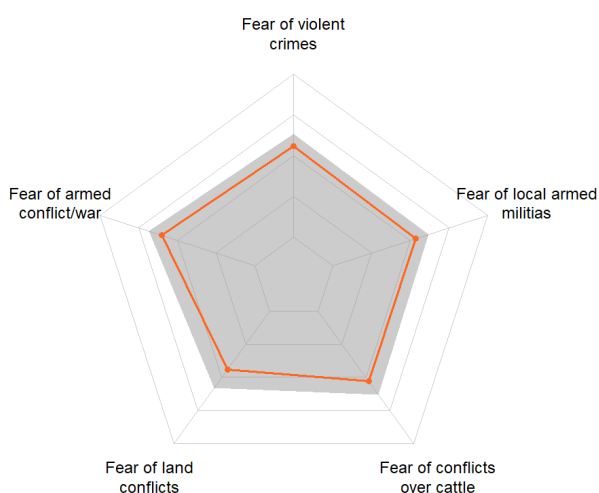
Figure 25. Security Fragility in Bor South



Yei: Lowest Security Fragility

Yei's low security fragility may represent how fluidly this dimension of fragility can change with political conditions each year. While Yei has experienced fierce fighting, 2024 was largely stable. Many respondents in Yei report feeling less frightened of armed conflict than respondents in other counties, even other counties in Central Equatoria state. Respondents in Yei are also the least likely of those in any county to report that they fear militias, self-defense vigilante groups, or cattle raiding in their community. This reflects the fact that much of the violence in Central Equatoria State since 2015 has been committed by organized armed groups, such as the NAS, SPLM-IO and SSPDF.²¹⁷ The Tumaini Dialogue Initiative in Kenya, through which the government sought to engaged the non-signatories of R-ARCSS in negotiations, may have supported this general sense of security in Yei. Some analysts argue that the Tumaini Dialogue was a political failure because it widened political rifts within the SPLM and SPLM-IO, failed to engage NAS, and undermined the R-ARCSS framework.²¹⁸ However, the conduct of the process (if not the outcome) may have – at least temporarily – diverted the attention away from the battlefield of some of the non-signatories of R-ARCSS operating in Central Equatoria State.

Figure 26. Security Fragility in Yei



²¹⁷ UNMISS Human Rights Division. "Annual Brief on Violence Affecting Civilians." (Juba, UNMISS, 2025). <https://www.un.org/sexualviolenceinconflict/wp-content/uploads/2025/04/report/auto-draft/2024-Annual-Brief.pdf>.

²¹⁸ United Nations Security Council, "S/2025/442 Report of the Panel of Experts on South Sudan", Section III.B; Joshua Craze, "On the Brink: The Politics of Violence in South Sudan," Small Arms Survey (2025), p8.

8. SOCIETAL FRAGILITY

Key Findings

- **Network embeddedness and social cohesion within the community:** Social cohesion is high in our sample, with over three quarters of people (77%) agreeing or strongly agreeing that people in their boma are “willing to help each other.” Similarly, three quarters of respondents (75%) agree or strongly agree that they would know who to seek help from in the case of an argument with a neighbour about a small issue. In other words, not only do people see their community as helpful to most community members, but they indicate that there are people within the community that would help them specifically.
- **Community/migrant relations:** Relations between local communities and migrants are on the whole quite positive; 72 per cent indicate that relations are either good or very good. There are no statistically significant differences between how returnees and host community members answer this question, nor between how IDPs and host community members respond.
- **Land Disputes:** A majority (57%) of respondents indicate that at least one land dispute has happened in their community.
- **Land Titling:** Of the 79 per cent of respondents who indicate that they own their land, 64 per cent do not possess ownership documents.

Overview of Societal Fragility in South Sudan

Societal Fragility is a concept that captures societal divisions, inequalities, and social cohesion. In South Sudan, decades of conflict have contributed to high levels of distrust between some social groups, particularly different ethnic groups. South Sudan presents low levels of trust at all levels of society and between different communities, ethnic and religious groups, and individuals.²¹⁹

Large scale displacement both internally within South Sudan and externally to surrounding countries was caused by decades of conflict prior to independence and the subsequent civil war. Flooding and ongoing violence also contributes to displacement. South Sudan currently has 1.8 million IDPs, 552,000 refugees, 63,000 returnees, and 18,000 stateless persons.²²⁰ Further, over 2 million South Sudanese are refugees abroad.²²¹ This makes it difficult to absorb new arrivals, including those fleeing the current conflict in Sudan. Lainya, Morobo, Yei, and Kajo-Keji have all seen increases in numbers of South Sudanese refugees from abroad returning (110,385), while Malakal and Rubkona are still grappling with a large number of internally displaced (Rubkona, 225,664; Malakal, 67,184) and increased arrivals from Sudan. Pibor is also facing a large IDP population of around 21,514 persons.²²² The situation is further complicated by the arrival of large numbers of Sudanese from South Sudan’s neighbour to the north, fleeing armed conflict there through northern waypoints in the southern country’s border region. In late 2024 migration rates were such that 2,000 new Sudanese were arriving in the town of Renk alone.²²³

For those who return whether from abroad or are displaced within South Sudan, integration is challenging and assistance needs are high. Barriers to long-term solutions include lack of identity documents that are necessary for accessing public services,²²⁴ HLP disputes, and property destruction.²²⁵ After the 2018 peace agreement, many IDPs and refugees who fled to other countries returned home to find that their houses had been claimed by

²¹⁹ USAID South Sudan, “USAID/South Sudan Transitional Justice Documentation Study: Final Report” (Aug. 2018): 46, 52.

²²⁰ UNHCR, “IGAD Region Population of concern to UNHCR as of 31 Mar 2025,” (Nairobi, Kenya, UNHCR, 2025).

²²¹ UNHCR, “South Sudan Refugee Crisis Explained,” (Juba, South Sudan, UNHCR, 2025).

²²² IOM, DTM, Round 16.

²²³ “South Sudan hits record one million new arrivals from Sudan crisis,” UNHCR (January, 2025).

²²⁴ David K. Deng, “Housing, Land and Property Disputes in South Sudan: Findings from a Survey Nimule, Torit, Wau and Yei,” South Sudan Law Society (2019): 10.

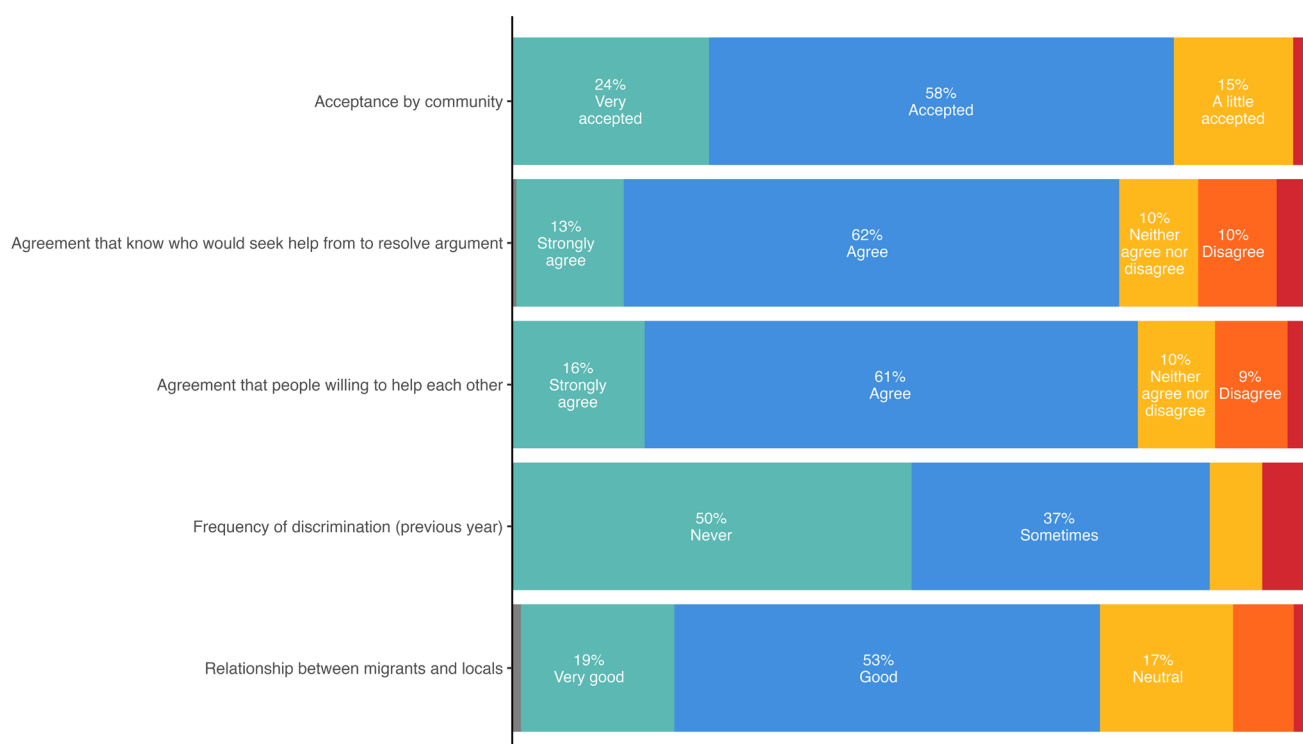
²²⁵ IOM DTM, “Return and Reintegration Survey in South Sudan’s Former Breadbasket” (Mar. 9, 2023), <https://dtm.iom.int/reports/south-sudan-return-and-reintegration-survey-south-sudans-former-breadbasket>.

other occupants, or were severely destroyed or damaged.²²⁶ However, property conflicts are not limited to IDP populations. A 2019 survey found that approximately one in five households (21%) had experienced an HLP dispute within the last three years,²²⁷ and this pilot survey found a similar rate of HLP disputes (23%, discussed below).

Although gender relations in South Sudan are complex and dynamic, women tend to be excluded from decision-making processes.²²⁸ The 2023 Women, Peace and Security Index (WPSI), ranked South Sudan 173 out of 177 in relation to inclusion, justice and security. In rural areas, prevailing cultural norms particularly marginalize women from political participation or decision-making.²²⁹ Women and girls' traditional roles include farming, collecting water and firewood, cooking, cleaning, brewing beer, and taking care of children; conversely, men and boys traditionally make familial decisions, manage cattle, hunt, fish, and make charcoal.²³⁰ This marginalization from resources and decision-making has led to significant imbalances by gender. Women's poor literacy rates are one example; only 16 per cent of females over fifteen can read, while 40 per cent of males the same age can.²³¹ This gap, driven by both social expectations and economic opportunities, further encourages girls and women to prioritize care-giving roles.²³² Women's underrepresentation in positions of authority, their economic dependence on men, and strong gender norms regarding roles of men and women are believed to contribute to widespread GBV, discussed further under the Human Fragility chapter.²³³

Index Component Results

Figure 27. Responses to Index Questions: Societal Fragility



²²⁶ Joshua Craze, "Why the return of displaced people is such a thorny issue in South Sudan," The New Humanitarian (Jan. 5, 2022), <https://www.thenewhumanitarian.org/analysis/2022/1/5/why-return-displaced-people-thorny-issue-South-Sudan>.

²²⁷ David K. Deng, "Housing, Land and Property Disputes in South Sudan Findings from a survey Nimule, Torit, Wau and Yei," South Sudan Law Society (2019): 1.

²²⁸ Country Profile: South Sudan." Georgetown Institute for Women, Peace and Security, October 19, 2023. <https://giwps.georgetown.edu/country/south-sudan/>.

²²⁹ CARE "Gender in Brief: South Sudan" (2020), <https://www.care.org/wp-content/uploads/2020/05/Gender20in20Brief20South20Sudan20.pdf>.

²³⁰ Ibid.

²³¹ Logo, Harriet Kuyang, Ojaswi Shah, and Charlotte Watson. "Gender Norms, Conflict Sensitivity and Transition in South Sudan." Conflict Sensitivity Resource Facility, July 2022.

https://www.csrf-southsudan.org/wp-content/uploads/2022/07/BAF-Gender-Conflict-Sensitivity-and-Transition-in-South-Sudan_14072022.pdf.

²³² Ibid.

²³³ Steven Chimwemwe Iphani, "A boy should be a fighter: Addressing harmful masculinities driving cattle-related violence," IOM South Sudan (2021), <https://publications.iom.int/system/files/pdf/A-Boy-Should-be-a-Fighter.pdf>.

Given longstanding tensions between ethnic groups in South Sudan, respondents' answers to societal fragility questions were surprisingly positive. 82 per cent of respondents feel accepted or very accepted in their community, and 50 per cent have not experienced any discrimination based on their gender, religion, ethnicity, disability, or place of origin. Further, despite extremely limited resources, more than three quarters (77%) agree or strongly agree that people in their boma are willing to help each other. A majority (75%) respondents indicate that they are socially embedded in their community, agreeing or strongly agreeing that they would know who to seek help from to resolve an argument with their neighbour. This may reflect the fact that many bomas are ethnically homogeneous, meaning that the key cleavages are between bomas rather than within them.

Further, when respondents were asked to describe “the relationship between individuals who are not originally from the community and the rest of the community members,” 72 per cent indicate that such relations are good or very good. There are not statistically significant differences in responses to this question between IDPs and host community members or between returnees and host community members.



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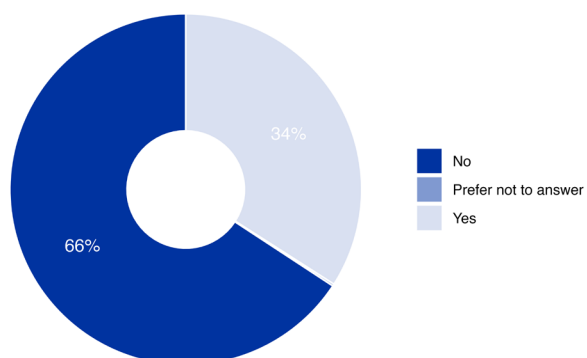
Key Issue: Housing, Land, and Property (HLP) Issues

An important consequence of political and legal fragility in South Sudan is land conflicts, which have been exacerbated by high levels of displacement during the conflict but do not only occur among IDP populations. For example, 57 per cent of the survey's respondents indicate that at least one land dispute has happened in their community.²³⁴ Further, over 200 respondents openly admit to taking their homes without the permission of the owners of those homes. Of the 79 per cent of respondents who indicate that they own their home, 64 per cent do not possess ownership documents.

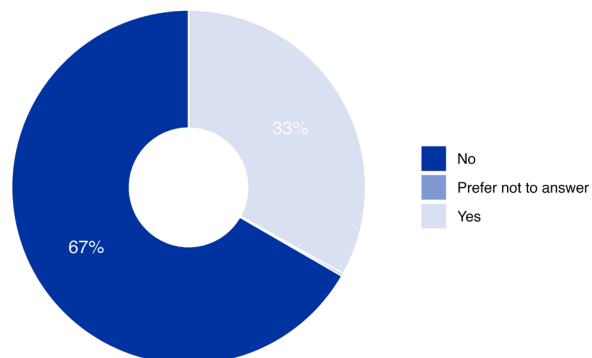
²³⁴ Property disputes are defined as boundary disputes, land grabbing, secondary occupations, multiple land title claims, or agrarian versus livestock farming conflicts. Secondary occupation occurs when someone settles on another's land while the original occupier is absent. This is distinct from land grabbing, which occurs when someone takes someone else's land by force.

Figure 28. Prevalence of HLP issues in community

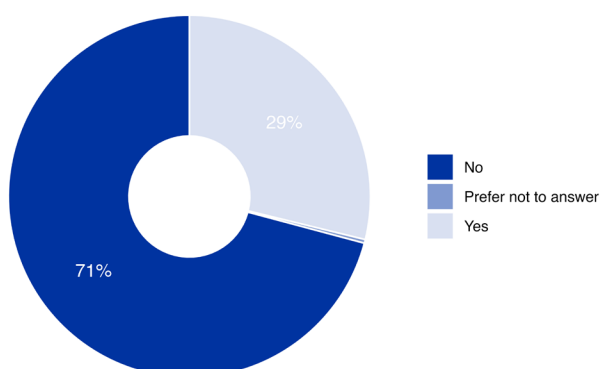
*"Have you heard about the following issues... happening in your community: **Boundary dispute**"*



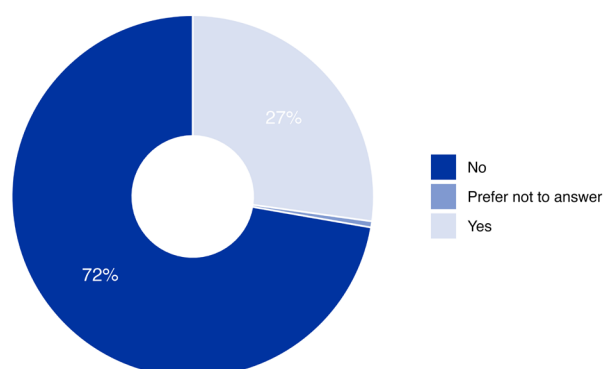
*"Have you heard about the following issues... happening in your community: **Land grabbing**"*



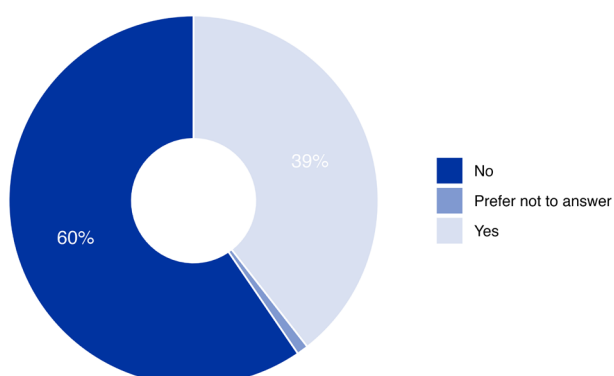
*"Have you heard about the following issues... happening in your community: **Secondary occupation**"*



*"Have you heard about the following issues... happening in your community: **Multiple land title claims**"*



*"Have you heard about the following issues... happening in your community: **Agrarian versus livestock farming conflicts**"*

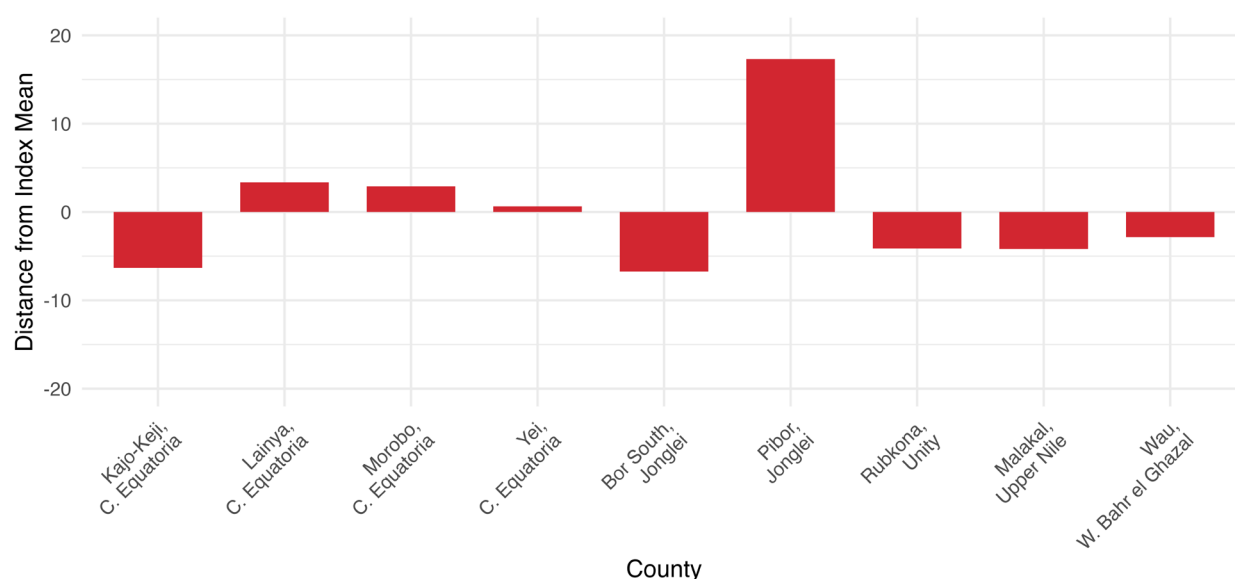


Key Issue: Women's Rights

The survey shows relatively open attitudes toward women's right to own lands, given the patriarchal context.²³⁵ 72 per cent of respondents agree that women should be able to own land independently. However, there is a statistically significant difference between men and women: 68 per cent of male respondents agree with the statement compared to 74 per cent of female respondents. Similarly, 73 per cent of respondents believe that women and men are equally capable of serving in governmental leadership positions. There is, for this question as well, a statistically significant difference between the responses of men and women: 74 per cent of women agree, but only 70 per cent of men do.

Index Results by Country

Figure 28. Societal Fragility Distance from Index Mean



Pibor: Highest Societal Fragility

Respondents in Pibor exhibit strikingly higher levels of societal fragility compared to other surveyed counties in South Sudan. The UN Panel of Experts on South Sudan cited local organizations as blaming severe hunger and insecurity on “violent asset-stripping... and chronic localized insecurity” for this severe hunger.²³⁶ This insecurity and asset-stripping flows from patterns of hostilities between the Murle, Lou Nuer and other groups that link to political disputes in Juba but unfold on the ground as high-fatality intercommunal hostilities. The Murle – the largest ethnic group in Pibor County – have a long and complex relationship involving both cooperation and conflict with state authority and the SPLM.²³⁷ They have organized for decades as a resistance movement against the state as the SSDM/A-Cobra Faction, formerly under David Yau-Yau, who negotiated a special administrative status for Pibor in 2014 as the GPAA. The Murle, through this cattle-camp-youth-based armed resistance have and continue to frequently engaged in conflicts with Lou Nuer armed youth, resulting in cattle theft, the abduction of women and children, and the widespread destruction of property.²³⁸ Trust between the Murle and other ethnic groups within Pibor is also low; neighbouring Toposa, Jiye and Anyiak groups in the south-eastern areas of Pibor County have been affected by intermittent raiding and cross-border violence.²³⁹ There are also tensions within the dominant

²³⁵ During the last round of this survey, although 73% of respondents agreed that child marriage was never acceptable, only 50% of respondents thought that women should be allowed to own land independently of their husbands. The change in responses is perhaps due to the team's efforts to recruit female enumerators; the presence of female enumerators may have allowed liberal female respondents to more openly voice their views and induced social desirability bias in more conservative male respondents.

²³⁶ United Nations Security Council, “S/2025/442 Report of the Panel of Experts on South Sudan,” para 121; Reach, Emergency Food Security Update: Pibor, November 2024.

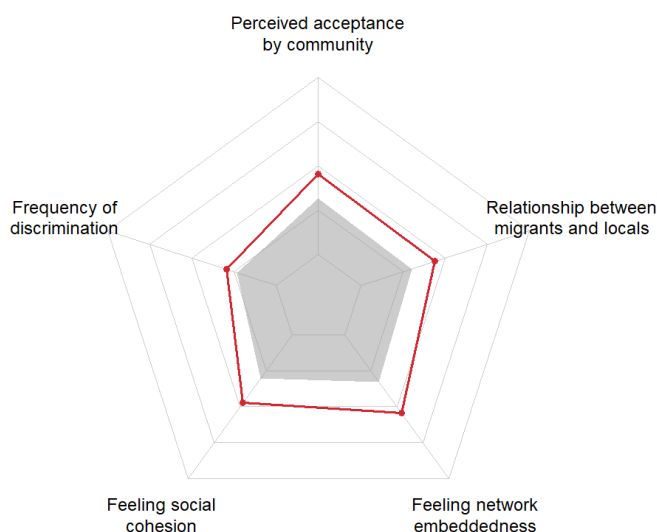
²³⁷ “Pibor County, Jonglei State.” Conflict Sensitivity Resource Facility, April 30, 2024. https://www.csrfsouthsudan.org/county_profile/pibor/. Diana Felix da Costa, “The politics of being Murle in South Sudan: state violence, displacement, and the narrativisation of identity,” *Journal of Eastern African Studies* (Sep. 2023).

²³⁸ “Intercommunal Violence in Jonglei-Pibor: UNMISS Supports Authorities in Returning Abductees.” United Nations Peacekeeping, January 23, 2023. <https://peacekeeping.un.org/en/intercommunal-violence-jonglei-pibor-unmiss-supports-authorities-returning-abductees>.

²³⁹ “Pibor County, Jonglei State.” Conflict Sensitivity Resource Facility, April 30, 2024. https://www.csrfsouthsudan.org/county_profile/pibor/. Diana Felix da Costa, “The politics of being Murle in South Sudan: state violence, displacement, and the narrativisation of identity,” *Journal of Eastern African Studies* (Sep. 2023).

ethnic group of Pibor; The Murle's strong age-based hierarchy may be driving growth in gang membership within Pibor, as young Murle people seek to gain power relative to elders.²⁴⁰

Figure 29. Societal Fragility in Pibor



Kajo-Keji and Bor South: Lower Societal Fragility

Social cohesion seems to be relatively high in both Kajo-Keji and Bor South counties. Especially in Kajo-Keji, most respondents feel they are accepted in their community. As previously mentioned, this may be reflective of the relatively monolithic ethnic composition in Kajo-Keji, where most people belong to the Kuku ethnic group. This may also be the reason that subjective discrimination is relatively low in both counties, and why network embeddedness is high: people seem to know who they can trust in both counties when it comes to dispute resolution. People in both counties also report fewer frictions between those originally from the community and those who are not. This is despite significant flood-induced migration in the Equatoria region.²⁴¹

Figure 30. Societal Fragility in Kajo-Keji

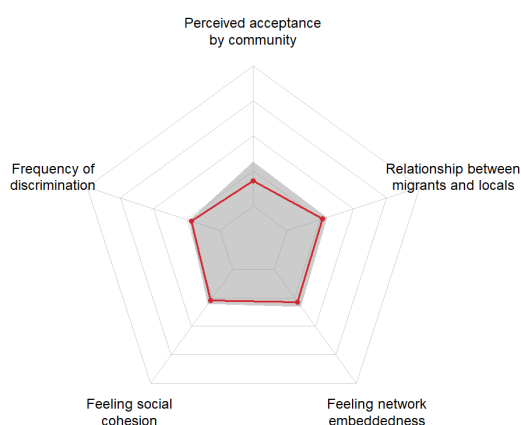
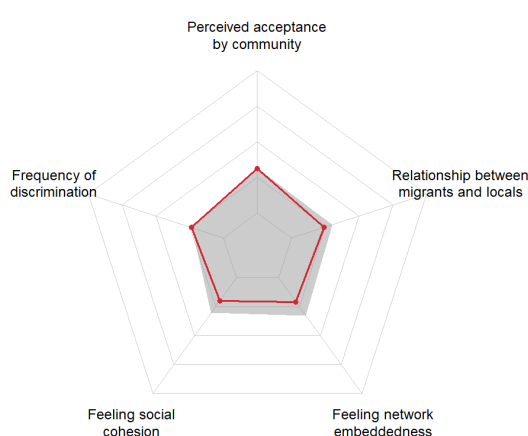


Figure 31. Societal Fragility in Bor South



²⁴⁰ Diana Felix da Costa, "Youth Gangs and Overcoming Waithood in a United Nations Protection of Civilians Site in South Sudan," (Journal of Intervention and Statebuilding, 2024).
²⁴¹ "Armed Violence Involving Community-based Militias in Greater Jonglei," (UNMISS, March 2021).

²⁴¹ James Kunhiak Muorwel, Jan Pospisil, Veronica Igale Monoja, "Caught Between Crises in South Sudan: Flood-Induced Migration of Dinka Bor Cattle Herders into the Equatoria Region," PeaceRep (2023).

9. IMPLICATIONS FOR POLICY & PROGRAMMING

Based on the results of the survey and focus group discussions, IOM developed the following policy and programmatic implications with input from the academic research team.

A. Investing in Agriculture-Driven Self-Reliance

The Fragility Study findings underscore that, although relatively few households consider themselves dependent on aid, they nonetheless experience high levels of economic fragility. This disparity indicates that current humanitarian assistance levels are insufficient to meet the basic needs of many South Sudanese individuals. Given growing disparities globally between need and humanitarian funding levels,²⁴² there is therefore a need to focus on investing in self-reliance.

The negative consequences of short-term humanitarian aid were highly salient in focus group discussions in which respondents frequently recalled receiving humanitarian support in the past — often short-term distributions of food, ration cards, shelter materials, or agricultural inputs — and expressed frustration when assistance ended or was delayed. In many cases, such interventions were perceived as ongoing commitments rather than temporary measures, and their termination generated disappointment and eroded trust.

These findings point to two challenges for humanitarian and development assistance to South Sudan: first, that current assistance models may not support communities in their transition toward self-sufficiency; and second, that opportunities for leveraging local resources and capacities remain underdeveloped. For example, agriculture — one of the country's most viable pathways to resilience — is often constrained by small, government-allocated plots that limit productivity. IOM's previous work has found that where agricultural interventions have been paired with sufficient land access, timely inputs, and capacity-building, they have shown clear potential to strengthen household self-reliance and reduce reliance on aid.



²⁴² OCHA, "Global Humanitarian Overview 2024" (Dec 11, 2023), <https://www.unocha.org/publications/report/world/global-humanitarian-overview-2024-enarfrs>. OCHA, "UN, partners unveil hyper-prioritized aid appeal amid 'cruel math' of brutal funding cuts" (June 16 2025), <https://www.unocha.org/publications/report/world/un-partners-unveil-hyper-prioritized-aid-appeal-amid-cruel-math-brutal-funding-cuts>

To address these challenges, future programming should:

- **Prioritize Self-Reliance-Oriented Agriculture:** Shift the focus from short-term aid delivery to long-term investment in agriculture as a resilience driver. Support community-based production systems, promote climate-smart practices, and advocate for improved land allocation policies that allow for larger, economically viable plots.
- **Harness Local Resources and Capacities:** Build interventions that maximize the use of locally available materials, skills, and markets to reduce external dependency and stimulate the green economy.
- **Strengthen Transition Pathways in Aid Delivery:** Design humanitarian inputs with a clear, built-in progression toward livelihood support – ensuring that assistance acts as a launchpad for self-reliance rather than a substitute for it.
- **Improve Timeliness and Alignment with Agricultural Cycles:** Ensure that seeds, tools, and other inputs reach communities in line with seasonal calendars to avoid wasted resources and missed opportunities for food production.

By rebalancing aid portfolios toward locally anchored, agriculture-driven solutions, programming can help communities move from reliance on external support toward sustainable, home-grown resilience.

B. Scaling Long-Term Investments to Build Sustainable Resilience

There is an urgent need to reorient support from short-term humanitarian interventions toward sustained, long-term investments that address the structural drivers of vulnerability. Repeated cycles of emergency relief — while essential in acute crises — have not created durable pathways out of fragility. Instead, they leave communities vulnerable to funding fluctuations and delivery gaps.

Global evidence from fragile and conflict-affected settings indicates that resilience-building interventions require a minimum of 5 to 10 years of sustained engagement to generate measurable, lasting outcomes. Multi-year programmes — particularly in agriculture, livelihoods and governance — allow for the gradual strengthening of local institutions, behaviour change, market linkages, and adaptation to shocks. Shorter project cycles (1–2 years), while useful for immediate relief, are insufficient to address systemic issues such as land tenure reform, climate-smart agricultural adoption, or market system development, which inherently require time to mature.

South Sudan possesses significant untapped potential in agriculture, renewable energy, and local market systems that, if developed over the long term, could serve as engines of economic growth and self-reliance. Expanding climate-smart agricultural production, improving land allocation, and building market linkages can strengthen household incomes, improve food security, and reduce reliance on external assistance. Coupled with investments in governance capacity, infrastructure, and conflict-sensitive development, such interventions can generate sustainable impact even in fragile and conflict-affected settings.

C. How the Fragility Index Can Support Area-Based Programming

With approximately 1,992,513 internally displaced persons and 2,963,408 returnees in South Sudan, responding to displacement and the vulnerabilities caused by this displacement is a critical priority for the humanitarian and development agenda.²⁴³ The fragility index can be used to better understand the area-based challenges that affect all populations, and where resources should be invested to enhance the recovery of areas that support all populations residing within them, regardless of status. The analysis can further strengthen approaches to support potential areas of integration for returnees whilst simultaneously addressing the existing needs within these areas.

Incorporating fragility analysis into project design can guide strategic resource allocation toward areas with the greatest potential for durable recovery, ensuring that investments strengthen services, infrastructure, and livelihoods for all residents – IDPs, returnees, and host communities alike. This approach also supports the identification of viable integration zones for returnees, allowing project teams to design interventions that foster social cohesion while simultaneously addressing existing gaps and vulnerabilities in host areas.

In practice, this means we can compare relative levels of a particular dimension of fragility across different counties. For example, Pibor county has the highest level of societal fragility, and Rubkona county has the highest level of political and legal fragility. Wau has the lowest level of economic fragility, and Yei has the lowest level of security fragility. This data would be helpful in the context of deciding which areas to allocate funds designated for a certain purpose; if the funds were designated for peacebuilding, for example, counties which have higher levels of security fragility may benefit more from those resources. Similarly, the results can help to identify areas of lower vulnerability where certain types of programming are more likely to succeed. For example, financing programmes for locally-led businesses are more likely to succeed in areas with lower economic fragility (e.g., Malakal and Wau) where beneficiaries are more likely to have the necessary skills, resources, and infrastructure to take advantage of the resources. However, development programming should also take into account other dimensions of fragility. For example, Malakal has a moderate level of security fragility driven in part by concerns about organized crime and gang violence,²⁴⁴ which present risks for business owners. Societal fragility issues in Malakal — including periodic conflicts between different ethnic groups, often driven by competition over land or other resources including cattle — have disrupted local markets in the past.²⁴⁵ The design and targeting of programming should take into account not only the most directly relevant dimension of fragility – for example, economic fragility in the context of programming to support small businesses) but also other dimensions of fragility (e.g., security, societal and environmental) that may interact with economic fragility.

Embedding displacement-sensitive planning into multi-sectoral, area-based programming — linking humanitarian response with development investments in health, education, livelihoods, governance and basic services. By designing projects around the shared needs of all communities within a fragile context, rather than along displacement status lines, programming can promote inclusion, reduce competition over limited resources, and contribute to sustainable recovery outcomes.

D. Leveraging the Fragility Index to Uncover Threats, Vulnerabilities, and Capacities Across Population Groups

The Fragility Index results can be analysed for variation in the same dimension of fragility between different sub-groups (e.g., women, IDPs, refugees, persons with disabilities) to inform status-based criteria for allocating aid and programming. For example, Table 2 presents the results of t-tests, a simple test of statistically significant differences between the means of two groups. These analyses indicate that IDPs are not significantly more fragile than host communities across most dimensions of fragility. IDPs are also not more fragile than returnees on any dimension of fragility.

²⁴³ IOM, DTM, Round 16.

²⁴⁴ Michael Daniel, “100 Youth Arrested on Gang Suspicion in Malakal,” Eye Radio (Sept. 9, 2024), <https://www.eyeradio.org/100-youth-arrested-on-gang-suspicion-in-malakal/>.

²⁴⁵ Radio Tamazuj, “Malakal traders confirm resumption of business, market stability,” (Jun. 28, 2024), <https://www.radiotamazuj.org/en/news/article/malakal-traders-confirm-resumption-of-business-market-stability>.

These T-tests also reveal systematic geographic disparities: people living outside county capitals face consistently higher economic and environmental fragility than urban residents, with more limited access to services, markets, and climate-resilient infrastructure. This finding underscores the need for Humanitarian-Development-Peace (HDP) programming to expand service delivery and livelihood opportunities in rural areas, strengthen rural climate adaptation measures, and ensure that targeting frameworks are not biased toward more accessible urban centres.

The Fragility Index could also provide data on variation in the vulnerabilities of individuals within a particular status-based group, helping improve the precision of targeting within that group. For example, it could help identify which women-headed households in a given area are most vulnerable to risks such as gender-based violence and would benefit from additional protection assistance.

Relatedly from above, the index should also support and guide in outlining the diverse roles and capacities of women and men hold within the communities and how they either mitigate barriers or accelerate the different levels of fragilities. As such, it is important to increase participation of women – as also active actors in decision-making like men to make holistic contributions to long-term local solutions to prevalent community challenges and concerns.



The index should be a cornerstone for developing concrete monitoring and evaluation frameworks that observe the extent to which milestones for addressing and overcoming complexities of fragilities are measured, alongside their impact against women and men. This includes defining indicators that are gender disaggregated – with corrective measures against the fragilities. Moreso, budgeting and allocation of resources to address these concerns that impact the status-based targeting in a gendered perspective should critically be considered.

The approach should also provide the extent to which human rights violations worsen and promote exclusion to equal access because of the different fragilities. This should inform measures to address the violations through engagements of relevant duty-bearers and rightsholders.

The index will be well-placed in refining the reporting and feedback mechanisms, following adaptations from emergencies contexts to a more structured and long-term approach. Besides, it will help create avenues for sharing relevant information – including early warnings, conflicts and climate trends – to the public or affected populations.

The fragility index will guide the formulation of laws and other legal frameworks that protect and respond to the needs, priorities and rights of women and men. For instance, climate-adaptation legal frameworks and contextualization of global and regional protocols that support the well-being in the context of climate change.

E. Leveraging the Fragility Index for Programme Assessments

The Fragility Index can be used to conduct household-level assessments of different types of vulnerability, enabling more precise targeting. While this study was conducted with random sample anonymous surveys to generate representative, population-level estimates without collecting sensitive personally identifying information, the same tool could be easily tailored for direct use in operational contexts. HDP actors could integrate the Fragility Index's 30-question module into existing registration or monitoring systems to produce detailed vulnerability profiles at the household level.

This approach would be particularly relevant for programmes that already rely on individual-level targeting, such as cash transfers, shelter support, and in-kind distributions. By identifying households with the highest fragility scores in specific dimensions, implementing agencies could prioritize assistance for those most in need. For example, the Index could pinpoint households most at risk of food insecurity, lacking access to safe water, or living in flood-prone areas – information that can inform the allocation of agricultural inputs, WASH interventions or climate adaptation support.

Moreover, a household-level application of the Index could help address gaps in the current aid landscape, where status-based targeting (e.g., solely focusing on IDPs or returnees) does not always align with actual vulnerability levels. Using the Index in this way would ensure that assistance reaches vulnerable households regardless of displacement status, gender, or location, and could also support longitudinal tracking of households over time to assess whether interventions are effectively reducing fragility.

F. Strengthening Local Governance and Justice Mechanisms

This study highlights low trust in formal institutions and high reliance on customary systems. This emphasizes the need to reimagine governance-focused peace programming so that it builds legitimacy and delivers tangible results at the community level. HDP actors should:

- **Embed Hybrid Governance Models:** Foster collaboration between statutory and customary authorities, establishing joint dispute resolution platforms that blend formal legal standards with community-trusted practices. This is particularly urgent for HLP disputes, where overlapping mandates and weak enforcement perpetuate insecurity.
- **Institutionalize Inclusive Participation:** Ensure that women, youth, and marginalized groups hold structured representation in both formal and informal governance processes, moving beyond ad hoc consultations to permanent roles in decision-making bodies.
- **Link Local Justice to Peacebuilding Outcomes:** Use HLP and other local dispute resolution processes as entry points to address wider political-legal fragility, embedding them in broader reconciliation and social cohesion efforts.
- **Invest in Civic Education and Safe Dialogue Spaces:** Pair technical governance support with community-led civic education campaigns and accessible, safe spaces for dialogue to strengthen democratic participation and counter political exclusion.
- **Create Local Accountability Loops:** Use sub-national Fragility Index data as a public accountability tool, enabling communities to track improvements (or deterioration) in governance performance over time and press for corrective action.

This approach moves beyond capacity building in isolation, positioning governance reform as a pathway to both immediate conflict mitigation and longer-term state-citizen trust-building.



G. Directions for Future Research

This proof-of-concept pilot of the Fragility Index has provided preliminary evidence of the tool's potential benefits for evidence-based design, implementation, and evaluation of transitional and recovery programming in South Sudan and other fragile contexts. More research is needed to enable the following next steps:

- Expanding the geographical coverage of the Fragility Index to additional counties in South Sudan and adapting the tool for use in other fragile countries;
- Estimating changes in different dimensions of fragility over time through subsequent rounds of data collection in the same counties;
- Supporting the dissemination and uptake of the Fragility Index and data for use by other humanitarian and development actors through IOM's D4S website, the Humanitarian Data Exchange, and other public platforms;
- Improving triangulation, complementarities, data-sharing, and learning between the Fragility Index and other data for solutions tools including the MIT, the SI, and the SMI, and the Durable Solutions Index (DSI);
- Improving random sampling procedures with population density estimates derived from satellite data in the absence of recent reliable census estimates;
- Further validation of the Fragility Index and mitigation of potential biases through additional testing by independent third-party research teams;
- Using the data to inform testable theories of change for programmatic interventions aimed at reducing or mitigating one or more dimensions of fragility;
- Using subsequent rounds of the Fragility Index as baseline and endline assessments to support rigorous impact evaluations of relevant programmatic interventions;
- Implementing fragility-based solutions to aid displaced persons and, in doing so, support host and returnee communities, given that data supports generally strong correlations between the fragilities of these groups.

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APPENDIX A

Index questions

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Human Fragility [Health]	Food Access	Q5.5 Over the past year, how often, if ever, have you or anyone in your family gone without the following items: never, just once or twice, several times, many times, or always? → Q5.5.1 Enough food to eat 0. Never 1. Just once or twice 2. Several times 3. Many times 4. Always -2. Prefer not to answer	NA	No
	Water Access	Q5.5 Over the past year, how often, if ever, have you or anyone in your family gone without the following items: never, just once or twice, several times, many times, or always? → Q5.5.2 Enough clean water for home use 0. Never 1. Just once or twice 2. Several times 3. Many times 4. Always -2. Prefer not to answer	NA	No
	Medical Treatment Access	Q5.5 Over the past year, how often, if ever, have you or anyone in your family gone without the following items: never, just once or twice, several times, many times, or always? → Q5.5.3 Necessary medicines or medical treatment 0. Never 1. Just once or twice 2. Several times 3. Many times 4. Always -2. Prefer not to answer	NA	No
	Shelter	Q7.1 What type of shelter are you currently living in? 1. No shelter 2. Rabooka 3. Tukul 4. Bricked Structure 5. Other, Specify^[1] -2. Prefer not to answer	NA	Yes
	Sexual violence	Q8.5 On a scale from 1 to 5, where 1 means “very unfrightened” and 5 means “very frightened,” how afraid are you of the following happening in your community? → Q8.5.6 Sexual violence 1. Very unfrightened 2. Unfrightened 3. Neither frightened nor unfrightened 4. Frightened 5. Very frightened -2. Prefer not to answer	NA	No

^[1] This answer was converted into NA

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Economic Fragility	Education	Q2.3 What is the highest level of schooling you have completed? - No formal schooling - Primary school - Secondary school - Vocational courses only - University - Prefer not to answer	NA	Yes
	Financial satisfaction	Q5.6 From 1 to 5, where 1 is “very dissatisfied” and 5 is “very satisfied,” how satisfied are you with your household’s financial situation? - Very dissatisfied - Dissatisfied - Neither dissatisfied nor satisfied - Satisfied - Very satisfied - Prefer not to answer	NA	Yes
	Ability to pay for expected large costs	Q5.7 On a scale from 1 to 5, where 1 is “very difficult” and 5 is “very easy,” how easy would it be for your family to pay a bride price currently? - Very difficult - Difficult - Neither difficult nor easy - Easy - Very easy - Prefer not to answer	Replacement of “bride price” with locally relevant expected large cost	Yes
	Ability to pay for unexpected large costs	Q5.9 How severe would you say the impact of an unexpected death would be on your household’s economic situation? - Not severe at all (no impact) - Not severe - Somewhat severe - Severe - Extremely severe - Prefer not to answer	Replacement of “unexpected death” with locally relevant unexpected large cost	No
	Dependency on aid	Q5.22 On a scale from 1 to 5, where 1 is “very difficult” and 5 is “very easy,” how easy would it be for your family to pay a bride price currently? - No dependency - Low dependency - Moderate dependency - High dependency - Full dependency - Prefer not to answer	NA	No

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Environmental Fragility	Climate change's impact on income	Q11.10A Many people in South Sudan have experienced the following: too much rain, too little rain, temperatures that are too high, temperatures that are too low, pests and crop diseases, and deforestation and erosion. How severe would you say the impact of any of these experiences have been on your household income, on a scale from 1 to 5 where 1 means “not severe at all and 5 means “extremely severe?” 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	Replacement of “South Sudan” with relevant country and list of climate hazards with locally relevant ones	No
	Climate change's impact on food consumption	Q11.10B And how severe would you say the impact of any of these experiences have been on your household food consumption? 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	NA	No
	Client change's impact on mobility	Q11.10C And how severe would you say the impact of any of these experiences have been on your household's ability to choose where you live? 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	NA	No
	Climate change's impact on health	Q11.10D And how severe would you say the impact of any of these experiences have been on your household's health? 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	NA	No
	Ability to act on early warning information to ensure household safety	Q11.11 Imagine that there was a flood coming to your town tomorrow. If you knew about this incoming flood before the water arrived, to what extent do you agree with the following statement: “I would be able to use this information to protect my household?” 1. Strongly disagree 2. Disagree 3. Neither agree nor disagree 4. Agree 5. Strongly agree -2. Prefer not to answer	Replace “flood” with locally plausible climate change-induced emergency	Yes

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Societal Fragility	Experiences of acceptance	Q6.1 How much do you feel welcomed or accepted in your current community? - Not at all accepted - A little accepted - Accepted - Very accepted - Prefer not to answer	NA	Yes
	Experiences of discrimination	Q6.2 In the past year, how often, if at all, have you been personally discriminated against based on any of the following: your gender, your religion, your ethnicity, any disability you might have, or your place of origin? - Never - Sometimes - Often - Always - Prefer not to answer	NA	No
	Social cohesion	Q6.3 To what extent do you agree with the following statement? “People in your boma are willing to help each other.” - Strongly disagree - Disagree - Neither agree nor disagree - Agree - Strongly agree - Prefer not to answer	Replace “boma” with locally relevant geographic boundary	Yes
	Network embeddedness	Q6.4 Imagine that you were in an argument with your neighbor about a small issue. To what extent do you agree or disagree with the following statement? “I know who I could seek help from to resolve this argument.” - Strongly disagree - Disagree - Neither agree nor disagree - Agree - Strongly agree - Prefer not to answer	NA	Yes
	Migrant-Local Cohesion	Q6.9 How would you describe the relationship between individuals who are not originally from the community and the rest of the community members? - Very poor - Poor - Neutral - Good - Very good - Prefer not to answer - Not applicable^[2]	NA	Yes

^[2] This answer was converted into NA

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Legal/ Political Fragility	Trust in state government	Q10.1 How much confidence do you have in the following institutions? → Q10.1.2 Your state government 1. No confidence 2. Some confidence 3. Confidence 4. A lot of confidence -2. Prefer not to answer	NA	Yes
	Trust in informal authorities	Q10.1 How much confidence do you have in the following institutions? → Q10.1.10 Informal authorities, such as councils of elders 1. No confidence 2. Some confidence 3. Confidence 4. A lot of confidence -2. Prefer not to answer	Replace “councils of elders” with locally relevant informal authorities	Yes
	Participation in democracy without fear	Q10.10 South Sudan’s next national elections are scheduled for December 2024 after being postponed several times. How safe or unsafe would you feel going to a polling station to cast your vote? 1. Very unsafe 2. Somewhat unsafe 3. Neither safe nor unsafe 4. Somewhat safe 5. Very safe -2. Prefer not to answer	Replace first sentence with relevant details about upcoming national election	Yes
	Freedom of speech	Q10.12 At present, how safe do you feel speaking publicly on politically sensitive issues? 1. Very unsafe 2. Somewhat unsafe 3. Neither safe nor unsafe 4. Somewhat safe 5. Very safe -2. Prefer not to answer	NA	Yes
	Trust in national government	Q10.13 Which of the following statements best describes the role that politicians in Juba play in your community? 1. Politicians in Juba make this community much less safe 2. Politicians in Juba make this community less safe 3. Politicians in Juba neither make this community more nor less safe 4. Politicians in Juba make this community more safe 5. Politicians in Juba make this community much more safe -2. Prefer not to answer	Replace “Juba” with national capital	Yes

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Security Fragility	Affected by crime	Q8.5 On a scale from 1 to 5, where 1 means “very unfrightened” and 5 means “very frightened,” how afraid are you of the following happening in your community? → Q8.5.2 Violent crimes 1. Very unfrightened 2. Unfrightened 3. Neither frightened not frightened 4. Frightened 5. Very frightened -2. Prefer not to respond	NA	No
	Affected by war	Q8.5 On a scale from 1 to 5, where 1 means “very unfrightened” and 5 means “very frightened,” how afraid are you of the following happening in your community? → Q8.5.3 Armed conflict / War 1. Very unfrightened 2. Unfrightened 3. Neither frightened not frightened 4. Frightened 5. Very frightened -2. Prefer not to respond	NA	No
	Affected by land conflict	Q8.5 On a scale from 1 to 5, where 1 means “very unfrightened” and 5 means “very frightened,” how afraid are you of the following happening in your community? → Q8.5.7 Land conflicts 1. Very unfrightened 2. Unfrightened 3. Neither frightened not frightened 4. Frightened 5. Very frightened -2. Prefer not to respond	NA	No
	Affected by resource conflict	Q8.5 On a scale from 1 to 5, where 1 means “very unfrightened” and 5 means “very frightened,” how afraid are you of the following happening in your community? → Q8.5.8 Conflicts over cattle 1. Very unfrightened 2. Unfrightened 3. Neither frightened not frightened 4. Frightened 5. Very frightened -2. Prefer not to respond	Replace “cattle” with locally relevant natural resource	No
	Affected by local armed militias	Q8.5 On a scale from 1 to 5, where 1 means “very unfrightened” and 5 means “very frightened,” how afraid are you of the following happening in your community? → Q8.5.9 Violence involving community-based militias, vigilante groups, or self-defense groups 1. Very unfrightened 2. Unfrightened 3. Neither frightened not frightened 4. Frightened 5. Very frightened -2. Prefer not to respond	NA	No

APPENDIX B

Sampled Bomas

State	County	Payam	Boma	Capital	In Original Sampling Plan	# of Respondents
Jonglei	Bor South	Baidit	Mathiang	0	Yes	105
Jonglei	Bor South	Jalle	Akuai-deng	0	Yes	103
Jonglei	Bor South	Bor	Panapet	1	Yes	113
Jonglei	Bor South	Bor	Langbar	1	Yes, backup	108
Central Equatoria	Kajo-keji	Nyepo	Kansuk	0	Yes	91
Central Equatoria	Kajo-keji	Nyepo	Lori	0	Yes, backup	102
Central Equatoria	Kajo-keji	Lire	Mekir	1	Yes	97
Central Equatoria	Kajo-keji	Lire	Likamerok	1	Yes	87
Central Equatoria	Kajo-keji	Lire	Mere	1	Yes, backup	21
Central Equatoria	Lainya	Mukaya	Dimu One	0	Yes	77
Central Equatoria	Lainya	Kopera	Korobe	0	Yes	100
Central Equatoria	Lainya	Mukaya	Biri	0	Yes, backup	27
Central Equatoria	Lainya	Lainya	Lokurubang	1	Yes	122
Central Equatoria	Lainya	Lainya	Bereka	1	Yes	88
Upper Nile	Malakal	Lelo	Obwa	0	Yes	51
Upper Nile	Malakal	Ogod	Wau	0	Yes, backup	77
Upper Nile	Malakal	Ogod	Padit	0	Yes	73
Upper Nile	Malakal	Malakal North	Hai Chathi	1	Yes	124
Upper Nile	Malakal	Malakal South	Tarawa	1	Yes, backup	26
Upper Nile	Malakal	Malakal Centre	Mederia	1	Yes	70
Central Equatoria	Morobo	Gulumbi	Kendila	0	Yes	104
Central Equatoria	Morobo	Lujulo	Yugufe	0	Yes	100
Central Equatoria	Morobo	Kimba	Kaya	1	Yes	100
Central Equatoria	Morobo	Kimba	Kimba Centre	1	Yes	100
Jonglei	Pibor	Lekuangle	Lekuangle	0	No	102
Jonglei	Pibor	Gumruk	Tangnyang	0	No	101

Jonglei	Pibor	Pibor	Kavachoch	1	No	101
Jonglei	Pibor	Pibor	Pibor	1	No	103
Unity	Rubkona	Nhialdiu	Nyaromne	0	No	121
Unity	Rubkona	Panhiany	Panhiany	0	No	80
Unity	Rubkona	Rubkona	Nying	1	Yes	29
Unity	Rubkona	Rubkona	Tong	1	Yes	50
Unity	Rubkona	Rubkona	Yoanyang	1	Yes, backup	120
Western Bahr el Ghazal	Wau	Bagari	Bringi	0	Yes	50
Western Bahr el Ghazal	Wau	Bagari	Bussere	0	Yes	152
Western Bahr el Ghazal	Wau	Wau North	Hai Kalvario	1	Yes	100
Western Bahr el Ghazal	Wau	Wau South	Nazareth	1	Yes	100
Central Equatoria	Yei	Mugwo	Longamere	0	Yes	94
Central Equatoria	Yei	Tore	Adio	0	Yes	100
Central Equatoria	Yei	Yei Town	Hai Sobe Zone	1	Yes	116
Central Equatoria	Yei	Yei Town	Pakkula	1	Yes	100

APPENDIX C

Index Reliability

Fragility Type	Cronbach's Alpha	Variance Explained by First Factor
Human	.45	49%
Economic	.34	38%
Environmental	.59	86%
Social	.53	53%
Legal/Political	.53	53%
Security	.86	48%

APPENDIX D

Malakal Case Study



Introduction

Malakal Town represents one of the most complex displacement situations in South Sudan. When this research was conducted in July 2024, five demographic groups lived in homogenous settlements in and around Malakal Town. While many of these groups shared displacement statuses and ethnocultural backgrounds, they faced stark differences in their assistance needs: some displaced groups were registered with the World Food Programme and could receive food assistance, while others had no access at all; some had government-provided shelter, while others from the same community had none at all; some had homes to reclaim, while others did not. The value of the fragility index is that it reveals the variations in challenges within groups that otherwise receive similar packages of support because other assessment tools place them in the same categories. Qualitative data about civilian experiences in Malakal illuminates how the fragility index reveals information that can support more tailored targeting of aid.

This case study evaluates data from eight focus groups in Malakal Town, observations of return processing at Malakal's transit center, and more than eight in-depth interviews with IOM staff implementing projects in the town. Malakal has been home to a multicultural population, including Shilluk, Dinka (mostly Padang), Nuer, Mabanese and Koma. It has also, at various times, been the center of land disputes among these groups, leading to violence, including between Padang Dinka and Shilluk

since July 2009. Malakal serves as a transit point for returnees from Sudan who arrive in the Nile Port to apply for onward transportation assistance (OTA). It is also the site of settlements of a largely Shilluk population on the UN PoC site, Nuer displaced by floods from northern Jonglei and Eastern Upper Nile who reside in a former fire station, and Nuer living in make-shift shelters on government-managed land (referred to as State Garden) who fled the PoC in June 2023 after confrontations with the Shilluk. In July 2024, the research team had in-depth discussions with community leaders and chiefs from all of these groups, as well as with state government officials.

This case study begins with a brief background on the politics of Malakal and how this history produced the displacement and settlement patterns that defined the town in 2024. The study then reflects on civilians' variable experiences of each of the six fragility dimensions.

Background

Prior to December 2013, Malakal served as a cosmopolitan center of politics and trade.¹ The Shilluk have claimed that Malakal sits at the center of their ancestral lands, while Malakal County borders lands traditionally claimed by the Nuer and Padang Dinka. Each of these groups have centers of community-based armed mobilization in easy striking distance of Malakal – Fashoda, for the Shilluk community-based militias, Baliet for Padang Dinka and Nasir for the Jikany section of the Nuer community.

Malakal is the gateway to South Sudan's most productive oilfields.² Some analysts argue that control of the South Sudanese state hinges on control of the Upper Nile oilfields, which in turn hinges on control of Malakal.³ Therefore, when the South Sudanese state fragmented in December 2013, the SPLA and SPLM-IO battled fiercely for control of Malakal. Civilians from many communities and ancestral lands – including, but not only, Nuer, Shilluk and Dinka – fled to the UN for protection as combat forces conducted widespread civilian killings.⁴

At the outset of the war, the Shilluk leader, Johnson Olony, who had mobilized a force called the Agwelek, struck

¹ Joshua Craze, "Displaced and Immiserated: The Shilluk of Upper Nile in South Sudan's Civil War, 2014-2019," Small Arms Survey, 2019, p23.

² Laura James, "Fields of Control: Oil and (in)Security in Sudan and South Sudan," Small Arms Survey, 2015, p38.

³ "The Malakal Attacks and Operationalizing the 28 States," USAID VISTAS Internal Report, March 9, 2016.

⁴ United Nations Mission in South Sudan, Conflict in South Sudan: A Human Rights Report (2014), para. 5.

an alliance with the government's SPLA.⁵ The hope was that this alliance against the SPLM-IO would be the most effective strategy for safeguarding Shilluk Lands. However, Kiir did not provide Olony with the political positions he had hoped to win in exchange for his support. In April 2015, a Padang Dinka miliary leader allegedly assassinated Olony's Deputy.⁶ Olony felt betrayed and mutinied with his Shilluk troops. In October 2015, President Salva Kiir expanded the number of states from 10 to 28, creating a Padang Dinka-controlled Eastern Upper Nile State containing all Upper Nile's oil reserves and giving the Shilluk two non-contiguous pockets of land divided by Malakal.⁷

These conflicts stoked unrest on the UN PoC site in Malakal. In January 2016, occasional fights occurred between Dinka and Nuer youth on the camp. Then, on 17 February 2016, armed men with SPLA uniforms entered the camp. Dinka women and children are reported to have departed the camp. Nuer and Shilluk community defense groups clashed with Padang Dinka youth militia.⁸ SPLA soldiers and Padang Dinka militia members entered the camp on 18 February and systematically burnt dwellings inhabited by Nuer and Shilluk community members, leaving Dinka and Darfuri dwellings intact. Following these events, the UN PoC had a predominantly Shilluk and Nuer population, while Dinka inhabitants had relocated into Malakal Town.⁹

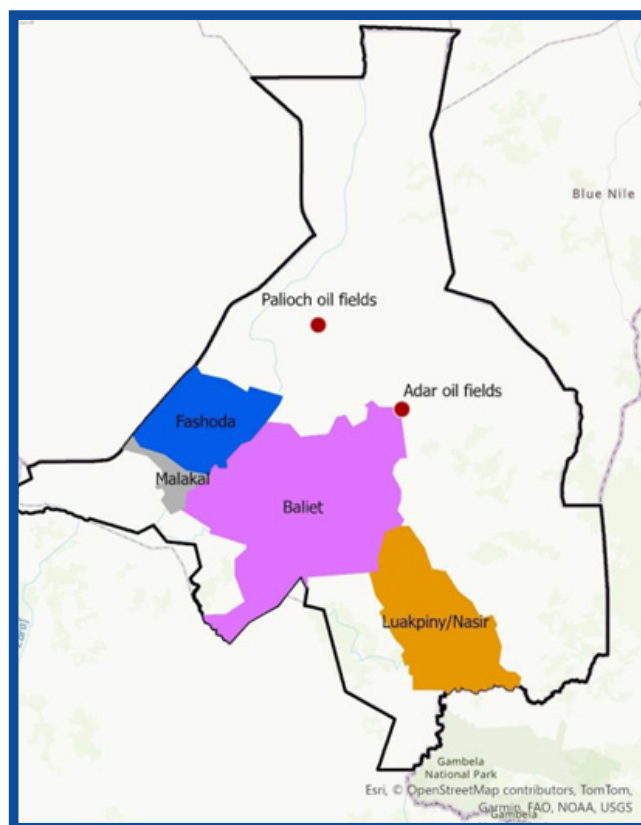
From July 2022, patterns of political division and population settlement in Malakal shifted again due to the arrival of displaced Nuer from conflicts in western Upper Nile State and northern Jonglei. Olony's (largely Shilluk) Agwelek forces clashed with SPLA-IO General Simon Gatwech's Kitgwang faction around the payams of Tong and Atar. In December 2022 alone, 1,400 civilians fleeing this violence entered Malakal's PoC.¹⁰ However, these disputes also led to distrust amongst Nuer and Shilluk on the PoC, inducing some Nuer to relocate from the camp into Malakal Town.¹¹

In 2023, the situation in Malakal became even more complicated: historic floods in northern Jonglei and Eastern Upper Nile pushed Nuer who had lost their homes into Malakal town. These flood-displaced populations began to take up residence in Malakal's former fire station. In

April 2023, the outbreak of civil war in Sudan led to large flows of South Sudanese returnees through Renk and Maban, then through Malakal as a central transit point. Then, in June 2023, a clash escalated between Nuer and Shilluk communities in the PoC, leading to the displacement of Nuer to Malakal town. These Nuer households stayed in a church compound and on the banks of the Nile. The local government then offered almost half of these Nuer households' temporary shelter on land referred to as 'State Garden.'

Thus, in 2024, the UN PoC housed a predominantly Shilluk population, with Nuer displaced located in different temporary shelter arrangements across Malakal, and a predominantly Padang Dinka population occupying permanent residences in the town. Meanwhile, returnees from Sudan were moving in large numbers through a transit center at Malakal's Nile port.

Figure 1. Centers of Community-based Arms Mobilization



⁵ Craze, "Displaced and Immiserated: The Shilluk of Upper Nile in South Sudan's Civil War, 2014-2019," p31.

⁶ "The Malakal Attacks and Operationalizing the 28 States."

⁷ Craze, "Displaced and Immiserated: The Shilluk of Upper Nile in South Sudan's Civil War, 2014-2019," p58.

⁸ "The Malakal Attacks and Operationalizing the 28 States"; UNMISS, "Press Release: UNMISS Condemns Violence in Malakal PoC Site," February 18, 2016.

⁹ Craze, "Displaced and Immiserated: The Shilluk of Upper Nile in South Sudan's Civil War, 2014-2019" p65.

¹⁰ "S/2023/294 Final Report of the Panel of Experts on South Sudan," United Nations Security Council, April 2023, para 75.

¹¹ South Sudan Biweekly Context Update, NGO Forum, 9 September 2022.

Fragility Components

The demographic and political history of Malakal underlines the diverse challenges to resettlement within displaced populations, including amongst groups that share a cultural background and language. People's expressions of their sources of fragility during the research team's discussions in Malakal provided stark illustrations of these variations.

Economic Fragility

Local government officials described how people in Malakal, across the board, were facing economic hardships due to hyperinflation.¹² However, of the groups consulted, the community leaders from the Nuer displaced in State Garden and the Nuer flood-displaced in Fire Brigade described the most acute barriers to livelihoods activities. In both informal settlements, community leaders shared how local gangs attacked women gathering firewood and made it too risky to move even short distances to plant plots of land or visit the market.¹³ The community leaders at Fire Brigade also described how their community was outside the remit of government assistance or the services of humanitarian organizations, and that they lacked the most basic tools, like shovels, to maintain sanitation or plant seeds for food.

Environmental Fragility

Malakal has provided a safe location for the flood displaced. As such, most populations who plan to remain in Malakal or see Malakal as their habitual residence view environmental fragility as less of a risk than in other parts of South Sudan. The Nuer flood-displaced in Fire Brigade described the limited options they faced for return and expressed deep fears about the floods that were still to come – noting that returns during the dry season would leave them vulnerable to being submerged in their homes again.

Human Fragility

Hunger was a theme in every focus group, including with Shilluk Leaders of the PoC, the PoC Community Committee, and Community Leaders in Fire Brigade and State Garden.¹⁴ Everybody the research team spoke to described shortfalls in their households' abilities to fulfil their calorific needs. Community leaders in State Garden expressed a distinction between those who had

registration cards with the World Food Programme (WFP) and could attend food distributions, and those who were not registered. The residents in State Garden included a combination of registered and unregistered households, and the community leaders described practices of sharing food and shelter within the community even when no single recipient household had enough. Very few, if any, households in Fire Brigade were registered with WFP – exposing the group to levels of hunger apparently more extreme than for residents of the PoC or elsewhere in Malakal Town.

Dinka chiefs in Malakal Town also made strong claims about their vulnerability to hunger.¹⁵ As one chief put it:

"We are experiencing poverty close to famine. And humanitarians are not being fair when they distribute aid. So, people are turning against each other for scraps. People in the POC receive food, people in town do not. People in town may cause trouble as a result. Our community is the most vulnerable but doesn't receive aid."

Political-Legal Fragility

HLP issues in Malakal Town present a central challenge to both resettlement and security. Local government officials outlined extensive efforts to develop a land registry and ensure an ordered, transparent process for adjudicating land claims, and asserted that some residents from the PoC had attempted to reclaim dwellings. However, discussions with IOM staff revealed that attempts to reclaim dwellings had led to violent intimidation, the dismantlement of houses and theft of building materials.

The Nuer displaced who were resident in State Garden had received shelter materials, and local government and humanitarian organizations had provided limited WASH facilities. But local government had not given land title to this population or documented its donation for use for temporary shelter. Indeed, IOM staff shared that Shilluk leaders had expressed grievances about once occupying the land and giving it to government for agricultural development.

The Shilluk leaders on the PoC expressed deep resistance to leaving the camp because of their assertion that the camp lay on Shilluk lands that they would lose if they moved. As one Shilluk Leader put it during the FGD on the camp:

¹² Group interview, local government officials, Malakal Town, 23 July 2024.

¹³ FGD, community leaders at Fire Brigade, Malakal Town, 24 July 2024; FGD, community leaders at State Garden, Malakal Town, 24 July 2024.

¹⁴ FGD, PoC Community Committee, Malakal PoC, 22 July 2024; Shilluk Leaders of the PoC, Malakal PoC, 25 July 2024.

¹⁵ FGD, Dinka Chiefs, Malakal Town, 26 July 2024.

"Where we sit now is the land of my ancestors. But right now, this land is not considered part of Malakal County. All these areas around here used to be the farms and houses of our people. Now we cannot claim it back."

Security Fragility

In 2024, Malakal PoC was the last to remain under UN management. Participants in FGDs on the camp, with Dinka chiefs in town and with the displaced at State Garden all shared accounts of the violence that had centered on the PoC camp. However, the greatest security challenge to the daily lives of many residents, displaced and settled, was the operation of local youth gangs. A Nuer community leader from State Garden shared that gang activity meant:

"We can't go and collect firewood because people will be raped. Young women or old women, it doesn't matter, they will be raped. You can't go to the market and get things for the children. It's terrible."

A Dinka chief described how the gangs operated between the PoC and town:

"Some of the youths in gangs were born here, some are migrants, but the majority are from here. The gangs don't discriminate in who they target; they attack for any items of value. They use sticks or knives, but if you have a gun they will try to take it. They rape women and beat people. They sell the items in the PoC. The items they take from people in the PoC they sell in town. The gangs are organized and different components are of mixed ethnicity."

Societal Fragility

The homogeneity of settlements – such as the PoC, which is predominantly Shilluk, and informal IDP settlements in town, which are Nuer – has the potential to entrench mistrust between the communities. These settlement patterns offer few opportunities for people from different groups to mix in their daily lives or collaborate on common community projects. However, focus group discussions revealed examples of internal and inter-group solidarity. At Fire Brigade, where residents were experiencing high levels of hunger and very limited assistance, community leaders expressed their support for the way humanitarian organizations were marshalling resources to South Sudanese returnees from Sudan. As one participant put it:

"When the war broke out in Sudan, IOM started helping returnees. We here feel this is very important, and we do not blame IOM."

IOM staff who had assisted with the creation of the State Garden settlement described the mutual aid practices that had developed amongst the Nuer displaced.¹⁶ When the Nuer had left the UN PoC in June 2023, they had constructed temporary camps in a church compound and on the banks of the Nile. The State Garden settlement was able to serve less than half the households that needed shelter and households were chosen by lottery. Although the shelters were designed for this limited number of households, many of those that were successful in the lottery housed those who had been unsuccessful and went to considerable lengths to share resources equitably.

Women amongst the Nuer displaced in State Garden and Fire Brigade experienced distinct rights risks. Accounts across FGDs suggested that gangs disproportionately targeted women, due both to their gender and to the types of activities they are generally responsible for – collecting firewood and gathering wild greens for food. A woman respondent at the Fire Brigade described:

"It is not only us being attacked. It is all women in town and the PoC. The gangs don't distinguish, it's anyone going to the bush. Some of us women can do livelihood activities in town, like selling nuts. But most of us are living on what we can collect from the bushes. Collecting from the bushes to keep our children alive."

A male participant at Fire Brigade explained the social barriers for men to assume responsibilities like collecting food in the bush:

"Socially there are some responsibilities for men and women. Men go to the bush to cut trees for construction, firewood and charcoal."

Implications for Fragility Index

Status-based measures of vulnerability have the potential to place the Nuer displaced in State Garden and Fire Brigade and the Shilluk population in the UN PoC in the same category, and designate them for the same kinds of assistance. However, qualitative evidence from Malakal demonstrates the different types of fragility each of these populations face across all six dimensions. In a place like Malakal, the fragility index offers a tool for tailoring assistance packages to groups in a way that effectively matches their distinct situations, in the same geographic location at the same temporal moment of crisis.

¹⁶ Interview with IOM staff member, Malakal, 24 July 2024.

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